Hirzebruch–Riemann–Roch Theorem

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August 22, 2026

Abstract

The Riemann Roch problem asks how many holomorphic, meromorphic, or algebraic sections a space admits — equivalently, what can be said about $H^0(M, \mathcal{F})$, and what the higher sheaf cohomology says when that group is smaller than one would expect. This paper is an exposition of the *Hirzebruch-Riemann-Roch Theorem*, which answers this for a complex vector bundle E over a projective variety M:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \smile \mathrm{td}(M).$$

The proof is organized around two generating functions — the χ_y-characteristic, which packages the Euler characteristics, and the T-characteristic T_y, which packages the Chern character and Todd class data. The two agree at $y = 1$ with $\chi(M, E)$, and at $y = -1$ with the spin index $\tau(M)$; that second agreement is the bridge which forces the theorem, and cobordism then reduces the statement to the case $M = \mathbb{C}\mathrm{P}^n$.

The exposition deliberately reverses the order of Hirzebruch's original treatment, introducing each tool at the point where the problem calls for it rather than in advance: Chern classes and the Chern character, via the splitting principle and K-theory; Dolbeault cohomology and Kähler geometry, to relate the analytic and algebraic Euler characteristics; the Todd class, as a multiplicative sequence; and the oriented cobordism ring together with the Hirzebruch signature theorem. A closing chapter sketches Grothendieck-Riemann-Roch and surveys the current generalizations in the literature.

Contents

This paper is an exposition on the *Hirzebruch-Riemann-Roch Theorem*, a generalization of the classical Riemann Roch theorem for curves and Riemann surfaces. The Classical Riemann Roch Theorem asserts that if D is a (Weil) divisor of a curve/Riemann surface C, the dimension of the vector space of functions on C satisfying D is dependent on $\deg(D)$ and the genus g. The goal of this paper is to show that the result generalizes to a projective variety M and show that the appropriate generalization of the divisor D is to consider vector bundles E on M.

In particular, this paper shall show that if M is a projective variety, E a complex vector bundle (equivalently a coherent sheaf), $\mathrm{ch}(E)$ the *Chern character* of E, and $\mathrm{td}(M)$ the *Todd class* of M, then:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \smile \mathrm{td}(M) \tag{1}$$

where $\chi(M, E)$ is the Euler characteristic for vector bundles and $\int_M$ is the evaluation of $\mathrm{ch}(E) \smile (M)$ at the fundamental class $[M]$. The high-level of the proof is to define two generating functions: one for the Euler-characteristic for (complex) vector bundles and the other for something that shall be called the *T-Characteristic* with respect to M and E which shall hold the information of $\mathrm{ch}(-)$ and $\mathrm{td}(-)$. Let $\chi_y(M, E)$ and $T_y(M, E)$ represent these two generating functions with dummy variable y. Then, it shall be shown that when $y = 1$:

$$T_1(M, E) = \chi_1(M, E) = \chi(M, E)$$

and when $y = -1$:

$$T_{-1}(M) = \chi_{-1}(M) = \tau(M)$$

where the bundle E was purposefully omitted, and τ is the *spin index* of M. The key connection is the spin index. Showing that both χ_y and T_y agree with τ when $y = -1$ shall be a driving force in this paper, namely it shall be the "bridge" which gives the equality in equation (1). Furthermore, by the use of cobordism, it shall be enough to show the result in the case where $M = \mathbb{C}\mathrm{P}^n$ for all n.

The proof of HRR requires many results from fields such as Hodge Theory, GAGA, cobordism theory, algebraic topology, and some other miscellaneous results. This paper will include the "core" results that do not require much build-up given the assumed necessary background (see box 0.0.1). The paper shall focus on the necessary background to define the aforementioned concepts, leaving some proofs as references to other textbooks (or to my notes whenever I have proven it elsewhere to save space in this paper).

The first chapter is to motivate the generalization. The goal is for someone with the necessary background who is comfortable with the Riemann Roch Theorem to be able to follow the generalizations and understand the choice of topics covered in this paper.

The second chapter reviews the necessary characteristic theory, giving a proof of the splitting principle and overviewing the construction of the Chern and Pontryagin Characters. The last section of the chapter goes over the definition of the Chern character and also sets up some motivation for the generalization of HRR to the Grothendieck Riemann Roch Theorem.

The third chapter goes into the cohomology of projective varieties. Wherever possible, the results are stated to work on almost-complex manifolds, but these can all be ignored without concern for future results in the paper. An important unique element that follows Hirzebruch's idea is the definition of the *virtual Euler characteristic* which is the Euler characteristic for a filtration of codimension 1 sub-varieties at each inclusion:

$$V_r \subseteq V_{r-1} \subseteq V_{r-2} \subseteq \cdots \subseteq V_1 \subseteq M$$

the key take-away will be to relate the Euler-characteristic to the virtual Euler-characteristic (theorem 3.4.6). This is for one of the reductions that Hirzebruch takes, namely he shall prove HRR over line bundles. As a codimension 1 sub-variety is associated to a line bundle, this shall allow for some important interplay between the space V_r and the line bundle $\mathcal{L}(V_r)$. The chapter also includes a bonus section for *vanishing theorems* which go over some special cases where the Euler characteristic $\chi(X, E)$ reduces to $H^0(X, E)$.

The fourth chapter introduces the Todd class and T-characteristic. The Todd class can be thought of as holding the appropriate geometric information and generalizes the genus g. The Chern character shall hold the algebraic information, and the T-characteristic shall combine these two. Then just like in the previous chapter, the virtual T-characteristic is defined and related to the T-characteristic.

The fifth chapter introduces the necessary cobordism theory to define the spin index and show it is a cobordism invariant, allowing for the reduction of the proof to $M = \mathbb{C}\mathrm{P}^n$ for $n \in \mathbb{N}$. Due to the length of the paper, the proof that the cobordism ring can be generated by Pontryagin numbers was omitted, but it is not a big leap with the background presented in this paper; a proof can be found in [16].

The sixth and final chapter shall prove the titular theorem. The first section goes over the classical Riemann Roch and also gives a simple generalization to vector bundles over an algebraic curve, mainly to motivate some results in the theory of vector bundle in general (namely the triviality of bundles when they exceed the dimension of the base space). The 2nd section combines the results of all previous chapters and prove the HRR, first over line bundles (the hard part) and then over vector bundles.

There are two more section in the last chapter covering generalizations of HRR. The 3rd section covers Grothendieck Riemann Roch and follows Vakil's notes on intersection theory [19]. As this is a paper on HRR, all the intersection theory build-up is skipped and it is simply presented for exposure (or for the reader who already knows intersection theory, it is an overview of the proof). The final section covers the various direction the HRR has been generalized, pushing the envelope in various direction as late as 2017.

Notation:

1. As much as possible $\mathbb{F}$ will represent a field (of characteristic 0). The notation k, K is already used when appropriate for curvature, canonical divisor, K-genus, K-theory, and so forth.

2. $\mathbf{F}(q)$ will represent the flag manifold of dimension q

3. If $H_i^* \times H^{n-i}$ is a perfect pairing, as the manifolds/varieties we are working with are nice enough we shall abuse notation due to De Rham's Theorem and write:

$$\int_Y c$$

   for the pairing of the cohomology class c with Y. If we want to make it more explicit, we shall write $\kappa_n[-]$ notation.

4. $A \sqcup B$ represents disjoint union

5. TM represents tangent bundles. $N_{M/N}$ represent the normal bundle of N with respect to M.

0.0.1 Background

1. Basics of Algebraic Geometry: Riemann-Roch, sheaf theory (in particular sheaves on projective space), divisor theory. Knowledge of sheaf cohomology is very useful but not strictly necessary.
2. GAGA will be assumed (there is no space to prove it). In chapter 1, this shall not be assumed and the line-bundle/divisor correspondence shall be sketched out.
3. complex analysis up to Ahlfors (knowledge of Riemann-Roch from a complex perspective is good too, see for ex. [28]).
4. Basic vector bundle theory and characteristic theory is necessary if the reader does not want to take the results in chapter 2 at face value (I cover the details in [12])
5. Riemann Geometry up to John M. Lee [18].
6. Algebraic Topology up to Hatcher [9] as well as one result he doesn't prove (Leray-Hirsch, see [13])

Hartshorne will sometimes be quoted, but the reader could get-by by using any other textbook in algebraic geometry. I will introduce the necessary Hodge Theory and cobordism theory, and only the characteristic theory that is necessary.

It should be noted for the reader that is following this paper as a supplement to Hirzebruch's original paper ([16]), my paper almost completely "reverses" the order of Hirzebruch's original book. Hirzebruch took a very "Bourbaki" approach where he does not motivate the proof of the necessary results and combines them only when it becomes strictly necessary, while I hope to motivate why the concepts are being covered as they are presented.

1

Motivation: Riemann Roch Problem

A natural question to ask in complex and algebraic geometry is

"how many holomorphic/meromorphic/algebraic/analytic/etc. functions or sections are there on a (suitable) space X"

By letting $\mathcal{F}(X)$ represent our choice of functions/sections (i.e. the global sections of a [quasi-coherent] sheaf $\mathcal{F}$), we shall denote this question as finding information about the 0th sheaf cohomological group

$$H^0(X, \mathcal{F})$$

which by [3] is equal to $\mathcal{F}(X)$. We denote $\mathcal{F}(X)$ in terms of sheaf cohomology because this group shall sometimes be trivial, and the reason for its trivialness can be described by higher order sheaf cohomology groups:

$$H^1(X, \mathcal{F}), H^2(X, \mathcal{F}), ...$$

Exactly how to extract this information is known as the *Riemann Roch Problem.* To motivate the direction the problem took, let us answer this question in some simple cases, build up to how the Riemann Roch question is answered for algebraic curves, and then motivate how the (classical) Riemann Roch Theorem would be generalized to a result about sections on vector bundles.

1.1 Riemann Roch Problem

The simplest case would be to deal with polynomial functions over $X = \mathbb{C}$ which are determined by a finite set of roots (counting multiplicity) and a constant:

$$p(z) = c(z - a_1) \cdots (z - a_n)$$

and similarly for rational functions. If we ask for holomorphic functions over $\mathbb{C}$, $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{H}(\mathbb{C}))$, then the Weierstrass Factorization Theorem (see [32, section 4.2]) tells us that $f \in \mathcal{H}(\mathbb{C})$ factors as an infinite polynomial times a unit $e^{g(x)}$ along with correction factors for the linear terms:

$$z^m e^{g(z)} \prod_k^{\infty} \left[\left(1 - \frac{z}{a_k}\right) e^{p_k(z)} \right]$$

where

$$p_k(z) = \frac{z}{a_k} + \frac{1}{2}\left(\frac{z}{a_k}\right)^2 + \cdots + \frac{1}{m_k}\left(\frac{z}{a_{m_k}}\right)^m$$

is the correction factor and the a_k are roots that form a non-accumulating sequence; in particular there are analytic concerns. A meromorphic function f on $\mathbb{C}$ can be determined by two entire functions $f = g/h$ (again see [32, section 4.2]); this is possible as for poles of f we can define a function h with the appropriate roots and multiplicities. In other words, determining meromorphic functions becomes a classification of non-accumulating sequences in $\mathbb{C}$. The situation requires even more analytic information in hyperbolic space D^2 (think of the existence of $\sum_n z^{n!}$ and $1/(1-z)$). To understand the algebraic behaviour, limiting the scope to *compact spaces* such as Riemann surfaces or projective varieties gets much better as any holomorphic/meromorphic functions must have finitely many roots/poles by the Identity Theorem (that if two functions agree on a set of accumulation points, they are equal), and so we can focus our efforts within the algebraic setting. This setting is plenty rich enough for study and often offers a good starting point to generalize to the non-compact case[1]. In fact, by the efforts of Serre, Chow, and Grothendieck, the category of analytic functions on (suitable) compact spaces is equivalent to the category of algebraic functions on these spaces ([33], [23]), meaning many compact spaces are "intrinsically" algebraic.

Let us try finding holomorphic/meromorphic functions on $X = \mathbb{C}\mathrm{P}^1$ and see if they are determined by the points of $\mathbb{C}\mathrm{P}^1$ just like in the case of $\mathbb{C}$. Immediately, $\mathbb{C}\mathrm{P}^1$ has only constant holomorphic functions by an easy generalization of Liouville or the maximum modulus principle on manifolds. By the Residue Theorem, the sum of the number of roots and poles of a meromorphic function must be 0.

Let us represent this formally. Let

$$D = \sum_i n_i p_i \qquad n_i \in \mathbb{Z},\ p_i \in \mathbb{C}\mathrm{P}^1$$

be a finite formal sum; we may think of it as the free (abelian) group $F^{\mathbb{Z}}(\mathbb{C}\mathrm{P}^1)$. Call D a *divisor*, and let $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$ represent this group. A divisor can be thought of as an object that remembers which roots/poles with which multiplicity we are interested in. Define:

$$\mathrm{Div}(f) = \sum_{p \in \mathbb{C}\mathrm{P}^1} \nu_p(f) p$$

[1] These cases are still being studied, see for example [1]

where $\nu_p(f)$ is the order of the zero/pole, and is negative if it is a pole. This forms a subgroup of $\mathrm{Div}(\mathbb{C}\mathrm{P}^1)$; represent it as $\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$ ("P" for principal), and elements of this subgroup are called *principal divisors*. This subgroup represents the divisors that are *representable* as a meromorphic function. Let us find which divisors can be represented. Let:

$$\deg D = \sum_i n_i$$

Notice that $\deg(-)$ is certainly a group homomorphism. By our earlier remark:

$$\deg(\mathrm{Div}(f)) = 0$$

which is a restriction applied to any meromorphic function on $\mathbb{C}\mathrm{P}^1$. It is easy to see that this restriction would work on any compact Riemann surface (see [32, chapter 6]). If $\deg(D) = 0$, by our earlier remark on the use of the Residue Theorem, we have that there exists an f such that $\mathrm{Div}(f) = D$, showing this condition characterizes meromorphic functions $f \in \mathcal{M}(\mathbb{C}\mathrm{P}^1)$ (but it shall not on other Riemann surfaces, as demonstrated bellow).

Let us see to what extent the set of all divisors fail to represent a meromorphic function. Consider the (*Weil*) *divisor class group*:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) = \frac{\mathrm{Div}(\mathbb{C}\mathrm{P}^1)}{\mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)}$$

for any $[D] \in \mathrm{Cl}(\mathbb{C}\mathrm{P}^1)$, $\deg([D])$ is well-defined as $\deg(D) = \deg(D + f)$ for any $f \in \mathrm{PDiv}(\mathbb{C}\mathrm{P}^1)$. If $\deg(D) = \deg(D')$, then $\deg(D) - \deg(D) = \deg(D - D') = 0$, and hence $D - D' = \mathrm{Div}(f)$. Therefore, we have that:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong \mathbb{Z}$$

which can be thought of as measuring the obstruction in a meromorphic function being defined by its points/roots.

Let us translate this intuition into sheaf cohomology, that is let us translate this into a measure of obstruction of local information having global information. The natural result we may want to think is that:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1})$$

However, $H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1})$ is trivial. This because there is no obstruction to extending a function *additivity*, in particular if $\mathbb{C}\mathrm{P}^1 = U_0 \cup U_1$ is the usual covering and if $f_{01} \in \Gamma(U_0 \cap U_1)$ then $f_{01} = g_0 - (-g_1)$ (decompose f_{01} into a Laurent series, split the Laurent series into the principal and non-principal part, change coordinates in the principal part so that it becomes holomorphic on U_1). Instead, the obstruction is captured in the multiplicative structure[2]:

$$\mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \cong H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1})$$

The higher order sheaf cohomology groups shall be trivial as the dimension of $\mathbb{C}\mathrm{P}^1$ is 1 (over $\mathbb{C}$), and hence we'll have:

$$\begin{aligned} H^0(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) &\cong \mathbb{C} \\ H^1(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}) &\cong \mathrm{Cl}(\mathbb{C}\mathrm{P}^1) \\ H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}_{\mathbb{C}\mathrm{P}^1}) = H^2(\mathbb{C}\mathrm{P}^1, \mathcal{O}^*_{\mathbb{C}\mathrm{P}^1}) &\cong \{0\} \\ &\vdots \end{aligned}$$

[2] These two are linked via a naturally exponential sequence $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 0$, the detail are left for [3]

Doing a similar translation for cohomology for previously mentioned case of $(X, \mathcal{F}) = (\mathbb{C}, \mathcal{O}_\mathbb{C})$ where $\mathcal{O}_\mathbb{C}$ is the sheaf of algebraic holomorphic functions, it is not hard to see that:

$$\mathrm{Cl}(\mathbb{C}) = \frac{\mathrm{Div}(\mathbb{C})}{\mathrm{PDiv}(\mathbb{C})} \cong 0$$

which shall give us:

$$\begin{aligned} H^0(\mathbb{C}, \mathcal{O}_\mathbb{C}) &\cong \mathbb{C}[x] \\ H^1(\mathbb{C}, \mathcal{O}_\mathbb{C}^*) &\cong \{0\} \\ H^2(\mathbb{C}, \mathcal{O}(\mathbb{C}) &\cong \{0\} \\ &\vdots \end{aligned}$$

as every divisor can be represented by a principal divisor. We can interpret this as there being no impediments to using local information to construct meromorphic functions. Notice that the ring $\mathbb{C}[x]$ is a UFD; this is the algebraic characterization of a trivial class group (details in [3]), which intuitively can be thought of as their being a unique way of constructing a function from local information.

This answers the Riemann Roch problem for holomorphic/meromorphic functions for $\mathbb{C}$ and $\mathbb{C}\mathrm{P}^1$. Let us look at the next simplest nontrivial example. Let $X = \mathbb{C}/\Lambda$ where $\Lambda \subseteq \mathbb{C}$ is a lattice spanned by the $\mathbb{R}$-linearly independent elements $(1, \tau)$ and let $\mathcal{F} = \mathcal{M}_X$ (all rational functions on X); X is called an *elliptic curve*, and so we shall label it as E. As E is compact, if $f \in \mathcal{M}_E$ then $\mathrm{Div}(f) = 0$. However, if $\deg(D) = 0$, this does not imply there exists an $f \in \mathcal{M}_E$ such that $\mathrm{Div}(f) = D$. To see this, let's analyze $\mathcal{M}_E$ a little more closely. As $\mathbb{C}/\Lambda$ is the quotient by Λ, any meromorphic function must respect the Λ structure:

$$f(z + n + m\tau) = f(z) \qquad \forall n + m\tau \in \Lambda$$

such a function is called *doubly periodic* or *elliptic function* (as they are defined on elliptic curves). Using this periodic condition, it is simple to see that $\mathrm{Div}(f) = 0$ by looking at:

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f'(z)}{f(z)} dz = 0$$

where D is the fundamental domain[3], namely the opposing edges cancel out. We can modify this integral to find other properties, for example:

$$\frac{1}{2\pi i} \int_{\partial D} z \frac{f'(z)}{f(z)} dz = n + m\tau \in \Lambda$$

which is nonzero as z is not doubly periodic. Replacing z by other algebraic functions all end up in Λ, and so we see that if $D = \mathrm{Div}(f)$:

$$\sum_i n_i p_i \equiv 0 \bmod \Lambda$$

where here $\sum_i n_i p_i \notin F^{\mathbb{Z}}(E)$ but $\sum_i n_i p_i \in E$, namely we are treating p_i as a complex number multiplied by n_i and summed. Certainly there is no such restriction on a general divisor $D \in \mathrm{Div}(E)$

[3] If there are zero/poles on the path, we simply shift around appropriately

of degree 0. We thus have shown that there is a surjective map:

$$\mathrm{Cl}^0(E) = \frac{\mathrm{Div}^0(E)}{\mathrm{PDiv}(E)} \twoheadrightarrow \mathbb{C}/\Lambda \tag{1.1}$$

where $\mathrm{Div}^0(E)$ are all divisor of degree 0. It is in fact an isomorphism, which comes from constructing an elliptic functions using the Weierstrass $\wp$-function, the details are in [32, chapter 5]. Thus, we get the short exact sequence:

$$0 \to \mathbb{C}/\Lambda \to \mathrm{Div}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

often called the *Jacobi-Abel* exact sequence as $\mathbb{C}/\Lambda$ is a Jacobian variety, and a short exact sequence:

$$0 \to \mathrm{Cl}^0(E) \to \mathrm{Cl}(E) \xrightarrow{\deg} \mathbb{Z} \to 0$$

All together, we get a short exact sequence and a commutative diagram:

$$0 \to \mathrm{PDiv}(E) \to \mathrm{Div}^0(E) \to E \to 0$$

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Div}^0(E) \arrow[r, hook] \arrow[d, two heads, "\pi"] & \mathrm{Div}(E) \arrow[r, "\deg"] \arrow[d, two heads, "\pi"] & \mathbb{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \mathrm{Cl}^0(E) \arrow[r, hook] & \mathrm{Cl}(E) \arrow[r, "\deg"] & \mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

In terms of cohomology, as the surjective map in equation (1.1) is an isomorphism:

$$\begin{aligned}
H^0(E, \mathcal{O}_E) &\cong \mathbb{C} \\
H^1(E, \mathcal{O}_E^*) &\cong E \times \mathbb{Z} \\
H^2(E, \mathcal{O}_E) = H^2(E, \mathcal{O}_E^*) &\cong \{0\} \\
&\vdots
\end{aligned}$$

Continuing on to other Riemann surfaces, we shall lose the isomorphism given in equation (1.1), the obstruction to the map being an isomorphism is the beginning of the study of Jacobian varieties (see [32, section 6.4]). The situation becomes even more complicated going to higher dimensions. The above strategy using divisors to find cohomology information becomes the beginning of the study of Weil and Cartier Divisors[4]; the study of Cartier/Weil divisors can be thought of as the study of the obstruction of an object from being constructed from codimension 1 objects, namely the study of $H^1(X, \mathcal{O}_X^*)$. This however does not comprehensively answer the problem of finding holomorphic/meromorphic functions on X. Overall, we get some useful information about the obstruction to finding global sections, but not necessarily concrete ways of finding $H^0(X, \mathcal{O}_X)$.

To answer the Riemann Roch problem, a more granular approach to find which functions (of various kinds) exist on a space X must be taken. As divisors keep track of root/pole information in 1-dimensional case, they are a great tool in the classifying functions that satisfy the conditions given by a divisor. To make this precise, define a partial ordering on divisors $\mathrm{Div}(X)$ where (for now) X is

[4]and more generally chow rings (see [3]). In section 6.3, we briefly cover how this characterizes rational cohomology under suitable algebraic assumptions

either a Riemann surface or a smooth projective curve: let $D_2 \geq D_1$ if each coefficient of $D_2 - D_1$ is non-negative. Then we can consider:

$$\mathcal{O}_X(D)(X) = \{f \in \mathcal{M}_X(X)^* \mid \mathrm{Div}(f) + D \geq 0\}$$

In other words, if $D = \sum_i n_i p_i$, then $\mathcal{O}_X(D)(X)$ is all meromorphic functions f where it has a pole at p_i of order at most n_i if $n_i > 0$ and roots at p_i of order at least $|n_j|$ if $n_j < 0$ (and $\mathcal{O}_X(D)$ is a [coherent] sheaf). From this, we can ask the Riemann Roch problem for $(X, \mathcal{O}_X(D))$. This is answered by the famous *Riemann-Roch Theorem* where Riemann proved the Riemann-inequality, and Roch provided the necessary work to make it an equality, see [28]. Due to its generalizations, it is now known as *Riemann-Roch for curves* and is the equality:

$$\dim H^0(X, \mathcal{O}_X(D)) - \dim H^1(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g \tag{1.2}$$

where g is the genus of the Riemann surface or smooth projective curve (see [34] or [3] for defining the genus of a curve via scheme theory, or for a direct computational proof using normalization see [32]). In the original phrasing of the Riemann Roch theorem, the group $H^1(X, \mathcal{O}_X(D))$ is replaced with $H^0(X, \mathcal{O}(K_X - D))^\vee$ where K_X is the canonical divisor of X; these groups are isomorphic by Serre Duality (see [3, section 6.3]). Then this space is identified with the space of meromorphic 1-forms (either through some simple techniques from complex analysis, see Brill-Noether Reciprocity in either [28] or [32], or by combining of Poincaré duality with De Rham's Theorem, see either [18] or [30]). Switching to the traditional notation $l(D) = \dim H^0(X, \mathcal{O}_X(D))$ and $i(D) = \dim H^0(X, \mathcal{O}_X(K_X - D))^*$ we get:

$$l(D) - l(K_X - D) = \deg(D) + 1 - g \qquad \text{(equiv.)} \qquad l(D) - i(D) = \deg(D) + 1 - g$$

This result will be proven in theorem 6.1.1. As we phrased the Riemann-Roch problem in the beginning, this formula would then be asking for the value of $l(D)$. There are cases where $l(K_X - D)$ disappears (for example if $D = 0$ or $D = K_X$), and there are many cases we can compute it easily enough. It is simple to compute the right hand side but in general it is harder to compute the left hand side. Thus, this formula becomes a new constriction that helps us find $l(D)$, namely it provides a lot of information towards answering the Riemann Roch question for $(C, \mathcal{O}_C(D))$. There are a set of theorems known as *vanishing theorems* that specify under which conditions $l(K_X - D)$ (or more generally, higher cohomology groups) vanish (covered in section 3.5, but for all the details see [26]); under these conditions we recover $l(D)$ without a "fudge factor"[5].

This formula works for Riemann surface/projective curves. Let us work towards the intuition behind the generalization. The notation on left hand side in equation (1.2) can be compactified by using the Euler-Poincaré characteristic (or simply the Euler characteristic): $\chi(X, \mathcal{O}_X(D)) = \sum_{i=0}^{\infty} (-1)^i \dim H^i(X, \mathcal{O}_X(D))$ so that:

$$\chi(X, \mathcal{O}_X(D)) = \deg(D) + 1 - g$$

As X is a curve $H^i(X, \mathcal{O}_X(D)) = 0$ for $i > 1$[6] giving the equality. The right hand side shall be drastically different, split into special invariants of vector bundles known as the *Chern character* and *Todd class* where the Chern character can be thought of as capturing some "algebraic" or "local"

[5]The term "fudge factor" was common vocabulary in the Italian school of the 19th century in the context of intersection theory. There is a connection in its use here and intersection theory due to the work of Fulton ([20]), but there is no space to cover the details in this paper

[6]This is a very common fact and can be proven in multiple different ways. Algebraic proof: [3, section 28.3], topological proof:[13, chapter 3], geometric proof:[12, chapter 3]

information given by D (this will generalize the $\deg(D)$) while the Todd class shall capture the "geometric" or "global" information (this will generalize the $1+g$). In fact, the Euler Characteristic can easily be generalized to be an invariant on vector bundles by using sheaf cohomology instead of singular cohomology, and hence the Riemann Roch theorem generalizes to a question about the existence of sections on vector bundles over compact manifolds, namely if $(X, \mathcal{F})$ is a compact manifold along with a vector bundle $\mathcal{F}$, then we want to compute:

$$H^0(X, \mathcal{F})$$

Note that there is a bijective correspondence between vector bundle and locally free sheaves (see [3, chapter 4] for the details), and hence we may treat $\mathcal{F}$ as either a vector bundle or a sheaf; the details for line bundles is given in section 1.2.1. The result of this will be:

$$\chi(X, \mathcal{F}) = \int_X \mathrm{ch}(X) \smile \mathrm{td}(X) = \int_X \mathrm{ch}(X)\mathrm{td}(X)$$

where the cup product $\smile$ is often omitted. This equation is precisely the result of the *Hirzebruch-Riemann Roch Theorem*.

1.2 Motivating the Generalization

Let us now motivate each component of the Hirzebruch-Riemann-Roch Theorem in more detail. The first section motivates how the divisor D can naturally be thought of as a line bundle, leading to the generalization to vector bundles. Then, the motivation for Euler characteristic is covered to show how it inherently captures both geometric and arithmetic information. In the final section, we shall motivate how the use of characteristic theory will naturally give the generalization of $\deg(D)$ and the genus.

1.2.1 How do Vector Bundles enter the Picture

To motivate how vector bundles will enter the picture, it is worthwhile seeing how $\mathcal{O}_X(D)$ can be interpreted as a line bundle. To that end, recall that the projective curve $X = \mathbb{P}^1$ as defined in classical algebraic geometry only has constant global sections. Namely, if $U_0, U_1 \cong \mathbb{A}^1$ is the usual cover with global sections (i.e. functions) $\mathcal{O}_X(U_0) \cong \mathcal{O}_X(U_1) = \mathbb{F}[x]$ and the compatibility condition on $U_0 \cap U_1$ is $f(z) = g(1/z)$, namely:

$$\begin{tikzcd} U_0 & U_1 \\ U_0 \cap U_1 \arrow[u, hook] \arrow[r, dash, "\cong"] & U_0 \cap U_1 \arrow[u, hook] \end{tikzcd} \quad \text{i.e.} \quad \begin{tikzcd} \mathbb{A}^1 & \mathbb{A}^1 \\ \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \arrow[r, dash, "\cong"] & \mathbb{A}^1 \setminus \{0\} \arrow[u, hook] \end{tikzcd}$$

gives:

$$\begin{array}{ccccccc} \mathcal{O}_X(U_0) & & \mathcal{O}_X(U_1) & & \mathbb{F}[x] & & \mathbb{F}[y] \\ \downarrow & & \downarrow & \text{i.e.} & \downarrow & & \downarrow \\ \mathcal{O}_X(U_0 \cap U_1) & \xrightarrow{\cong} & \mathcal{O}_X(U_0 \cap U_1) & & \mathbb{F}[x, 1/x] & \xrightarrow{\cong} & \mathbb{F}[y, 1/y] \end{array}$$

where $x \mapsto 1/y$ and $1/x \mapsto y$, Then if $F([x' : y']) \in \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$ is a global function where $x = x'/y'$ and $y = y'/x'$ (when $y', x' \neq 0$ respectively), then $F([x : 1]) \to f(x) \in \mathcal{O}_X(U_0)$ and $F([1 : y]) \to g(y) \in \mathcal{O}_X(U_1)$ are the restrictions of the function to U_0 and U_1 respectively, and by the above diagram it must be that these functions map to the same element in the restriction, namely:

$$f(x) \to f(x) = g(1/x) = g(y) \leftarrow g(y)$$

but it is impossible that $f(x) = g(1/x)$ in $\mathbb{F}[x]$ unless f, g are the same constant. Thus, it must be that the only global functions on the projective line are constants:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = \mathbb{F}$$

The fundamental limitation here is that the restriction $f(x) = g(1/x)$ is very strong. What if we can loosen this restriction? For example what if

$$f(x) = xg\left(\frac{1}{x}\right)$$

Then we can in fact form linear homogeneous functions! For example consider

$$F([x' : y']) = ax' + by'$$

Then F maps to $f(x) = ax + b$ and $g(y) = a + by$. Then:

$$f(x) = ax + b = x\left(a + b\frac{1}{x}\right) = xg\left(\frac{1}{x}\right)$$

which works! More generally, if the condition $f(x) = x^n g(1/x)$ allows for degree n homogeneous polynomials, and the homogeneity can happen for any number of variables. Denote the degree n-homogeneous polynomials for $\mathbb{P}^k$ by $\mathcal{O}_{\mathbb{P}^k}(n)(\mathbb{P}^k)$.

Now, recall that a n-dimensional vector bundle $\pi : E \to X$ over a field $\mathbb{F}$ can be defined by taking open sets $U_i \subseteq X$ such that $\pi^{-1}(U_i) \to U_i \times F$ and ensuring that for any intersection $U_i \cap U_j$ we have a gluing map:

$$g_{ij} : \pi^{-1}(U_i \cap U_j) \to \mathrm{GL}_n(\mathbb{F})$$

such that $g_{jk}g_{ij} = g_{ik}$ (the cocycle condition). When limiting to line bundles, the gluing map becomes:

$$g_{ij} : \pi^{-1}(U_i \cap U_j) \to \mathbb{F}^*$$

Intuitively, this give a point-wise gluing conditions on the bundle: if $v \in U_i$ and $w \in U_j$ that are the same point in $U_i \cap U_j$, then:

$$v = A_{ij}w$$

Now, let us correspond $\mathcal{O}_{\mathbb{P}^1}(n)$ to a line bundle $\mathcal{L}(n)$. Let $\mathcal{L}(n)$ be locally trivial on U_1, U_2: $\Gamma(U_i, \mathcal{L}(n)) = \mathcal{O}_{\mathbb{CP}^1}(n)(U_i)$. On their intersection define the gluing map:

$$g_{ij} : \mathcal{O}_{\mathbb{P}^1}(U_i \cap U_j) \to \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j) \cong \mathbb{F}\left[x, \frac{1}{x}\right] \setminus \{0\}$$

where we take $\mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$ in place of $\mathbb{F}^*$ as these are the units (i.e. isomorphisms) on this intersection. For a given f, g to restrict to the same functions on the intersection, we now "twist" g to be:

$$f(x) = x^n g(1/x) \qquad x^n \in \mathcal{O}^*_{\mathbb{P}^1}(U_i \cap U_j)$$

The word "twist" comes from both complex analysis where z^n twists $\mathbb{C}$ onto itself around the branching point, and from algebraic geometry where the non-triviality of bundles is partly captured by characteristic classes that represent the obstruction to some non-trivial global trivialization via twisting (we shall cover this intuition in chapter 2).

This gives how we can construct a line bundle $\mathcal{L}(n)$ for sheaves $\mathcal{O}_{\mathbb{CP}^1}(n)$. The converse works as well: the details of the correspondence is given in [3, section 4.1] (especially how to transition from points of a bundle to section of a sheaf). Overall, we see that $\mathcal{O}_{\mathbb{P}^1}(n)(\mathbb{P}^1)$ is the global sections of a line bundle; more generally $\mathcal{O}_{\mathbb{P}^1}(n)$ remember all the sections over each open set U, as $\mathcal{O}_{\mathbb{P}^1}(n)$ is a sheaf.

With this intuition, we can see how to interpret $\mathcal{O}_X(D)$ as a line bundle. Choosing an affine cover U_i for X, we can define:

$$D|_{U_i} = \sum_{p \in U_i} n_p p$$

which is the representation of the divisor D in U_i. Then for each affine U_i, choose $s_i \in \mathcal{M}_X(U_i)$ so that $\langle s_i \rangle_{\mathbb{F}} = \mathcal{O}_X(D)(U_i)$ is a generator as a $\mathbb{F}$-algebra of the local sections and:

$$\mathrm{Div}(s_i) = -D|_{U_i}$$

So that s_i has roots/poles of the appropriate order. Such a section exists, for $U_i \cong \mathbb{A}^1$ (as X is a curve) so $\mathcal{O}_X(U_i) \cong \mathbb{F}[x]$ and so by our earlier discussion we can certainly find an s_i with exactly zeros and poles at $-D|_{U_i}$ and it is an exercise in ring-theory to show this is a generator (think of choosing a uniformization parameter for a local ring).

Let us label the collection of these meromorphic functions $\mathcal{O}_X(D)(U_i)$. To find an $f \in \mathcal{O}_X(D)(X)$ (i.e. a global section), we need to insure we can glue together nicely on intersection. Consider (nonempty) $U_i \cap U_j$ and take the generators $s_i \in \mathcal{O}_X(D)(U_i)$ and $s_j \in \mathcal{O}_X(D)(U_j)$. It is not hard to see that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. Then certainly on $U_i \cap U_j$:

$$\mathrm{Div}(s_i) = \mathrm{Div}(s_j) = -D|_{U_i \cap U_j}$$

and so, as we are considering meromorphic functions, we can consider:

$$\mathrm{Div}\left(\frac{s_j}{s_i}\right) = \mathrm{Div}(s_j) - \mathrm{Div}(s_i) = 0$$

As s_j/s_i has no zeros/poles on $U_i \cap U_j$, this means that it is holomorphic and nowhere vanishing on $U_i \cap U_j$. But then:

$$g_{ij} = \frac{s_j}{s_i} \in \mathcal{O}^*_X(U_i \cap U_j)$$

Showing that we have the desired transition functions to define a line bundle! Thus, we may glue them to form all global sections $\mathcal{O}_X(D)(X)$ for every D, hence $\mathcal{O}_X(D)$ can be interpreted a line bundle. In most cases (and in most spaces that will be considered in this paper), every line bundle $\mathcal{L}$ is induced by a divisor D; section 3.2 demonstrates this correspondence for Kähler manifolds, and for the correspondence more generally on (the appropriate type of) scheme see [3, section 4.3].

1.2.2 Motivating Euler Characteristic

The origins of the Euler characteristic stem from an observation by Leonhard Euler about the relation between the vertices, edges, and faces of a convex Polyhedron. If V, E, F represents the respective quantities, then:

$$V - E + F = 2$$

For example, a cube has 8 vertices, 12 edges, and 6 faces, so $8 - 12 + 6 = 2$. Similarly, a closed connected planar graph holds the same property (counting the outside of the planar graph as a face). This sum is called the *Euler Characteristic* of the polyhedron. If the polyhedron was not convex, this relation need not hold, however it is possible to have a non-convex polyhedron whose Euler characteristic is 2. This shows that the Euler characteristic captures some rough information about the geometry of the shape.

Generalizing this, we can form CW-complex C on a topological space X and define the Euler characteristic as:

$$\chi(X, C) = \sum_{i=0}^{\infty} (-1)^i k_i$$

where k_i is the number of CW-complexes per dimension. One of the crowning results of algebraic topology in the 21st century is the invariance of this number of choice of CW complex. This led to the development of *singular cohomology* with the homology group $H_i(X, \mathbb{Z})$ and the *Betti numbers* $b_i = \mathrm{rk}(H_i(X, \mathbb{Z}))$ and:

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i b_i = \sum_{i=0}^{\infty} (-1)^i \mathrm{rk}(H_i(X, \mathbb{Z}))$$

As this is a homological constant, it is sometimes called the *Euler-Poincaré* characteristic to distinguish it from the Euler characteristic for polyhedra, but this shall rarely be done in this paper. As cohomology groups are invariant up to homotopy on X, the Euler characteristic is invariant to homotopy. It is often computationally easier to work with the cohomology group, as the spaces we work with are smooth manifolds by the Poincaré duality [13, section 4.3] (or its generalization, Serre duality [3, section 5.3]), we have the equivalent sum:

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i \mathrm{rk}(H^i(X, \mathbb{Z}))$$

How should we interpret the alternating nature of the sum when summing over homology/cohomology? The key idea is to notice that the alternating sum mimics the pattern when invoking the inclusion-exclusion principle. Let us go back to Euler's original $V - E + F$ formula and take for example a cube. Then we can think of each face as including the vertices and edges, and the edges as including the vertices. By counting all the faces, we have in fact double counted the edges. We thus subtract the sum of the edges. But this shall at the same time remove all the vertices from the sum, and hence we add them back in. This is exactly the inclusion-exclusion principle.

When converting to a cohomological perspective, we recover the inclusion-exclusion principle from the perspective of *Čech cohomology.* In particular, the intuition is to cover your space X in open sets, define the desired type of functions on each open set, and measure the obstruction to gluing these functions on intersections, intersection of intersections, and so on and so forth:

$$\begin{tikzcd}
\check{C}^0(\mathcal{U},\mathcal{F}) \arrow[r, "d^0"] \arrow[d, equal] & \check{C}^1(\mathcal{U},\mathcal{F}) \arrow[r, "d^1"] \arrow[d, equal] & \check{C}^2(\mathcal{U},\mathcal{F}) \arrow[r, "d^2"] \arrow[d, equal] & \cdots \\
\prod_i \mathcal{F}(U_i) \arrow[r, "\delta^0"] & \prod_{i<j} \mathcal{F}(U_i \cap U_j) \arrow[r, "\delta^1"] & \prod_{i<j<k} \mathcal{F}(U_i \cap U_j \cap U_k) \arrow[r, "\delta^2"] & \cdots
\end{tikzcd}$$

Representing this visually:

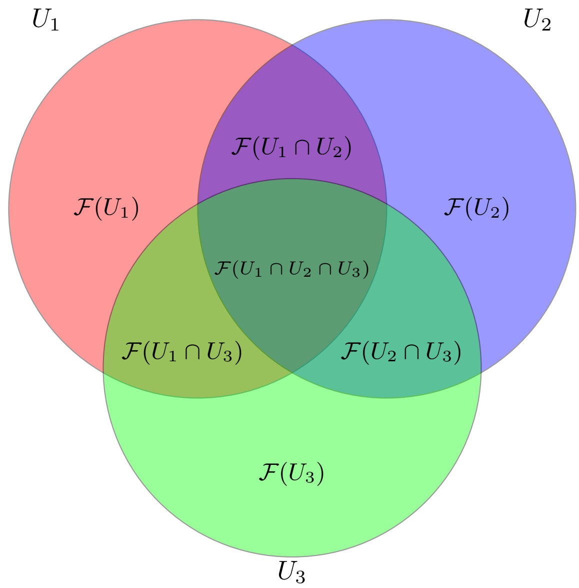

Figure 1.1: Venn diagram for an open cover with three sets U_1, U_2, U_3 and their sections.

How should we interpret what invariance is captured by the Euler characteristic? The case of the convex and non-convex polyhedron hints at a notion of curvature, and indeed by the Gauss-Bonnet theorem a Riemann surface X (without boundary) with Riemann metric ω which induces a Gaussian curvature K is equal to the Euler characteristic by the formula:

$$\chi(M) = \frac{1}{2\pi}\int_M K dA$$

This further generalizes to higher-dimensional Riemann manifolds (as shall be mentioned shortly). Another interpretation from the theory of vector fields is that that number of zeros/poles as well as point with non-trivial curl are given by the topology of the surface, and hence they are represented by χ (see [13]).

These are very useful invariants, especially in 2 dimension where the classification of Riemann surfaces up to homotopy if first broken down into the case where the Euler characteristic is positive, zero, or negative (representing spherical, flat, and hyperbolic curvature respectively), and these values are strongly tied to the properties of vector fields on the surface. However, we see that the Euler characteristic can be defined on any space admitting a cohomology theory, suggesting the vector-field interpretation may be more suitable. The following section covers these ideas.

1.2.3 Motivating Characteristic Theory

By the discussion in section 1.2.1, $\deg(D)$ and $1-g$ should be interpreted as a property of a bundle. Let us return to the Gauss-Bonnet Theorem to find the motivation of this generalization. Recall that if M is a compact two-dimensional Riemannian manifold with empty and K is be the Gaussian curvature of M, then:

$$\chi(M) = \frac{1}{2\pi}\int_M K dA$$

It was proven by Chern (the *Chern Theorem* or *Chern-Gauss-Bonnet Theorem*) that this generalizes to an even dimensional closed Riemann manifold:

$$\chi(M) = \int_M e(\Omega)$$

where e is the *Euler class* (which is a characteristic class), and Ω is the 2-form representing curvature which can be embedded in:

$$\bigwedge^{2n} T^*M$$

More generally, if M is any smooth orientable n-dimensional manifold then:

$$\chi(M) = (e(TM), [M])$$

where $[M]$ is the fundamental class and $(-,-)$ is the cap product.

1.2.1 Key take-way: Euler Class and Top Chern Class
the Euler characteristic χ is already strongly linked to characteristic theory as it can be seen as the perfect pairing between the Euler class and the fundamental class. The details of the definition of the Euler class can be seen in [12], the important point to remember is that it is the given by taking the *top Chern class* of a manifold M and a modified tangent bundle E_{TM},

This definition does not consider non-trivial vector bundles, in particular when M is closed then $\chi(M)$ can be considered as the Euler characteristic over the trivial bundle $\mathbb{1}$ as:

$$H^i(M, \mathbb{1}) = H^i(M, \mathbb{R})$$

By considering different bundles E, using sheaf cohomology we can define:

$$\chi(M, E) = \sum_i^{\infty} (-1)^i \dim H^i(M, \mathcal{O}_X(E))$$

The extra non-trivial algebraic information within the cohomology groups introduced by E motivates the focus on Chern classes for the generalization.

Thus, let us motivate what the Chern classes represent. Let $K_{\mathbb{C}}(X)$ be the collection of isomorphism classes of $\mathbb{C}$-vector bundles over X, and $H^*(X)$ the cohomology ring of X. First off, a Chern class

c is a natural transformation $K_{\mathbb{C}}(-) \Rightarrow H^*$, showing that the Chern class c is a sum of ith Chern classes, one for each H^i. Next, as H^i vanishes when $i > \dim(M)$, there can be at most finitely many ith Chern classes. As we are mapping into the cohomology ring, the Chern class will represent some obstruction information. Given a complex vector bundle E of dimension n, it is natural to ask whether it is possible to define k frames $1 \leq k \leq n$ on E that are linearly independent. The existence of a nontrivial ith Chern class shall represent an obstruction to having a $n-i$ globally linearly independent frames, in particular it will not be possible to extend $n-i$ linearly independent frames to a $n-i+1$ linearly independent frame. As this is a vector-bundle dependent invariant, the Chern classes represent some algebraic complexity of E in terms of the cohomology of M.

Given a Chern class $c_i \in H^{2i}(M, \mathbb{Z})$ (the $2i$ shall be explained in chapter 2), what would evaluating c_i represent? Given a representative T of a homology class $H_{2i}(M, \mathbb{Z})$, we can define the *Chern number* $c_i(E)$ at T to be:

$$\langle c_i(E), [T] \rangle = \int_T c_i(E)$$

The Chern number counts the number of zeros/poles of a generic section of the vector bundle E, restricted to the submanifold T (see [12]). In terms of the earlier curvature interpretation, it is around zero or poles (where poles can be considered as zero's in the reverse orientation) that we accumulate non-trivial curling/divergence. Let us apply this intuition where M is a Riemann surface and E a line bundle on M. Then $i = 1$ so that $c_1(\mathcal{L}(D)) \in H^2(M, \mathbb{Z})$, for a line bundle $\mathcal{L}(D)$ associated to D. Then by the above intuition, we see that:

$$\int_D c_1(\mathcal{L}(D)) = \deg(D)$$

In a sense, this is the algebraic part of the Riemann Roch theorem. If we let $D = 0$, then:

$$\chi(M) = 1 - g$$

To get "$1 - g$" from Chern classes, notice that as we can always define the tangent bundle TM of M, we can compute $c_1(TM)$. Then by the *Poincaré-Hopf Theorem* ([13] or [30]) we get that $c_1(TM) = 2 - 2g$. Thus we get:

$$\chi(M) = \frac{1}{2} c_1(TM)$$

It should be noted that in this case, the Euler class is equal to the top Chern class, $c_1(TM) = e(TM)$, and so in this low-dimension case we could have just applied the Gauss-Bonnet Theorem.

When moving up to higher dimensions, it is important to limit to smooth (projective) varieties, namely because the higher cohomological information on closed n-manifolds where $n \geq 2$ holds more information than there are geometric sub-manifolds; limiting to smooth varieties (and later proven by Grothendieck to proper morphisms) shall restrict the cohomology groups enough to use Chern classes in higher dimensions. To tie the geometric properties of manifolds to smooth varieties, we shall show in section 3.2 what conditions must be added on a manifold M so that it has to be a smooth variety. For now, we shall simply assume M is a smooth variety.

Note that the 1-dimensional case is exceptionally simple. For those who know the Riemann-Roch theorem for surfaces (ex. see Hartshorne [34, chapter 5.1, theorem 1.6])

$$\chi(D) = \chi(0) + \frac{1}{2} D.(D - K)$$

they can verify that:

$$\chi(M, \mathcal{L}(D)) = \frac{1}{2}c_1(\mathcal{L}(D))^2 + \frac{1}{2}c_1(\mathcal{L}(D))c_1(TM) + \frac{1}{12}(c_1(TM)^2 + c_2(TM))$$

showing that the ability to write χ in terms of TM and $\mathcal{L}(D)$ does extend in a non-trivial way to higher dimensions.

So the goal now is to find a connection between the Euler-characteristic and some equation of Chern classes that works in higher dimensions. Two important decomposition of the Euler characteristic is:

$$\chi(M \oplus M', E)) = \chi(M, E) \cdot \chi(M', E)$$

which is a standard cohomology proof when $E = \mathbb{1}$, and in theorem 3.3.6 we shall show that:

$$\chi(M, E \oplus E') = \chi(M, E) + \chi(M, E')$$

Thus, we shall need two invariance to take into account the different algebraic behaviour of the manifold and vector bundles over direct products. The *Chern character* shall be the right invariant that captures the algebraic information for any manifold M and vector bundle E: we shall show that:

$$\mathrm{ch}(E \oplus E') = \mathrm{ch}(E) + \mathrm{ch}(E')$$

for the geometric counter-part, it was *John Arthur Todd* who found the geometric idea before Chern classes were even discovered (`https://en.wikipedia.org/wiki/Todd_class`). Hirzebruch put them to good use connecting them to Chern classes in defining the *Todd class* which shall have the property that:

$$\mathrm{td}(M \oplus M') = \mathrm{td}(M) \cdot \mathrm{td}(M')$$

where inputting $M, M', M \oplus M'$ should be thought of as inputting the tangent bundle of these manifolds. If we were working with trivial bundles, then we can simply write:

$$\chi(M) = \int_M \mathrm{td}(M)$$

generalizing the Euler class. Then with the "correction term" given by the Chern character $\mathrm{ch}(-)$ to take into account the algebraic information to finally come up with:

$$\chi(M, E) = \int_M \mathrm{ch}(M)\mathrm{td}(M)$$

Notice how this can be seen as a great generalization of Chow's Theorem, where we are not limited to just looking at the curvature form, but had to limit to smooth varieties in order for the "data" to be well-defined. The Chow theorem can be recovered from the Hirzebruch-Riemann-Roch Theorem (see this link for an overview).

2

Necessary Vector Bundle Theory

This chapter will cover the necessary vector/fiber bundle theory that is used in Hirzebruch Riemann Roch. For an in-depth reading, there are my notes [12] that also include the algebraic K-theory background necessary to generalize the results of this chapter to Fulton's Intersection Theory. For a different reference, I recommend Hatcher's textbook on vector bundles [31] where he also covers geometric K-theory. Furthermore, this chapter shall freely use the bijective correspondence between locally free sheaves and vector bundles and that they share the same section/sheaf cohomology. I may convert between the language of sheaf theory and bundle theory without much warning.

Let $E \to X$ be vector bundle, where X is paracompact and finite dimensional (either as a manifold or topologically)[1]. A natural set of questions to ask is what are the possible (global) sections on E with respect to X. A classical example of this question is to ask if there are any non-vanishing global sections on TS^2 with respect to S^2; the *Hedgehog Theorem* (or infamously, the *Hairy Ball Theorem*) asserts that no such global section can exist [13]. As local sections are easy to produce by taking advantage of local trivialization, the question is what type of *obstructions* are there in producing global sections. This is thus a cohomological problem.

By the work of mathematicians such as Chern, Thom, Whitney, and many others, there are natural transformations from the Vector-bundle functor to the cohomology functor of the underlying space:

$$c : K_{\mathbb{F}}(-) \Rightarrow H^*$$

where $K_{\mathbb{F}}(X)$ is the isomorphism classes of $\mathbb{F}$-vector bundles over X, and $\mathbb{F}$ is a field (usually $\mathbb{R}$ or $\mathbb{C}$). Without much trouble, this can be generalized to principal G-bundles; this shall be done for the important case of Flag bundles and Grassmannians that will allow for both classification results and computational results. In this case, a *characteristic class* would be a natural transformation:

$$c : b_G(-) \Rightarrow H^*$$

[1] these conditions insure we get maps between the cohomological groups and vector bundles while avoiding pathologies

where $b_G(-)$ is the functor taking X to the isomorphism of principal G-bundles over X. Note that there can be more than one characteristic class. Uncurling the categorical language, these natural transformations associate (up to isomorphism[2]) to each principal G bundle $P \to X \in b_G(X)$ an element $c(P) \in H^*$ such that

$$c(f^*(P)) = f^*(c(P))$$

that is, the pullback of the vector bundle commutes with the pullback of cohomology classes. The characteristic classes will be the right notion to capture the cohomological information, in particular when working over real and complex vector bundles, the cohomology of the classifying space (BO for real, BU for complex) is given by the *Stiefel-Whitney* and *Chern* classes (see [12]):

$$H^*(BO; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}_2[w_1, ..., w_n] \qquad \text{and} \qquad H^*(BU; \mathbb{Z}) \cong \mathbb{Z}[c_1, ..., c_n] \tag{2.1}$$

In this paper, our base spaces are (complete) varieties[3], so we shall mainly focus on complex vector bundles and principal $\mathrm{GL}_n(\mathbb{C})$-bundles. The complex case works with *Chern class* and the *Pontryagin class*: the Chern class is the natural characteristic class of complex vector bundles (as seen in equation (2.1)), while the Pontryagin class can be thought of as the characteristic classes of the complexification of a real vector bundle.

The Chern class is central to the Hirzebruch Riemann Roch as they are directly used in defining the Chern character and the Todd class. On the other hand, the Pontryagin classes shall be useful in that they will allow us to fully classify all cobordant oriented manifolds, a result that will be necessary for only requiring checking χ agrees with $\int_X \mathrm{ch}(E)\mathrm{td}(T_X)$ on complex projective space.

Another characteristic class that shall not be covered but is important to know as it is one of the generalizations of the Euler characteristic is the *Euler class*. As the name suggests, the Euler class is linked to the Euler characteristic, in particular if M is an oriented closed n-manifold, TM the tangent bundle, $e(TM) \in H^n(M; \mathbb{Z})$, and $[M] \in H_n(M; \mathbb{Z})$ the fundamental class of M, then:

$$e(TM) \frown [M] = \chi(M)$$

or, given the De Rham isomorphism (see [30]), re-labeling the notation $e(TM)$ to the corresponding *Euler form*:

$$\int_M e(TM) = \chi(M)$$

If M where given a Riemann metric, then the classical generalization of the Gauss-Bonnet theorem for manifolds without boundary (the *Chern Theorem*) let's us replace TM with Ω, the curvature form, giving:

$$\int_M e(\Omega) = \chi(M)$$

In other words, the Chern Theorem finds an alternative way of evaluating the Euler class at the fundamental class that relies on the curvature form Ω. The Hirzebruch-Riemann-Roch Theorem can be seen as finding another alternative evaluation at the Euler class when the underlying space M has enough structure for its characterization to be done using the Chern classes.

Finally, the definition of Chern class and Pontryagin class taken here can be thought of as the "classical" definition. In algebraic geometry, there is natural "intersection theory" perspective of of the Chern class that maps the chow group $A_n(-)$ to $A_{n-k}(-)$. There is a natural way in which this definition is linked to the classical definition of Chern class via the Chern character, and the we shall briefly bring it up in the context of the generalization of the Hirzebruch Riemann Roch to the *Grothendeick Riemann Roch theorem*; see section 6.3.

[2] even up to homotopy, see [12]

[3] in section 6.3 we shall extend to proper schemes

2.1 Definitions and key Results

For reference, here is the definition of a vector bundle and Principal bundle we shall use (this differs from the definition Hirzebruch gives via gluing of a vector bundle with structure groups, but matches contemporary textbooks such as those by Hatcher and John/Jeffrey M. Lee):

Definition 2.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \to B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber is called *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is isomorphic to F, or in particular

$$\pi^{-1}(b) \cong b \times F$$

2. (local trivialization)For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes

$$\begin{tikzcd} \pi^{-1}(U) \arrow[r, "\sim"] \arrow[d, "\pi"'] & U \times F \arrow[dl, "\pi_1"] \\ U & \end{tikzcd}$$

When clear from context, the fiber bundle is usually denoted as E.

Definition 2.1.2: G-bundle

Let $E \to X$ be a fiber bundle and G a topological group that acts faithfully on the fibers. Then a G-*atlas* is a collection of local trivializations $\{U_i, \varphi_i\}_i$ such that for any index i, j on the overlap $U_i \cap U_j$ (if it is non-empty):

$$\varphi_j^{-1} \circ \varphi_i : (U_i \cap U_j) \times F \to (U_i \cap U_j) \times F$$
$$\varphi_j^{-1} \circ \varphi_i(p, f) = (p, g_{ij}(p)f)$$

with the added conditions (cocycle conditions):

1. $g_{11} = \mathrm{id}$
2. $g_{ij} = g_{ji}^{-1}$
3. $g_{ik} = g_{jk}g_{ij}$

Two G-atlases are equivalent if their union is a G-atlas. A G-*bundle* is a fiber bundle with an equivalence relation of G-atlases.

Definition 2.1.3: Principal G-Bundle

If $F = G$ is a topological group and $P \to X$ is a fiber bundle over X group where there is a free and transitive continuous right group action $P \times G \to P$ that preserves the fibers of P, then it is called a *principal G-bundle* (or *principal bundle* if G is clear from context. In other words, a principal G-bundle is a G-bundle where the fibers are G.

Each fiber is called a *G-torsor* (which is a space that where the action $G \curvearrowright F_x$ has trivial stabilizer, hence "is" the group G but does not intrinsically have a notion of identity element)

Definition 2.1.4: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle complex vector bundle.*

The following fact shall freely be used (proven in [12]):

Theorem 2.1.5: Homotopy Invariance Vector Bundles

Let $p : E \to B$ a vector bundle and $f_0, f_1 : A \to B$ be homotopic maps. Then if A is paracompact,

$$f_0^*(E) \cong f_1^*(E)$$

***Proof*:**
[12, section 1.2]

Theorem 2.1.6: Classifying Vector Bundles

Let X be a paracompact space. Then $K^n(X)$ is in correspondence with the set of homotopy classes of maps $X \to \mathrm{Grass}_n$, denoted $[X, \mathrm{Grass}_n]$. In other words:

$$K^n(X) \cong [X, \mathrm{Grass}_n]$$

In particular, if $E \to X$ is an n-dimensional vector bundle, it is isomorphic to $f^*(E_n)$ for some $f : X \to \mathrm{Grass}_n$ where E determines f up to homotopy.

***Proof*:**
[12, section 1.2]

This motivates the following natural definition:

Definition 2.1.7: Universal Bundle

The infinite Grassmannians Grass_k are called the *universal bundles*; for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted BO, BU,and BSp respectively.

2.2 The Splitting Principle

One of the most important results about vector bundles is that there is a pullback map which allows them to split into a direct sum of line bundles. As a consequence, the total Chern class (which is representable as a polynomial) can be split into product of Chern classes of line bundles (which can be represented by linear terms). The key cohomological result is the following fiber-local to global theorem proved by Leray and Hirsch:

Theorem 2.2.1: Leray-Hirsch Theorem

Let R be a commutative ring and $p : E \to X$ be a fiber bundle. In [13] it was shown $H^*(E; R)$ is a $H^*(X; R)$-module where

$$\alpha \cdot \beta := p^*(\alpha) \smile \beta$$

with $\alpha \in H^*(X)$ and $\beta \in H^*(E)$. If

1. for each fiber F the inclusion $F \hookrightarrow E$ induces a surjection on $H^*(-; R)$
2. $H^n(F; R)$ is a free R-module of finite rank for each n.

Then $H^*(E; R)$ is a free $H^*(X; R)$-module. A basis for the $H^*(X; R)$-module $H^*(E; R)$ can be obtained by choosing any set of elements in $H^*(E; R)$ that map to a basis for $H^*(F; R)$ under the map induced by the inclusion.

***Proof*:**
see [13].

Theorem 2.2.2: Splitting Principle

Let $\pi : E \to B$ be a complex vector bundle. Then there exists a space $F(E)$ and a map $p : F(E) \to B$ such that the pullback $p^*(E) \to F(E)$ splits as a direct sum of line bundles, and

$$p^* : H^*(B; \mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z})$$

is injective

***Proof*:**
Consider the pullback $\mathbb{P}(\pi)^*(E)$ of E via the map $\mathbb{P}(\pi) : \mathbb{P}(E) \to B$ where $\mathbb{P}(E)$ is the space of lines through the origin in all fibers of E. This pullback contains a natural one-dimensional subbundle

$$L = \{(\ell, v) \in \mathbb{P}(E) \times E \mid v \in \ell\}$$

An inner product on E pulls back to an inner product on the pullback bundle, and thus a splitting of the pullback as a sum $L \oplus L^\perp$ with the orthogonal bundle $L^\perp$ having dimension one less than E. Then by the Leray-Hirsch theorem to $\mathbb{P}(E) \to B$, $H^*(\mathbb{P}(E); \mathbb{Z})$ is the free $H^*(B; \mathbb{Z})$-module with basis $1, x, \ldots, x^{n-1}$ and in particular the induced map $H^*(B; \mathbb{Z}) \to H^*(\mathbb{P}(E); \mathbb{Z})$ is injective since one of the basis elements is 1.

This construction can be repeated with $L^{\perp} \to \mathbb{P}(E)$ in place of $E \to B$, until we get the splitting, completing the proof.

The following will be important when applying the Euler characteristic to $F(B)$ for the Hirzebruch Riemann Roch:

Theorem 2.2.3: Splitting Principal Over Varieties

Let B be a projective variety. Then $F(B)$ is also a projective variety

***Proof*:**
[4]. A weaker version of this theorem is cited later in the paper (see theorem 3.2.14).

2.3 Chern and Pontryagin Classes

The axiomatic approach taken by Grothendeick shall be taken:

Theorem 2.3.1: Existence and Uniqueness of Chern Classes

Let $E \to X$ be a complex vector bundle. Then there exists a sequence of functions $c_1, c_2, ...$ where $c_i(E) \in H^{2i}(X; \mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $c_i(f^*(E) = f^*(c_i(E))$ for a pullback $f^*(E)$
2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \cdots \in H^*(X; \mathbb{Z})$
3. $c_i(E) = 0$ if $i > \dim E$
4. Given $E \to \mathbb{C}\mathrm{P}^\infty$ is the canonical line bundle, $c_1(E)$ is a generator of $H^2(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ specified in advance.

***Proof*:**
see [12] for a full proof. The key idea is to invoke the Leray-Hirsch Theorem for existence and the splitting principle for uniqueness. In fact, we get a bit more out of the Leray-Hirsch theorem: by the classification of vector bundles (theorem 2.1.6) there is map $f : E \to \mathbb{C}\mathrm{P}^\infty$ which induces a cohomology pullback $f^* : H^*(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \to H^*(\mathbb{P}(E); \mathbb{Z})$, in particular $f^* : H^1(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \to H^1(\mathbb{P}(E); \mathbb{Z})$ which pullback the generator $\alpha \in H^1(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z})$ to an element $f^*(\alpha) = x$. Then the elements $1, x, ..., x^n$ are the generators of $H^*(\mathbb{P}(E), \mathbb{Z})$ as a free $H^*(X; \mathbb{Z})$-module, and if $\alpha\beta := \alpha \smile \beta$ then we get a unique relationship:

$$x^n - c_1(E) + \cdots (-1)^n c_n(E) \cdot 1 = 0$$

which gives a *relationship* between the Chern classes.

Uniqueness is given by the splitting principle, namely if there was another such relationship, then as the above equation must pullback and split, the Chern classes must match.

For the choice of normalization, note that for property (4), the canonical line bundle over $\mathbb{C}\mathrm{P}^1$ could have been used instead of $\mathbb{C}\mathrm{P}^\infty$ as $\mathbb{C}\mathrm{P}^1 \hookrightarrow \mathbb{C}\mathrm{P}^\infty$ induces an isomorphism

$$H^1(\mathbb{C}\mathrm{P}^\infty; \mathbb{Z}) \cong H^1(\mathbb{C}\mathrm{P}^1; \mathbb{Z})$$

and we know by some cohomology theory that we can choose any hyperplane for the fundamental class of $H^1(\mathbb{C}\mathrm{P}^1; \mathbb{Z})$, hence $c_1(E) \in H^1(\mathbb{C}\mathrm{P}^1; \mathbb{Z})$ becomes a choice of hyperplane.

Note that this definition differs slightly from what Hirzebruch presents: he replaces the third axiom with the following:

(3) If $\xi_1, \ldots, \xi_q$ are continuous $\mathbf{U}(1)$-bundles over X then

$$c(\xi_1 \oplus \cdots \oplus \xi_q) = c(\xi_1) \smile \cdots \smile c(\xi_q).$$

The real equivalent has more subtly due to the fact that real manifolds and vector bundles have a choice of orientation, and hence the natural cohomological setting to work in is over $\mathbb{Z}/2\mathbb{Z}$-coefficient and classify $H^*(BO; \mathbb{Z}/2\mathbb{Z})$ where we take $\mathbb{Z}/2\mathbb{Z}$ to get rid of choice of orientability. This uses *Stiefel-Whitney* classes which we did not touch on in this paper. To classify $H^*(BO; \mathbb{Z})$, we essentially take the complexification of a real vector bundle, which gives us the following best possible result:

Definition 2.3.2: Pontryagin Class

Let $E \to B$ be a real vector bundle. Complexity it by adding the almost complex structure to $E \oplus E$:

$$E^{\mathbb{C}} = E \oplus E \qquad i(x, y) = (-y, x)$$

Then a *Pontryagin class* $p_i(E) \in H^{4i}(B; \mathbb{Z})$ is given by:

$$p_i(E) = (-1)^i c_{2i}(E^{\mathbb{C}}) \in h^{4i}(B; \mathbb{Z})$$

To characterize Pontryagin classes, we state the following:

Proposition 2.3.3: Characterizing Grassmannian Cohomology

All torsion of $H^*(BO, \mathbb{Z})$ is 2-torsion; if T is the torsion subgroup then:

$$\frac{H^*(BO, \mathbb{Z})}{T} \cong \mathbb{Z}[p_1, ..., p_k] \qquad n = 2k \text{ or } 2k+1$$

***Proof*:**
This is not important for our current work, so the proof is deferred to [12] or [31]

What shall be important to use is the use of Pontryagin classes in the theory of cobordism in section 5.

Pontryagin and Chern classes relate through the following relation:

Theorem 2.3.4: Chern and Pontryagin Relation

The Chern classes c_i of the almost complex manifold X are related to the Pontryagin classes p_i of X (regarded as a differentiable manifold) by:

$$p = \sum_{i=0}^{\infty}(-1)^i p_i = \sum_{i=0}^{\infty} c_i \sum_{j=0}^{\infty}(-1)^j c_j.$$

***Proof*:**
untangle and use definition 2.3.2.

2.4 Chern Character and K-Theory

Fix a base space X and let $K(X) = K_{\mathbb{C}}(X)$ be the set of isomorphism classes of complex vector bundles on X. Let $c(-) : K(X) \to H^*(X;\mathbb{Z})$ be the Chern class. Recall that

$$c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$$

In particular, it brings the addition in $K(X)$ to multiplication in H^*. The collection $K(X)$ can be completed to form a group, also denoted $K(X)$, and it is known as the *Grothendieck K-group*. The tensor gives it a natural ring structure. By the above construction, we see that $c(-)$ is *not* a ring homomorphism for it is not the case that $c(E \otimes F) = c(E) + c(F)$ (hint: E, F must be at least 2-dimensional for the counter-example, see [12]). From the Chern class $c(-)$, we can construct a new map known as the *Chern character*

$$\text{ch}(-) : K(X) \to H^*(X;\mathbb{Z}) \otimes \mathbb{Q}$$

which *will* be a ring homomorphism, provided we add rational coefficients.

By the splitting principle, it suffices to work with line bundles. The total Chern class is:

$$c(L) = 1 + c_1(L)$$

where $c_1(L) \in H^2(X;\mathbb{Z})$. Then it is a worth-while exercise to show that for *line bundles*[4] (not for general vector bundles!):

$$c_1(L_1 \otimes L_2) = c_1(L) + c_1(L)$$

which as addition. To fix this into multiplication, we can take:

$$\text{ch}(L) = e^{c_1(L)} = 1 + c_1(L) + \frac{c_1(L)^2}{2!} + \frac{c_1(L)^3}{3!} + \cdots$$

where the power should be taken as repeated cup product. This fixes the multiplicative problem as:

$$\text{ch}(L_1 \otimes L_2) = e^{c_1(L_1)+c_1(L_2)} = e^{c_1(L_1)} \cdot e^{c_1(L_2)}$$

[4] use the correspondence between line bundles and cohomology classes, see [12]

But this introduces rational coefficients. As $H^*(X;\mathbb{Z})$ can have torsion, ex. $H^*(\mathbb{C}\mathrm{P}^1;\mathbb{Z}) \cong \mathbb{Z}[h]/(h^2)$ (see [13, appendix]), many of the terms here disappear and it is not well-defined. Thus, we must define:

$$H^*(X;\mathbb{Q}) := H^*(X;\mathbb{Z}) \otimes \mathbb{Q}$$

which in [29] it is shown that the torsion is eliminated, and hence $\mathrm{ch}(L) \in h^*(X;\mathbb{Q})$ is well-defined. Then, for an arbitrary vector bundle E, we can find a unique splitting

$$E = L_1 \oplus L_2 \oplus \cdots \oplus L_n$$

and define:

$$\mathrm{ch}(E) = \mathrm{ch}(L_1) + \mathrm{ch}(L_2) + \cdots + \mathrm{ch}(L_n)$$

where each $\mathrm{ch}(L_i) = t_i$ are the *Chern root* of $c(E)$. By construction:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2)$$

Giving the desired algebraic properties. However, $\mathrm{ch}(E)$ is dependent on the Chern classes $c_1(L_i)$ where i goes over the Chern roots, as opposed to $c_j(E)$ where j increased in dimension. This is easily fixed with a bit of combinatorics. Namely, $c_j(E)$ is given by:

$$c_j(E) = \sigma_k(c_1(L_1), ..., c_1(L_n))$$

where σ_k is the kth elementary-symmetric polynomial. Define the *power sums* to be:

$$s_k(x_1, ..., x_n) = \sum_i^n x_i^k$$

Then the *Newton identities* (see this link) show that (via elementary computations):

$$\begin{aligned}
s_1 &= \sigma_1 \\
s_2 &= \sigma_1^2 - 2\sigma_2 \\
s_3 &= \sigma_1^3 - 3\sigma_2\sigma_1 + 3\sigma_3 \\
s_3 &= e_1^4 - 4e_2e_1^2 + 4e_3e_1 + 2e_2^2 - 4e_4 \\
&\vdots
\end{aligned}$$

From these we get:

$$\mathrm{ch}(E) = \sum_{i=1}^n \left(\sum_{k=0}^\infty \frac{t_i^k}{k!} \right) = \sum_{k=0}^\infty \frac{1}{k!} \left(\sum_{i=1}^n s_k(c_1(E), ..., c_n(E)) \right)$$

Writing out the first few terms we get:

$$\mathrm{ch}(-) = \dim_{\mathbb{C}}(-) + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2 + 3c_3) + \cdots$$

giving us a power series purely in terms of the Chern classes of E! Note that the Chern character in facts captures more information than the Chern classes as the Chern classes are *stable*, meaning they are invariant when taking the direct sum with trivial bundles, however $\mathrm{ch}(-)$ has $\dim_{\mathbb{C}}(-)$ in its 0th position, in particular it remembers the rank of the bundle[5] To connect this to the K-group, we require the following:

Proposition 2.4.1: Extension and Tensors With Chern Character

1. let E'', E, E' be vector bundles such that:

$$0 \to E'' \to E \to E' \to 0$$

then:

$$\mathrm{ch}(E) = \mathrm{ch}(E') + \mathrm{ch}(E'')$$

And similarly

$$\mathrm{ch}(E \otimes \eta) = \sum_{i,j} e^{(\gamma_j + \delta_j)t} = \sum_{i,j} e^{\gamma_j t} e^{\delta_j t} = \left(\sum_i e^{\gamma_j t}\right)\left(\sum_j e^{\gamma_j t}\right) = \mathrm{ch}(E)\,\mathrm{ch}(\eta).$$

2. Let E_1, E_2 be vector bundles. Then:

$$\mathrm{ch}(E_1 \otimes E_2) = \mathrm{ch}(E_1) \cdot \mathrm{ch}(E_2)$$

where $\cdot$ is the cup product.

***Proof*:**
exercise.

Thus, there is a natural map:

$$\mathrm{ch} : K(X) \to H^*(X; \mathbb{Q})$$

This shows that $\mathrm{ch}(-)$ is the right notion for capturing the cohomological information of vector bundles which respects the algebraic structure of $K(X)$. In chapter 4, we shall show that $\mathrm{ch}(E)$ capture the bundle information, while $\mathrm{td}(X)$ (define in chapter 4) shall capture the relevant information from the underlying space.

A natural question to ask is whether $f : X \to Y$ is a continuous map, then does $\mathrm{ch}(-)$ commutes with f^*, namely if:

$$\begin{tikzcd} K(Y) \arrow[r, "f^*"] \arrow[d, "\mathrm{ch}"'] & K(X) \arrow[d, "\mathrm{ch}"] \\ H^*(Y;\mathbb{Q}) \arrow[r, "f^*"] & H^*(X;\mathbb{Q}) \end{tikzcd}$$

However this is not the case. What fixes the commutativity is the introduction of the Todd class. It is in fact a little more complicated as we would usually be taking coherent sheaves and working with the proper pushforward rather than the pullback, and so the details are left for section 6.3:

[5] As a fun aside, if $X = \mathbb{C}\mathrm{P}^n$, then the Hilbert polynomial can be recovered by the Chern character, and vice-versa; see [6]

$$\begin{tikzcd} K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(Y) \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TY)"] \\ A_*(X;\mathbb{Q}) \arrow[r, "f_*"] & A_*(Y;\mathbb{Q}) \end{tikzcd}$$

where $A_*(-,\mathbb{Q})$ is the *chow ring* with rational coefficients.

3

Cohomology of Varieties – new Euler Characteristics

The Euler characteristic is in a sense "too rough" of an invariant for varieties. We require to capture some of the complex structure to find the right invariants. In particular, we shall refine the usual De-Rham/singular cohomology into *Dolbeault* cohomology which shall capture the obstruction to a function/local-section being *holomorphic*. This shall form a new collection cohomology groups (depending on the degree) that will allow us to define new Euler-characteristics that generalize the topological Euler Characteristic.

We start by developing some basic Hodge theory. Some key facts that shall be shown are that the Dolbeault cohomology groups are finite dimensional by using the fact that elliptic operators have finite-dimensional kernels (see theorem 3.1.8), that the deRham theorem has a natural generalization to the Dolbeault' Theorem, that all forms have a harmonic representative, and that line bundle are in bijective correspondence with $(1,1)$-cohomology classes; the reader who is comfortable with this result can skip over most the technical aspects of this section.

Then, as all smooth projective manifold can be embedded into $\mathbb{C}\mathrm{P}^n$ have an algebraic equation by Chow's theorem ([33]), there is a natural compatibility between the Riemann metric, symplectic form, and complex structure. This shall define Kähler manifolds, which are almost a variety; after defining what it means to be positive (essentially, insuring the tangent bundle is very ample, think of $\mathcal{O}(n)$ vs $\mathcal{O}(-n)$ for $n \in \mathbb{N}$) we shall show that all positive closed Kähler manifolds are a (projective) smooth variety. This will let us translate the results of Dolbeault cohomology (ex. the finiteness of the cohomology groups) over to algebraic geometry. Note that in the proof of Grothendieck Riemann Roch, this is replaced with the fact that $H^i(M, \mathcal{F})$ is finite-dimensional for M projective and $\mathcal{F}$ coherent (see my notes [3] or Hartshorne [34]).

We then define new Euler-characteristics, and generalize to defining the *virtual characteristic* which is the Euler characteristic of submanifolds that remembers which submanifolds it came from. We shall end off by mentioning some important Vanishing theorems that would allow us to compute $H^0(M, \mathcal{F})$.

3.1 Dolbeault Cohomology

We shall assume the basic knowledge of differential geometry that can be found in [30] or [7], and of complex analysis that can be found in [32] or [24]. This section shall be very rapid to cover the key ideas necessary to define the Euler characteristics, sometimes omitting some proofs that are better covered in a full text in Hodge Theory.

Let M be a real manifold. Then we can take the fiber-wise tensor product $TM \otimes \mathbb{C}$ to complexify the manifold. If $J : TM \to TM$ is an endomorphism such that $J^2 = -1$[1], then $TM \otimes \mathbb{C}$ can be decomposed into:

$$TM \otimes \mathbb{C} \cong T^{1,0}M \oplus T^{0,1}M$$

where

$$T^{1,0}M = \{x - iJx \mid x \in TM\} \qquad \text{and} \qquad T^{0,1}M = \{x + iJx \mid x \in TM\}$$

Thus, there is a natural decomposition of tangent bundles into holomorphic and anti-holomorphic components. The choice of J on $TM \otimes \mathbb{C}$ makes (M, j) an *almost complex manifold*. All complex manifolds are almost complex, but the converse is not the case (see the Newlander-Nirenberg theorem in [4]). Much (but not all) of the work in this section works just as well on almost-complex manifolds; this paper will stick to complex manifolds but the reader should feel free to work with almost-complex manifolds.

For a moment, let M be a one-dimensional complex manifold. As holomorphic mappings respects complex and anti-complex structure, any 1-form with complex coefficients can be uniquely with respect written as a sum:

$$fdz + gd\bar{z}$$

Let $\Omega^{1,0}$ represent holomorphic 1-forms, and $\Omega^{0,1}$ represent anti-holomorphic 1-forms.The spaces $\Omega^{1,0}$ and $\Omega^{0,1}$ are both stable under holomorphic coordinate changes (use Cauchy-Riemann). A similar pattern takes shape in higher dimensions. Let M be n-dimensional. Like before, any 1 form with complex coefficients can be written as:

$$\sum_{j=1}^{n} (f_j dz^j + g_j d\bar{z}^j)$$

Define:

$$\Omega^{p,q} = \underbrace{\Omega^{1,0} \wedge \cdots \wedge \Omega^{1,0}}_{p \text{ times}} \wedge \underbrace{\Omega^{0,1} \wedge \cdots \wedge \Omega^{0,1}}_{q \text{ times}}$$

Then we know that for every $r \in \mathbb{N}$, every smooth r-form on M (with $\mathbb{C}$-coefficients) can be written as the sum of (p, q)-forms where $p + q = r$ (as the basis of these forms are wedge product of the lower dimensions, see [30], [13]).

Now, let E^k be the space of all complex differential forms of total degree k. Then

$$E^k = \Omega^{k,0} \oplus \Omega^{k-1,1} \oplus \cdots \oplus \Omega^{1,k-1} \oplus \Omega^{0,k} = \bigoplus_{p+q=k} \Omega^{p,q}$$

[1]Note this forces M to be even-dimensional

if M is a complexified manifold, then:

$$\Omega^k(M,\mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}$$

Note that the direct sum decomposition is stable under holomorphic coordinate changes, hence it determines a vector bundle decomposition. From this direct product, there are natural projections for each direct summand:

$$\pi^{p,q} : E^K \to \Omega^{p,q}$$

Now, define $d : \Omega^r \to \Omega^{r+1}$ in the natural way where:

$$d(\Omega^{p,q}) \subseteq \bigoplus_{r+s=p+q+1} \Omega^{r,s}$$

Definition 3.1.1: Dolbeault Operators

$$\partial = \pi_{p+1,q} \circ d \qquad \text{and} \qquad \overline{\partial} = \pi_{p,q+1} \circ d$$

In coordinates, let:

$$\alpha = \sum_{|I|=p,|J|=q} f_{IJ} dz^I \wedge d\bar{z}^J \in \Omega^{p,q}$$

where I and J are multi-indices. Then:

$$\partial\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial z^\ell} dz^\ell \wedge dz^I \wedge d\bar{z}^J$$

and

$$\bar{\partial}\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial \bar{z}^\ell} d\bar{z}^\ell \wedge dz^I \wedge d\bar{z}^J.$$

When M is a complex manifold (equiv. it has a complex structure), then by the *Newlander–Nirenberg theorem*:

$$d = \partial + \bar{\partial}$$
$$\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

3.1.2 For those who know Hodge Theory Note that these operators require M to be complex, not just almost complex. There are similar results for almost-complex manifold, see [4] or [17].

Thus, we see there is some interesting cohomology that can occur; this defines *Dolbeault cohomology*. The symbol ∂ suggests that they are dual to d, and indeed the Poincaré lemma generalizes to complex manifolds.

Let us define the Dolbeault cohomology. Notice that $\overline{\partial} f = 0$ if f is holomorphic, therefore it is in some sense the natural choice; the other choice (∂) keeping track of the anti-holomorphic information. Thus, define:

Definition 3.1.3: Dolbeault Cohomology

$$H^{p,q}_{\overline{\partial}}(M) = \frac{\ker(\overline{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\overline{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

Where the $\overline{\partial}$ is dropped if it clear from context: $H^{p,q}(M)$.

As

$$H^{p,q}_{\overline{\partial}}(M) = \overline{H^{p,q}_{\partial}(M)}$$

We see that the difference of choice is a difference in conjugation, and hence we stick to $\overline{\partial}$ for the holomorphic relation.

For our purposes, we will naturally want to generalize this to the Dolbeault cohomology of vector bundles. let E be a holomorphic vector bundle on a complex manifold M. Then take the map:

$$\overline{\partial}_E : \Gamma(E) \to \Omega^{0,1}(E)$$

taking section of E to $(0,1)$-forms with values in E. This will satisfy the Leibniz rule with respect to $\overline{\partial}$, and is thus a $(0,1)$-connection on E. Then we can do the same trick from Riemann Geometry and extended it to the exterior covariant derivative, namely;

$$\overline{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E)$$

which acts on on a section $\alpha \otimes s \in \Omega^{p,q}(E)$ by:

$$\overline{\partial}_E(\alpha \otimes s) = (\overline{\partial}\alpha) \otimes s + (-1)^{p+q} \alpha \wedge \overline{\partial}_E s$$

This naturally extends linearly to any section of $\Omega^{p,q}(E)$ and $\overline{\partial}_E^2 = 0$, hence it defines the Dolbeault cohomology with coefficient in E. It is standard to check that the Dolbeault cohomology groups do not depend on the choice of Dolbeault operator $\overline{\partial}_E$ compatible with the holomorphic structure of E [21], and so:

Definition 3.1.4: Dolbeault Cohomology For Vector Bundles

$$H^{p,q}(M,(E,\bar{\partial}_E)) = \frac{\ker(\bar{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E))}{\operatorname{im}(\bar{\partial}_E : \Omega^{p,q-1}(E) \to \Omega^{p,q}(E))}$$

which is unambiguously denoted as:

$$H^{p,q}(M,E)$$

if $\mathbb{1}$ is the trivial bundle, then we get back the usual Dolbeault cohomology.

Definition 3.1.5: Hodge Number

$$h^{p,q}(M,E) = \dim H^{p,q}(M,E)$$

If $E = \mathbb{1}$ then:

$$h^{p,q}(M) = \dim H^{p,q}(M,\mathbb{1})$$

Let Ω^p be the sheaf of holomorphic p-forms on M. Then the following is the natural generization of De Rham's theorem for Dolbeault cohomology:

Theorem 3.1.6: Dolbeault's Theorem

$$H^{p,q}(M) \cong H^q(M, \Omega^p)$$

and if we are working over a vector bundle:

$$H^{p,q}(M, E) \cong H^q(M, \Omega^p \otimes E)$$

Furthermore:

$$H^{0,q}(M, E) \cong H^q(M, \Omega^0 \otimes E) = H^q(M, E)$$

***Proof*:**
This proof is very similar to the De Rham case. Note it can also be proven by taking a resolution of $\Omega^{p,q}$ using *fine sheaves* (sheaves that admit partitions of unity)[a] and applying the $\overline{\partial}$-lemma (see [21] or [16, section 15.4]).

[a] flasque (flabby) sheaves are not fine, but are the next best thing they are *soft*: every section in a closed subset can be extended to the whole sheaf

For future reference, if E is a complex vector bundle, we shall denote its conjugate by $\overline{E}$ (defined in the natural way) which is anti-complex and anti-isomorphic to E. We can also define the dual bundle E^* in the natural way. Note that per-fiber, the dual can be given by Hermitian of the inverse of the matrix.

Hodge Theorem: Finiteness of Cohomology Groups

Let M now be compact and complex. Let us build-up to showing that each class in $H^{p,q}(M, \mathbb{C})$ is representable by a harmonic form. This will allow us to quote result from the theory of elliptic operators that will show these vector spaces are finite dimensional, and in theorem 6.2.1 will be key in showing the Euler characteristic is equal to the spin index. For notational simplicity, let:

$$A^{p,q}(W) = \Gamma(V, \Omega^{p,q}(W))$$

and:

$$A^{p,q} = A^{p,q}(\mathbb{1})$$

where $\mathbb{1}$ is the trivial bundle representing holomorphic functions. We shall define the Hodge operator that will allow us to relate the different $A^{p,q}$. If W is a complex vector bundle, we can define W^* and a (perfect) pairing:

$$A^{p,q}(W) \times A^{r,s}(W^*) \to A^{p+r,q+s}(\mathbb{1}) \qquad (\alpha, \beta) \mapsto \alpha \wedge \beta$$

If $W = \mathbb{1}$, this is the usual wedge (or exterior) product. The product is anti-commutative in the sense that:

$$\alpha \wedge \beta = (-1)^{(p+r)(q+s)} \beta \wedge \alpha$$

Choosing $r = n-p$, $s = n-q$ we get the top form $\alpha \wedge \beta \in A^{n,n}(\mathbb{1})$. We can thus define:

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge \beta$$

By Stokes theorem, if $\alpha = \overline{\partial}\gamma$ for $\gamma \in A^{p,q-1}(W)$ and $\overline{\partial}\beta = 0$, (or if $\overline{\partial}\alpha = 0$ and $\beta = \overline{\partial}\gamma$) then:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta = \int_M \overline{\partial}(\gamma \wedge \beta) = \int_M d\,(\gamma \wedge \beta) = 0$$

Thus, we get a well-defined pairing:

$$H^{p,q}(M, E) \times H^{n-q,n-q}(M, E^*) \to \mathbb{C} \qquad (\alpha, \beta) \mapsto \langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta$$

Now, by some linear algebra and differential geometry we see that $\overline{TM} \cong T^*M$, and so:

$$\begin{aligned}\bigwedge^p TM \otimes \bigwedge^q TM &\cong \bigwedge^p TM \otimes \bigwedge^n T^*M \otimes \bigwedge^n TM \otimes \bigwedge^q T^*M \\ &\cong \bigwedge^{n-p} T^*M \otimes \bigwedge^{n-q} TM \\ &\cong \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}\end{aligned}$$

There is thus a natural duality operator:

$$* : \bigwedge^p TM \otimes \bigwedge^q TM \to \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}$$

and $* \circ *$ becomes multiplication by $(-1)^{p+q}$.

Now, by vector-bundle theory we know we can without loss of generality reduce W to a unitary group bundle ($U(q)$-bundle); see [12]. There is then the natural hermitian anti-isomorphism:

$$\psi : W \to W^* \qquad \psi^{-1} : W^* \to W$$

If κ is the anti-isomorphism from $\bigwedge^r TM \otimes \overline{\bigwedge^s TM}$ to itself via conjugation, Then define:

$$\star = \psi \otimes (\kappa \circ *) \qquad \widetilde{\star} = \psi^{-1} \otimes (\kappa \circ *)$$

This gives:

$$\begin{aligned}\star &: W \otimes \bigwedge^p TM \otimes \overline{\bigwedge^q TM} \to W^* \otimes \bigwedge^{n-p} TM \otimes \overline{\bigwedge^{n-q} TM} \\ \widetilde{\star} &: W^* \otimes \bigwedge^r TM \otimes \overline{\bigwedge^s TM} \to W \otimes \bigwedge^{n-r} TM \otimes \overline{\bigwedge^{n-s} TM}.\end{aligned}$$

If $r = n-p$, $s = n-q$, then $\widetilde{\star} \circ \star$ is multiplication by $(-1)^{p+q}$. Now, these produce natural anti-isomorphisms of:

$$\begin{aligned}\star &: \Omega^{p,q}(W) \to \Omega^{n-p,n-q}(W^*) \\ \widetilde{\star} &: \Omega^{r,s}(W^*) \to \Omega^{n-r,n-s}(W)\end{aligned}$$

Using these, we can define the "co-" of $\overline{\partial}$:

$$\delta : \Omega^{p,q}(W) \to \Omega^{p,q-1}(W) \qquad \delta = -\widetilde{\star} \circ \overline{\partial} \circ \star$$

With this, if $\alpha, \beta \in A^{p,q}(W)$ then we can define:

$$(\alpha, \beta) = \langle \alpha, \star\beta \rangle = \int_M \alpha \wedge \star\beta$$

It is easy to see that $(\alpha, \alpha) \geq 0$ and $(\alpha, \alpha) = 0$ if and only if $\alpha = 0$. We also get the natural adjoint:

$$(\alpha, \delta\beta) = \langle \overline{\partial}\alpha, \beta \rangle$$

Indeed given:

$$\langle \alpha, \delta\beta \rangle = -\int_M \alpha \wedge \star\widetilde{\star}\overline{\partial}\beta = (-1)^{p+q+1} \int_M \alpha \wedge \overline{\partial} \star \beta$$

Then:

$$\begin{aligned}
(\overline{\partial}\alpha, \beta) - (\alpha, \delta\beta) &= \int_M (\overline{\partial}\alpha \wedge \star\beta + (-1)^{p+q} \alpha \wedge \overline{\partial} \star \beta) \\
&= \int_M \overline{\partial}(\alpha \wedge \star\beta) \\
&= \int_M d(\alpha \wedge \star\beta) \\
&= 0 && \text{Stokes Thm.}
\end{aligned}$$

We now reach one of the central results in Hodge theory. Define the *Hodge operator* (or Laplace-Bertram operator):

$$\Delta = \delta\overline{\partial} + \overline{\partial}\delta$$

This defines an operator:

$$\Delta : A^{p,q}(W) \to A^{p,q}(W)$$

Let

$$\ker \Delta = B^{p,q}(W)$$

which could be interpreted as all complex harmonic forms, namely forms that satisfy:

$$\delta(\alpha) = \overline{\partial}(\alpha) = 0$$

The use of the elliptic operator is that it gives way to the following decomposition:

$$A^{p,q}(W) = \overline{\partial}A^{p,q-1}(W) \oplus \delta A^{p,q+1}(W) \oplus B^{p,q}(V, W)$$

Notice that we naturally get:

$$Z^{p,q}(W) = \overline{\partial}A^{p,q-1}(W) \oplus B^{p,q}(V, W)$$

Thus, when taking the Dolbeault cohomology group:

Theorem 3.1.7: Hodge Theorem

$$H^{p,q}(V,W) \cong \frac{Z^{p,q}(W)}{\bar{\partial}A^{p,q}(W)} \cong B^{p,q}(V,W)$$

In other words, these forms are naturally harmonic! Note that in general $h^{p,q}(M) \neq h^{q,p}(M)$; if M is a Kähler manifold this will be the case.

As Δ is an elliptic operator, we get by elliptic operator theory that each of these spaces is *finite dimensional* (see [4]). We thus also get:

$$H^r(M,\mathbb{C}) \cong \bigoplus_{p+q=r} B^{p,q}$$

and the Betti number is the sum of Hodge numbers:

$$b_r(M) = \sum_{p+q=r} h^{p,q}(M)$$

For future reference, an element of $H^{p+q}(M,\mathbb{Z})$ or $H^{p+q}(M,\mathbb{R})$ is said to be of type (p,q) if it is of type (p,q) when regarded as an element of $H^{p+q}(M,\mathbb{C})$.

All of this is summarized in the following two theorems:

Theorem 3.1.8: Kodaira Finite Dimension Theorem

The Dolbeault cohomology group $H^{p,q}(M,E)$ is finite dimensional isomorphic to the vector space of complex harmonic forms of type (p,q) with coefficients in E. Furthermore:

$$H^p(M,E) = H^{0,p}(M,E)$$

and is zero for $p,q > n$.

***Proof*:**
For more details, see for example [4] or [25].

Theorem 3.1.9: Serre-Hodge Duality Theorem

Let E be a complex vector bundle over the compact manifold M. Then the bilinear form $\langle -,-\rangle$ is a dual pairing of $H^{p,q}(M,E)$ with $H^{n-p,n-q}(M,E^*)$. If $K = \bigwedge^n TM$ is the canonical line bundle, Then $H^p(M,E)$ and $H^{n-q}(M,K\otimes E^*)$ are dual vector spaces.

***Proof*:**
The work above is most of it, but the details can be found in [4] or [25].

3.2 Geometric-Algebraic Bridge: Kähler Manifold

Let M be a compact complex Riemann manifold. A Hermitian metric on M is defined as:

$$ds^2 = 2\sum g_{\alpha\beta}(z,\overline{z})(dz^\alpha \cdot d\overline{z}^\beta)$$

where $\overline{g_{\alpha\beta}} = g_{\beta\alpha}$. From a Hermitian metric ds^2, we get a natural exterior differential form:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

This form can be re-written as a real differential form by taking real coordinates x^α, $1 \leq \alpha \leq 2n$ and

$$z^\alpha = x^{2\alpha-1} + ix^{2\alpha}$$

Definition 3.2.1: Kähler Metric

Let ds^2 be a Hermitian metric on the compact complex Riemann manifold M and let:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

Then if $d\omega = 0$, ds^2 is called a *Kähler metric.*

If ds^2 is a Kähler metric, ω represents an element of $H^2(M,\mathbb{R})$ (using the De-Rham isomorphism); this is called the *fundamental class of the Kähler metric.*

Definition 3.2.2: Kähler Manifold

Let M be a compact complex Riemann manifold with Riemann metric ds^2. Then given a choice of ω that induces ds^2 to be Kähler metric, (M,ω) is called a *Kähler manifold.*

On a Kähler manifold, Δ commutes with the conjugation $\alpha \mapsto \overline{\alpha}$ between $B^{q,p}$ and $B^{p,q}$ which induces an isomorphism between these groups, and so:

Theorem 3.2.3: Symmetry of Hodge Number On Kähler Manifold

Let M be a Kähler manifold. Then:

$$h^{p,q}(M) = h^{q,p}(M)$$

***Proof*:**
details in [4] or [25].

Note that these numbers do not depend on the choice of Kähler metric ($d\alpha = 0$ for a harmonic (p,q)-form regardless of metric) and hence it is enough for a Kähler metric to exist.

Now, being a Kähler manifold is very close to being a projective variety. Recall from Hartshorne that a sheaf on a scheme is very ample[2] if has "enough projectives" (see Hartshorne's exposition at [34]

[2] or ample, meaning it is very ample after tensoring with an appropriate $\mathcal{O}(n)$

or my exposition at [3]). Not all Kähler manifolds have enough projectives, but it is a surprisingly simple condition that insures that a Kähler manifold can embed into projective space.

Definition 3.2.4: Real Positivity

Let $x \in H^{1,1}(M, \mathbb{R})$. Then x is said to be *positive* if x can be chosen as the fundamental class of a metric on M. Kähler

Proposition 3.2.5: Properties of Real Positivity

Let M be a Kähler manifold. Then

1. At least one element of $H^{1,1}(M, \mathbb{R})$ is positive
2. The zero element of $H^{1,1}(M, \mathbb{R})$ is not positive
3. If $x > 0$ and $y > 0$ then $x + y > 0$ $(x, y \in H^{1,1}(M, \mathbb{R}))$
4. If $x > 0$ and $r > 0$ then $rx > 0$ $(x \in H^{1,1}(M, \mathbb{R}),\ r \in \mathbb{R})$
5. if $x, y \in H^{1,1}(M, \mathbb{R})$ and $x > 0$, then there exists a positive integer g (depending on x, y) such that $gx - y > 0$.

***Proof*:**
exercise.

Definition 3.2.6: Integral Positivity

Let M be a Kähler manifold and $x \in H^{1,1}(M, \mathbb{Z})$. Then x is said to be *positive* if it is positive when regarded as an element of $H^{1,1}(M, \mathbb{R})$.

Definition 3.2.7: Hodge Manifold

A Kähler manifold M is called a *Hodge Manifold* if $H^{1,1}(M, \mathbb{Z})$ contains at least one positive element.

Another way of saying this is that M admits a Kähler metric whose fundamental class is in the image of the natural homomorphism $H^2(M, \mathbb{Z}) \to H^2(M, \mathbb{R})$.

Example 3.2.8: Complex Space is Hodge Manifold

let $M = \mathbb{C}\mathrm{P}^n$. Then

$$H^{1,1}(\mathbb{C}\mathrm{P}^n, \mathbb{Z}) = H^2(\mathbb{C}\mathrm{P}^n, \mathbb{Z}) \cong \mathbb{Z}$$

and so if $x \in H^{1,1}(\mathbb{C}\mathrm{P}^n, \mathbb{R}) = H^2(\mathbb{C}\mathrm{P}^n, \mathbb{R})$, there is a real number $r > 0$ such that rx lies in the image of the aforementioned inclusion homomorphism. In particular, the positive elements of $H^2(\mathbb{C}\mathrm{P}^n, \mathbb{Z})$ are the integral multiples of the cohomology classes of oriented planes.

Theorem 3.2.9: Hodge Manifold Are Algebraic

Let M be a Kähler manifold. Then M embeds into $\mathbb{C}\mathrm{P}^n$ for some n if and only if it is a Hodge Manifold.

***Proof*:**
see [25].

Similarly, of L is a line bundle over M, then it is *projectively induced* if for some embedding of M into $\mathbb{C}\mathrm{P}^n$, L is the restriction of M to a line bundle H with cohomology class h_n (h_n representing the cohomology class of characterized by a hyperplane). Mirroring the result in divisor theory, a projectively induced line bundle can always be given by a divisor:

Theorem 3.2.10: Bertini Theorem

Let F be a projectively induced line bundle over the projective variety M. Then a divisor D induces F, namely $F = \mathcal{L}(D)$.

This is actually a special case of the general Bertini theorem that you find when googling "Bertini Theorem".

***Proof*:**
[25]

Next, just like in sheaf cohomology over schemes, higher-order cohomology shall vanish (given the vectr bundle is sufficiently "twisted" to be very ample):

Theorem 3.2.11: Kodaira Higher Order Vanishing Theorem

Let F be a complex line bundle over a Kähler manifold M, and let F' be a positive line bundle over M. Then

$$H^i(M, F \otimes (F')^k)$$

vanish for all $i > 0$ and k sufficiently large.

***Proof*:**
[25]

Theorem 3.2.12: Characterizing Line Bundles On projective variety

Let M be a Hodge Manifold, L an analytic line bundle over M. Then there are projectively induced line bundles A, B where $L = A \otimes B^{-1}$. Thus, we have:

$$F = \mathcal{L}(A) \otimes \mathcal{L}(B)^{-1}$$

***Proof*:**
see [16, p.141, thm 18.2.5].

Finally, the following results show when pullbacks can be considered algebraic:

Theorem 3.2.13: Line Bundles Are Algebraic Over Projective Space

let L be a $\mathbb{C}$-fiber bundle over M with $\mathbb{C}\mathrm{P}^r$ as fibers and $\mathrm{PGL}_{r+1}(\mathbb{C})$ as the structure group. Then it is an projective variety.

***Proof*:**
[26]

Theorem 3.2.14: Borel Vanishing Theorem

Let L be a complex fibre bundle over the projective variety M with a projective variety as fibre and a connected structure group. Assume that the first Betti number of F is zero. Then L is itself algebraic.

***Proof*:**
[16, p.141]

As Hodge manifold and projective variety are interchangeable, this paper shall freely use any analytic properties given by its correspondence to Hodge manifolds.

3.3 Euler Characteristic and Generalization

In the final proof, we shall work with smooth projective varieties and locally free coherent sheaves, but it will do not harm at the start of this section to let M be a smooth compact manifold and E a complex vector bundle; we shall re-introduce more assumptions as needed. As we have a double complex, we can define more than one Euler characteristic. First, recall there is the natural:

$$\chi(M,E) = \sum_{i=0}^{\infty} \dim H^i(M,E) = \sum_{i=0}^{n} \dim H^i(M,E)$$

If $E = \mathbb{1}$, we shall denote it as χ_{Top} and call it the *Topological Euler characteristic*. By theorem 3.1.8 and theorem 3.1.9 there is a natural grading so that we can define:

$$\chi^p(M,E) = \chi\left(V, E \otimes \bigwedge^p TM\right) = \sum_{q=0}^{n} (-1)^q h^{p,q}(M,E)$$

Immediately we get:

$$\chi^0(M,E) = \chi(M,E) \qquad \text{and} \qquad \chi^p(M,E) = 0 \text{ when } p < 0 \text{ or } p > n$$

and if $E = \mathbb{1}$ then:

$$\chi^p(M, \mathbb{1}) = \chi^p(M) = \sum_{q=0}^{n} (-1)^q h^{p,q}(M)$$

We can collect these characteristic in a generating function:

Definition 3.3.1: χ_y-characteristic

$$\chi_y(M, E) = \sum_{p=0}^{n} \chi^p(M, E) y^p \qquad \text{(sim.)} \qquad \chi_y(M) = \sum_{p=0}^{n} \chi^p(M) y^p$$

The term $\chi_y(M, E)$ is the χ_y-characteristic of E and $\chi_y(M)$ is called the χ_y-genus of M

By construction:

$$\chi_0(M, E) = \chi^0(M, E) = \chi(M, E) \qquad \text{(sim.)} \qquad \chi_0(M) = \chi^0(M) = \chi(M)$$

Inspired by Hartshorne [34], call:

$$\chi(M) = \sum_{q=0}^{n} (-1)^q h^{0,q}(M)$$

the *arithmetic genus* of M. By Serre Duality:

Proposition 3.3.2: χ_p-characteristic and Serre Duality

$$\chi^p(M, E) = (-1)^n \chi^{n-p}(M, E^*)$$
$$\chi(M, E) = (-1)^n \chi(M, K \otimes E^*)$$

***Proof*:**
apply Serre duality.

Let us now assume M is Kähler. Let $g_q = h^{q,0}$. Then by the symmetry, the arithmetic genus $\chi(M)$ of a compact Kähler manifold is equal to:

$$\chi(M) = \sum_{i=0}^{n} (-1)^i g_i$$

Proposition 3.3.3: Euler Characteristic is Multiplicative

Let M, N be Kähler. Then:

$$\chi_y(M \times N) = \chi_y(M) \cdot \chi_y(N),$$

***Proof*:**
If M and N are Kähler manifolds then by Künneths formula:

$$h^{p,q}(M \times N) = \sum_{\substack{r+u=p \\ s+v=q}} h^{r,s}(M)h^{u,v}(N).$$

Let y, z be indeterminates and associate to each Kähler manifold M the polynomial $\prod_{y,z}(M) = \sum_{p,q} h^{p,q} y^p z^q$. Then the above is equivalent to:

$$\prod_{y,z}(M \times N) = \prod_{y,z}(M) \cdot \prod_{y,z}(N).$$

Let $z = -1$ in. Then $\prod_{y,-1}(-) = \chi_y(-)$ an so:

Next, let us understand $\chi_y(M)$ for two specific values of y. Let us start with $y = -1$. Then:

$$\chi_{-1}(M) = \sum_{p=0}^{n}(-1)^p \chi^p(M) = \sum_{p,q}(-1)^{p+q} h^{p,q}(M) = \sum_{i}^{n}(-1)^i \dim H^i(M) = \chi_{\mathrm{Top}}(M)$$

is equal to the usual Euler characteristic.

The most important value to prove will be for when $y = 1$. In this case, the χ_1-characteristic will be equal to a value called the *(spin) index* of M. When defining a generating function T_y that will hold the Chern character and Todd class information, $y = -1$ shall represent $\mathrm{ch}(E) \smile \mathrm{td}(M)$, and when $y = 1$ it shall *also* be equal to the index of M, and the proof of the Hirzebruch Riemann Roch boils down to showing that this forces the coefficients of T_y and χ_y to be the equal and hence they must also agree when $y = -1$. The index results shall be proven in chapter 5 and the specific theorem for the character when $y = 1$ is proven in theorem 6.2.1.

Let us finish with the special case M a Kähler manifolds and $h^{p,q} = 0$ when $p \neq q$. Then $\chi_y(M)$ is (essentially) equal to the Poincaré polynomial:

$$P(t; M) = \sum b_r t^r$$

where b_r is the rth Betti number of M. Indeed, we get:

$$\chi^p(M) = \sum_q (-1)^1 h^{p,q} = (-1)^p h^{p,p} = (-1) b_{2p}$$

therefore:

$$\chi_{-t^2}(M) = P(t; V) = \sum_r b_r t^r$$

3.3.4 Why do we care about this special case? Because complex projectives varieties $\mathbb{C}\mathrm{P}^n$ and flag manifolds $F(n)$ both are Kähler manifolds with this property. The case for $\mathbb{C}\mathrm{P}^n$ comes from elementary algebraic topology. For the flag manifold $F(n)$, recall that $H^*(F(n), \mathbb{Z})$ by the splitting principle is generated by element $\gamma_i \in H^2(F(n); \mathbb{Z})$. Then by characteristic theory there are complex $\mathbb{C}^*$-bundles E_i over $F(n)$ where $c_1(E_i) = \gamma_i$.

The result in the above remark is an iff. To prove the "only if" of the above remark, we need theorem 3.3.5 where we have that each γ_i is of type $(1,1)$, and therefore the cohomology classes of $F(n)$ are of type (p,p). Let us build-up to prove this theorem. Take the usual exact sequence:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_M \to \mathcal{O}_M^* \to 0$$

where $\underline{\mathbb{Z}}$ is the constant sheaf, $\mathcal{O}_M$ is the sheaf of holomorphic functions, and $\mathcal{O}_M^*$ is the sheaf of never-zero holomorphic functions (invertible holomorphic functions). Then we get an exact cohomology sequence:

$$H^1(M, \mathcal{O}_M^*) \xrightarrow{\delta_*^1} H^2(M, \mathbb{Z}) \to H^2(M, \mathcal{O}_M^*)$$

Then on a Kähler manifold we have $H^2(M, \mathcal{O}_M^*) = H^2(M, \mathbb{1}) \cong B^{0,2}(M)$. Thus, an element $a \in H^2(M, \mathbb{Z})$ is mapped to a nonzero element of $H^2(M, \mathcal{O}_M)$ if and only if a is of type $(1,1)$. Then from characteristic theory for vector bundles we know that if $\xi \in H^1(M, \mathcal{O}_M^*)$ associated to a complex $\mathbb{C}^*$-bundle, then:

$$\delta_*^1(\xi) = c_1(\xi)$$

If L is a complex line bundle over M and ξ is the associated $\mathbb{C}^*$-bundle, then $c_1(\xi)$ is the *cohomology class* of L. Then we get:

Theorem 3.3.5: Characterizing Line bundles on Kähler Manifolds

Let M be a compact Kähler manifold. Then an element $a \in H^2(M, \mathbb{Z})$ is the cohomology class of a complex line bundle over M if and only if it is of type $(1,1)$.

***Proof*:**
Hirzebruch references [16, p.127, Thm 15.9.1]. The reader can also prove the more general correspondenc between cohomology classes of H^2 and line bundles (see [12] for the details) and then apply the harmonic theory developed in this section.

3.3.1 Euler Characteristic on Intersections

The Hirzebruch Riemann Roch will reduce to being proven on line bundles. As line bundles on projective varieties are associated to codimension 1 subvarieties/submanifolds, we can take a filtering of M of codimension 1 subvarieties to represent the line bundles. This will lead to decomposing χ in terms of the characteristic of all possible filterings. To that end, we shall develop some more properties of χ_y. Let:

$$0 \to E' \to E \to E'' \to 0$$

be a short exact sequence of complex vector bundles. This naturally induces a short exact sequence of sheaves [3]:

$$0 \to \mathcal{O}_M(E) \to \mathcal{O}_M(E) \to \mathcal{O}_M(E'') \to 0$$

Thus, without loss of generality we may use common results from sheaf theory, in particular the results from Hartshorne [34] shall be quoted freely (for my own notes with proofs see [3]). What is important is that we get that we get that each new Euler-characteristic is additive on exact sequences:

Theorem 3.3.6: Generalized Euler Characteristic Additive

Let
$$0 \to \mathcal{O}_M(E) \to \mathcal{O}_M(E) \to \mathcal{O}_M(E'') \to 0$$
be a short exact sequence of complex vector bundles over M. Then:
$$\chi(M,E) = \chi(M,E') + \chi(M,E'')$$
$$\chi^p(M,E) = \chi^p(M,E') + \chi^p(M,E'')$$
$$\chi_y(M,E) = \chi_y(M,E') + \chi_y(M,E'')$$

***Proof*:**
Proved in [3], also an exercise in Hartshorne, see [34, section 2.5])

Theorem 3.3.7: Splitting Principle and Euler Characteristic

Let E be a complex vector bundle over a compact complex manifold M, E splits into $A_1, A_2, ..., A_q$. Let E' be another complex vector bundle over M. Then:
$$\chi(M, E' \otimes E) = \sum_i^q \chi(M, E' \otimes A_i)$$

***Proof*:**
Use induction and the above theorem

Thus, χ on vector bundles split additivity, similarly to how the Chern character $\mathrm{ch}(-)$ behaves.

As Weil divisors are in correspondence to line bundles on projective varieties (for algebraic line bundle see [34], [3], and recall theorem 3.2.12 for the right generalization to analytic line bundles), and Weil divisor are codimension 1 sub-(varieties/manifolds), they shall be the ideal object for the aforementioned filtering. From now on, all divisor are smooth/regular Weil divisors. As usual let E be a complex vector bundle over a complex manifold M, and let D be a divisor of M. Then it is associate to a line bundle $\mathcal{O}_M(D) = \mathcal{L}(D)$. We can now tensor $E \otimes \mathcal{L}(D)$, and consider $(E \otimes \mathcal{L}(D))_D$ to be the restriction to D of the vector bundle $E \otimes \mathcal{L}(D)$. We can take the sheaf $\mathcal{O}_M((E \otimes \mathcal{L}(D))_D)$ over D. By Extending by zero, this is a sheaf on M (in Hartshorne, this is denoted by the pushforward on the inclusion map $j_*(D)$ [34], where j_* can also be interpreted as the Gysin map as described by Fulton [20][3]). By the corresponding of vector bundles and locally free sheaves, we may without loss of generality consider $\mathcal{O}_M(E)|_D$ a sheaf or a vector bundle.

Theorem 3.3.8: Extending via Divisors

let M be a complex manifold , D a Cartier divisor of M, E a complex vector bundle over M. Then there is a short exact sequence:
$$0 \to \mathcal{O}_M(E) \to \mathcal{O}_M(E \otimes \mathcal{L}(D)) \to \mathcal{O}_M(E)|_D \to 0$$

[3]This will be important for section 6.3

***Proof*:**
This is a standard proof, this can be found in pure-sheaf form in Hartshorne [34, chapter 2] or my notes [3, chapter 5].

When it is not confusing, we shall denote E_D to be the complex vector bundle over D. We shall also write:

$$\chi(D,E) := \chi(D,E_D)$$
$$\chi^p(D,E) := \chi^p(D,E_D)$$
$$\chi_y(D,E) := \chi_y(D,E_D)$$

Theorem 3.3.9: Splitting Euler Characteristic Using Divisor

Let M be a compact complex manifold, D a divisor, E a complex vector bundle over M. Then:

$$\chi(M,E) = \chi(M, E \otimes \mathcal{L}(D)^{-1}) + \chi(D,E)$$

In particular, if $E = \mathbb{1}$, then:

$$\chi(M) = \chi(M, \mathcal{L}(D)^{-1}) + \chi(D)$$

***Proof*:**
combine above results

We want to extend this inductively so that χ_y can be written in terms of χ^p. First, recall from Riemann Geometry that we get the following short exact sequence:

$$0 \to T(D) \to T(M)|_D \to N_{M/D} \to 0$$

where $N_{M/D}$ is the normal bundle of D with respect to M. It is not hard to see it's fibers are of dimension 1, and a little more work shows it is isomorphic to $\mathcal{L}(D)_D$, and so:

$$0 \to T(D) \to T(M)|_D \to \mathcal{L}(D)_D \to 0$$

We then get a natural exact sequence:

$$0 \to \bigwedge^p (T(D)) \to \bigwedge^p (T(M)|_D) \to \bigwedge^{p-1} (T(D)) \otimes \mathcal{L}(D)_D \to 0$$

Dualizing we get:

$$0 \to \bigwedge^{p-1} (T^*(D)) \otimes \mathcal{L}(D)_D^{-1} \to \bigwedge^p (T^*(M)|_D) \to \bigwedge^p (T^*(D)) \to 0$$

Now if E is a complex vector bundle over M, we can tensor the sequence. Then by the above theorems:

$$\chi\left(D, E \otimes \bigwedge (T^*(M))\right) = \chi^{p-1}(D, E \otimes \mathcal{L}(D)^{-1}) + \chi^p(D,E)$$

Replacing E with $E \otimes \bigwedge^p (T^*(M))$, manipulating we get:

Definition 3.3.10: Four-term Formula

$$\chi^p(M,E) = \chi^p(M, E\otimes \mathcal{L}(D)^{-1}) + \chi^p(D,E) + \chi^{p-1}(D, E\otimes \mathcal{L}(D)^{-1})$$

This is true for $p > 0$, and for $p = 0$ interpret the last term on the right hand side as being 0. Furthermore, $\chi^p(D,E) = 0$ for $p = n = \dim M$ and if $p > n$ then all four terms are 0. Letting y be a indeterminate, the above equation holds with each term multiplied by y^p. Summing over $p \geq 0$ gives:

Definition 3.3.11: Four-term Formula – Generating Function

$$\chi_y(M,E) = \chi_y(M, E\otimes \mathcal{L}(D)^{-1}) + \chi_y(D,E) + y\chi_y(D, E\otimes \mathcal{L}(D)^{-1})$$

Now, by repeating the application of equation in definition 3.3.10, we can write $\chi^p(D,E)$ as a linear combination of integers of the form $\chi^q(M,A)$ where A is will be an appropriate complex vector bundle over M. For example:

$$\chi^0(D,E) = \chi^0(M,E) - \chi^0(M, E\otimes \mathcal{L}(D)^{-1}) \tag{3.1}$$

To calculate the last term, we use Serre duality and use:

$$\chi_1(M,E) = \chi_1(M, E\otimes \mathcal{L}(D)^{-1}) + \chi_1(D,E) + \chi_1(D, E\otimes \mathcal{L}(D)^{-1})$$

and replace E with $E\otimes \mathcal{L}(D)^{-1}$ in equation (3.1) to get:

$$\chi^1(D,E) = \chi^1(M,E) - \chi^1(M, E\otimes \mathcal{L}(D)^{-1}) - \chi^0(M, E\otimes \mathcal{L}(D)^{-1}) + \chi^0(M, E\otimes \mathcal{L}(D)^{-2})$$

Recursively continuing this, we get:

$$\chi^p(D,E) = \sum_{i=0}^{p} (-1)^i [\chi^{p-i}(M, E\otimes \mathcal{L}(D)^{-i}) - \chi^{p-i}(M, E\otimes \mathcal{L}(D)^{-(i+1)})]$$

This gives an important equation. These relation's do not necessarily hold for a general complex compact manifold, in particular it is not guaranteed the right hand side cancels out, while the left hand side will always be zero for $p > n$. If M is Kähler, or can be embedded into projective space, the relations hold, as shall be in the next section.

Combining these together in a generating function, we get:

$$\chi_y(D,E) = \sum_{i=0}^{\infty} (-y)^i [\chi_y(M, E\otimes \mathcal{L}(D)^{-i}) - \chi_y(M, E\otimes \mathcal{L}(D)^{-(i+1)})] \tag{3.2}$$

3.4 Virtual χ_y-characteristic

In equation (3.2), the right hand side is a complicated equation. The goal of this section is to show that the formula can be represented in terms of manifolds given by line bundles. By the splitting principle and the correspondence between divisors and line bundles, this goal is well-defined.

The idea behind the following section goes as follows. Let's say $V \hookrightarrow M$ is $\mathbb{C}$-codimension 1. Then what is the relation between $\chi(M, E)$ and $\chi(V, E)$? What if we have a sequence $V_r \hookrightarrow V_{r-1} \hookrightarrow \cdots \hookrightarrow V_1 \hookrightarrow M$ codimension 1 at each inclusion. Can we relate $\chi(M, E)$ to V_r in terms of the information given by $V_1, ..., V_r$? Ideally, we can define some:

$$\chi((V_1, ..., V_r), E) = \cdots$$

which will give us the Euler characteristic of V_r given the information of the intersection. Converesely, it would be good if there exists functions F_i such that:

$$\chi(M, E) = \sum_i F_i(V_1, .., V_r, E)$$

which allows us to link the property of $\chi(M, E)$ to simpler sub-manifolds. We shall do exactly this in this section.

The greatest advantage of this is that as each of these manifolds are codimension one with respect to the next one, so we can associate to them line bundles, and the line bundles are in bijective correspondence with cohomology classes (as M is projective). Thus, given a set of line bundles $\{L_1, ..., L_r\}$, we get a filtering $V_1 \subseteq V_2 \subseteq \cdots \subseteq V_r \subseteq M$ allowing to reduce the eventual proof of Hirzebruch Riemann Roch to a proof on line bundles; we can also think of each V_r as the intersection of M with some hyper-plane (use theorem 3.2.10 and some divisor theory). As it will turn out, the result is independent of order, and hence it is well-defined to take the set $\{L_1, ..., L_r\}$ instead of the tuple $(L_1, ..., L_r)$.

By using the splitting principle, given TM of M splits into $A_1, ..., A_n$, We aim to find an equation for $\chi(M, TM)$ in terms of the line bundles.

A quick word must be given in terms of the reformulation of this in the Grothendieck Riemann Roch. The inclusion $j : V \to M$ will be used to pullback a cohomology class $j^* : H^*(M) \to H^*(V)$ giving a contra-variant theory. However, using Fulton's intersection theory we instead would take $j_* : A_*(V) \to A_*(M)$; this is the *Gysin map*. As the Grothendieck Riemann Roch is a purely algebraic proof, Gysin map will be the "correct" generalization to scheme theory. Note that the Grothendieck Riemann Roch will note require to keep track of the set of line bundles $\{A_1, ..., A_n\}$ as the proof technique hinges on a very different approach (see section 6.3).

3.4.1 Arithmetic

Let us represent the arithmetic of the intersection. Let M be a smooth manifold, $\{L_1, ..., L_r\}$ be associated line bundles. Define:

$$R = \mathbb{Z}[e, \ell_1, \ell_2, ..., \ell_r, \ell_1^{-1}, \ell_1^{-2}, ..., \ell_r^{-1}]$$

where each ℓ_i is a free variables that should be thought of as corresponding to a line bundle L_i of M. Define a homomorphism $R \to \mathbb{Z}[[y]]$ by defining on the generators:

$$h^M(e^m \cdot \ell_1^{m_1} \cdots \ell_r^{m_r}) = \chi(M, E^m \otimes L_1^{m_1} \otimes \cdots \otimes L_r^{m_r})$$
$$h_y^M(e^m \cdot \ell_1^{m_1} \cdots \ell_r^{m_r}) = \chi_y(M, E^m \otimes L_1^{m_1} \otimes \cdots \otimes L_r^{m_r})$$

where the powers should be taken as taken the tensor product. Then by some elementary algebra this extends to a $\mathbb{Z}[[y]]$-module $R[[y]]$ with homomorphism extending to $R[[y]]$; label it h as well. By definition:

$$h^M(1) = \chi(M) \qquad \text{and} \qquad h_y^M(1) = \chi_y(M)$$

By using the results of the previous section, given a divisor D, we can define a $\mathbb{Z}[[y]]$-homomorphism

$$h_y^D(e^m \cdot \ell_1^{m_1} \cdots \ell_r^{m_r}) = \chi_y(D, E^m \otimes L_1^{m_1} \otimes \cdots \otimes L_r^{m_r})$$

where like before the line bundles have a natural restriction to $\mathcal{L}(D)$. By equation (3.2):

$$\chi_y(D,E) = h_y^D\left(e\frac{1-d^{-1}}{1+yd^{-1}}\right)$$

and by theorem 3.3.9:

$$\chi(D) = h^D(1-d^{-1}) \qquad \text{and} \qquad \chi(D,E) = h^D(e(1-d^{-1}))$$

By taking successive divisors of divisor, we multiply by further factors of $\frac{1-d^{-1}}{1+yd^{-1}}$, motivating the following definition:

Definition 3.4.1: Virtual Characteristic

$$\chi_y(\{L_1,...,L_r\},E)_M := h_y\left(e\prod_{i=1}^r \frac{1-\ell_i^{-1}}{1+y\ell_i^{-1}}\right)$$

If $E = \mathbb{1}$, it shall simply be denoted

$$\chi_y(\{L_1,...,L_r\})_M := \chi_y(\{L_1,...,L_r\},\mathbb{1})_M$$

By the construction of the virtual character, it is clear it is not dependent on the order of the filtering but on the choice of line bundles associated to them, and that it respects the same property as the usual χ_y character:

$$\chi_y(\{L_1,...,L_r\},E)_M = \sum_{p=0}^{\infty} \chi^p(\{L_1,...,L_r\},E)_M y^p$$

By taking $\chi_M = \chi_M^0$ we can write:

$$\chi(\{L_1,...,L_r\},E)_M = h\left(e\prod_{i=1}^r(1-\ell_i^{-1})\right)$$

An important concrete example would then be:

Proposition 3.4.2: Euler Characteristic – Split Man and Bundle

$$\chi(\{L\})_M = \chi(M) - \chi(M, L^{-1})$$

Notice that we are subtracting; this motivates the term "virtual" as we are treating L^{-1} as an "inverse" vector bundles, a notion well-defined in K-theory.

Let us now find χ_y in terms of virtual characteristics. By definition of h, we immediately get:

$$\chi_y(D,E) = \chi_y(\{\mathcal{L}(D)\},E)_M \tag{3.3}$$

As $\chi_y(D,E)$ is a polynomial in $n-1$ terms, so is the right hand side. We shall extend this for larger when $N = D_1 \cap \cdots \cap D_r$ which shall give a $n-r$ degree polynomial when M is a nonsingular variety (theorem 3.4.8). First, the restriction of any of D for any set $\{L_1,...,L_r\}$ must be computed:

Theorem 3.4.3: Virtual Characteristic Divisor Restriction

Let $M, E, L_1, ..., L_r$ as usual, and D a divisor of M, $\mathcal{L}(D) = L_1$. Then:

$$\chi_y(\{L_1, ..., L_r\}, E)_M = \chi_y(\{(L_1)_D, ..., (L_r)_D\}, E_D)_D$$

***Proof*:**
Let:

$$R(x) = \frac{1 - x^{-1}}{1 + yx^{-1}}$$

Then by definition

$$\chi_M((L_2)_D, \dots, (L_r)_D|, E_D)_D = h^M\left(e\prod_{i=2}^{r} R(\ell_i)\right)$$

It follows from our work above that:

$$h^M(e^m\ell_1^{h_1}\dots\ell_r^{h_r}) = \hat{h}_M(e^m\mu\ell_1^{h_1}\dots\ell_r^{h_r}R(\ell_1))$$

Then by letting $t = R(\ell_1)$ we get:

$$h^M\left(e\prod_{i=2}^{r} R(\ell_i)\right) = h^M\left(e\prod_{i=1}^{r} R(\ell_i)\right) = \chi_M(L_1, \dots, L_r|, E)_M$$

Corollary 3.4.4: Virtual Characteristic and Trivial Bundle

Let $M, E, L_1, ..., L_r$ as usual, and assume one $L_i = \mathbb{1}$ is the trivial bundle. THen:

$$\chi_y(\{L_1, ..., L_r\}, E)_M = 0$$

***Proof*:**
immediate consequence

Theorem 3.4.5: Algebra of Product of Line Bundles

Let M be a compact complex manifold, E a complex vector bundle, $L_1, ..., L_r, A, B$ complex line bundles. Then:

$$\begin{aligned}\chi_y(\{L_1, \cdots, L_r, A\otimes B\}, E)_M = \\ &\chi_y(\{L_1, \cdots, L_r, A\}, E)_M \\ &+ \chi_y(\{L_1, \cdots, L_r, B\}, E)_M \\ &+ (y-1)\chi_y(\{L_1, \cdots, L_r, A, B\}, E)_M \\ &- y\chi_y(\{L_1, \cdots, L_r, A, B, A\otimes B\}, E)_M\end{aligned}$$

***Proof*:**
Let M be a compact complex manifold, E a complex analytic vector bundle over M, and L_i, A, B complex analytic line bundles over M. Then the following functional equation holds for the virtual χ_y-characteristic:

$$\chi_y(\{L, A \otimes B\}, E)_M =$$
$$\chi_y(\{L, A\}, E)_M + \chi_y(\{L, B\}, E)_M + (y-1)\chi_y(\{L, A, B\}, E)_M - y\chi_y(\{L, A, B, A \otimes B\}, E)_M$$

By definition:

$$\chi_y(L_1, \dots, L_r|_M, E) = h_y^M(e \cdot \prod_{i=1}^{r} R(\ell_i))$$

where the function $R(x)$ is given by:

$$R(x) = \frac{x-1}{x+y}$$

Starting with the case of a single line bundle L. Let the term u correspond to the common parts of the bundles in the equation:

$$u = e \cdot R(\ell)$$

Now, we can express each term in the theorem's main equation using this notation:

1. $\chi_y(F, A \otimes B|_M, E) = h_y^M(u \cdot R(ab))$
2. $\chi_y(F, A|_M, E) = h_y^M(u \cdot R(a))$
3. $\chi_y(F, B|_M, E) = h_y^M(u \cdot R(b))$
4. $\chi_y(F, A, B|_M, E) = h_y^M(u \cdot R(a)R(b))$
5. $\chi_y(F, A, B, A \otimes B|_M, E) = h_y^M(u \cdot R(a)R(b)R(ab))$

The theorem states:

$$h_y^M(u \cdot R(ab)) = h_y^M(u \cdot R(a)) + h_y^M(u \cdot R(b)) + (y-1)h_y^M(u \cdot R(a)R(b)) - y \cdot h_y^M(u \cdot R(a)R(b)R(ab))$$

Since h_y^M is a homomorphism, it is linear with respect to polynomials in y. We can thus factor out $h_y^M(u \cdot \dots)$ from the right-hand side:

$$h_y^M(u \cdot R(ab)) = h_y^M\left(u \cdot [R(a) + R(b) + (y-1)R(a)R(b) - yR(a)R(b)R(ab)]\right)$$

Therefore, the theorem is proven if the following algebraic identity holds:

$$R(ab) = R(a) + R(b) + (y-1)R(a)R(b) - yR(a)R(b)R(ab)$$

and indeed:

$$\begin{aligned}
&R(a)+R(b)+(y-1)R(a)R(b)-yR(a)R(b)R(ab)\\
&=\frac{a-1}{a+y}+\frac{b-1}{b+y}+(y-1)\frac{(a-1)(b-1)}{(a+y)(b+y)}-y\frac{(a-1)(b-1)}{(a+y)(b+y)}\frac{ab-1}{ab+y}\\
&=\frac{(a-1)(b+y)+(b-1)(a+y)+(y-1)(a-1)(b-1)}{(a+y)(b+y)}-y\frac{(a-1)(b-1)(ab-1)}{(a+y)(b+y)(ab+y)}\\
&=\frac{ab+ay-b-y+ab+by-a-y+(y-1)(ab-a-b+1)}{(a+y)(b+y)}-\dots\\
&=\frac{2ab+a(y-1)+b(y-1)-2y+(y-1)(ab-a-b+1)}{(a+y)(b+y)}-\dots\\
&=\frac{2ab+ay-a+by-b-2y+yab-ya-yb+y-ab+a+b-1}{(a+y)(b+y)}-\dots\\
&=\vdots\\
&=\frac{ab-1}{(a+y)(b+y)(ab+y)}\left[(a+y)(b+y)\right]\\
&=\frac{ab-1}{ab+y}\\
&=R(ab)
\end{aligned}$$

Then, by the product formula in the definition of the virtual characteristic, notice we get the equation (letting $\mathcal{L}=L_1,...,L_r$):

$$\chi(\{\mathcal{L},A\otimes B\},E)_M=\chi(\{\mathcal{L},A\},E)_M+\chi(\{\mathcal{L},B\},E)_M-\chi(\{\mathcal{L},A,B\},E)_M$$

as we sought to show.

Let us now say that every vector bundle splits on M. Then TM splits into line bundles $A_1\oplus\cdots\oplus A_m$. Then $\bigwedge^p TM$ splits into $\binom{m}{p}$ line bundles of the form:

$$A_{i_1}^{-1}\otimes\cdots\otimes A_{i_p}^{-1}\qquad (i_1<\cdots<i_p)$$

By theorem 3.3.7:

$$\begin{aligned}
\chi^p(M,E)&=\chi(M,E\otimes\bigwedge^p TM)\\
&=\sum_{i_1<\cdots<i_p}\chi(M,E\otimes A_{i_1}^{-1}\otimes\cdots\otimes A_{i_p}^{-1})
\end{aligned}$$

By the definition of

$$\chi_y(M,E)=\sum h_y^M\left(e\prod_{i=1}^m(1+ya_i^{-1})\right)$$

This shall now give us a decomposition of χ in terms of the virtual character

Theorem 3.4.6: Virtual Character Decomposition

Let M be a complex manifolds where TM splits completely into linear terms $A_1, ..., A_m$. Let E be a complex vector bundle. Then:

$$(1+y)^m \chi(M,E) = \sum_{l=0}^{m} y^l \sum_{i_1<\cdots<i_l} \chi_y(\{A_{i_1}, ..., A_{i_l}\}, E)_M$$

***Proof*:**
Writing in term h_y^M:

$$\begin{aligned}
\sum_{l=0}^{m} y^l \sum_{i_1<i_2<\cdots<i_l} h_y^M(eR(a_{i_1})\dots R(a_{i_l})) &= h_Y^m\left(\sum_{l=0}^{m} y^l \sum_{i_1<i_2<\cdots<i_l} eR(a_{i_1})\dots R(a_{i_l})\right)\\
&= h_y^M\left(e\prod_{i=1}^{m}(1+yR(a_i))\right)\\
&= h_y^M\left(e\prod_{i=1}^{m}(1+y)(1+ya_i^{-1})^{-1}\right)\\
&= (1+y)^m h_y^M\left(e\prod_{i=1}^{m}(1+ya_i^{-1})^{-1}\right)
\end{aligned}$$

Thus, we get

$$h_y^M(e^m a_1^{h_1} a_2^{h_2}\dots a_r^{h_r}) = h^M\left(e^m a_1^{h_1} a_2^{h_2}\dots a_r^{h_r}\prod_{i=1}^{m}(1+ya_i^{-1})\right)$$

Then, letting:

$$t = \prod_{i=1}^{m}(1+ya_i^{-1})$$

gives:

$$\begin{aligned}
(1+y)^m h_Y^m\left(w\prod_{i=1}^{m}(1+ya_i^{-1})^{-1}\right) &= (1+y)^m h\left(w\prod_{i=1}^{m}(1+ya_i^{-1})^{-1}(1+ya_i^{-1})\right)\\
&= (1+y)^m h(w) = (1+y)^m\chi(V,W)
\end{aligned}$$

as we sought to show

3.4.2 Relating Euler Characteristic to Chern Numbers

Finally, let us show that the virtual characteristic with r linear bundles is an $n-r$ polynomial, and that evaluating when $n=r$ works the same as evaluating M at the cup-product of the respective $\ell_i \in H^2(m,\mathbb{Z})$.

Lemma 3.4.7: 0-dimensional Case

Le $M = \{p_1, ..., p_k\}$ be a 0-dimensional manifold, E an n-dimensional vector bundle over M, $L_1, ..., L_r$ line bundles over M. Then:

1. $\chi_y(M, E) = nk$
2. $\chi_y(\{L_1, ...L_r\}, E) = 0$ for $r \geq 1$

***Proof*:**

1. $\chi_y(M, E) = \chi(M, E) = \dim H^0(M, E) = nk$
2. By Bass-Serre theorem (see [12][a]), every line bundle over a finite set of points is trivial, so by corollary 3.4.4
$$\chi_y(\{L_1, ..., L_r\}, E) = 0$$

[a] A special case of Bass-Serre is proven in theorem 6.1.2 which is a geometric proof as compared to the algebraic proof in the cited result

In general, recall when M is projective that $\chi_y(M, E)$ is a polynomial as the power series must terminate by definition. Let us prove by induction on the dimension n on M that the virtual χ_y-characteristic is also a polynomial.

Theorem 3.4.8: Induction Case

Let M be a complete variety of complex dimension n. Let E be a q-dimensional complex vector bundle, and let $L_1, ..., L_r$ be analytic line bundles over M. Then:

1. For $r > 0$, $\chi_y(\{L_1, ..., L_r\}, E)_M = 0$ for $r > n$.
2. For $r \leq n$, $\chi_y(\{L_1, ..., L_r\}, E)_M$ is a polynomial of degree $\leq n - r$ in y with integer coefficients
3. If $r = n$, then
$$\chi_y(\{L_1, ..., L_n\}, E)_M = \chi(L_1, ..., L_n\})_M = \int_M q \smile l_1 \smile l_2 \smile \cdots \smile l_n$$
where $l_i \in H^2(M, \mathbb{Z})$ are the cohomology classes of L_i.

We shall often use $\cdot$ instead of $\smile$ notation to de-clutter notation.

***Proof*:**

1/2. Lemma lemma 3.4.7 prove it for $\dim M = 0$. Suppose it is true for $\dim M < n$. By theorem 3.2.11 there are divisors D and S of M such that $\mathcal{L}(D) = L_1 \otimes \mathcal{L}(S)$. Then by theorem 3.4.6:

$$\begin{aligned}
&\chi_M(\{\mathcal{L}(D), L_2, \cdots, L_r\}, E) \\
&\quad = \chi_M(\{L_1, \cdots, L_r\}, E) + \chi_M(\{\mathcal{L}(S), L_2, \cdots, L_r\}, E) \\
&\quad + (y-1)\chi_M(\{\mathcal{L}(S), L_1, \cdots, L_r\}, E) - y\chi_M(\{\mathcal{L}(D), \mathcal{L}(S), L_1, \cdots, L_r\}, E)
\end{aligned}$$

We must show the 2nd term has degree $\leq n-r$. Note that induction hypothesis, together with theorem 3.2.11 shows 1st, 3rd, 4th and 5th term all have degree $\leq n-r$ and vanish for $r > n$. Note that if $r = 1$ then 1st term is $\chi_M(D, E_D)$ and 3rd term is $\chi_M(S, E_S)$; these terms are polynomials of degree $\leq n-1$ by the definition of the (non-virtual) χ_M-characteristic. Therefore 2nd term is a polynomial of degree $\leq n-r$ and zero for $r > n$.

3. By theorem 3.2.11 there are divisors D and S of M such that $\mathcal{L}(D) = L_1 \otimes \mathcal{L}(D)$.

   Then for $n \geq 2$, (1) gives:

$$\chi(\{\mathcal{L}(D), L_2, \ldots, \ell_n\}, E) = \chi(\{L_1, L_2, \ldots, \ell_n\}, E) + \chi(\{\mathcal{L}(D), L_2, \ldots, \ell_n\}, E)$$

   and for $n = 1$

$$\chi(\{\mathcal{L}(D)\}, E) = \chi(\{L_1\}, E) + \chi(\{\mathcal{L}(D)\}, E) \tag{3.4}$$

   Then by lemma lemma 3.4.7 the $n = 1$ case gives

$$\chi(\{L_1\}, E) = qd - qs = q\ell_1[M_1]$$

   where d, s are the number of points of D, S. Therefore (2) is true for $\dim M = 1$. Now for the sake of induction supposed it is proved for $1 \leq \dim M < n$. We now apply the induction hypothesis to equation (3.4). First, recall that if $h \in H^2(M, \mathbb{Z})$ is the cohomology class representing the divisor D then $c_1([D]) = h$. If $j : D \hookrightarrow M$ is an inclusion (treating D as a codimension 1 submanifold), then for any $x \in H^{n-2}(M, \mathbb{Z})$:

$$\int_D j^*(x) = \int_M h \smile x = \int_M hx$$

   Then by theorem 3.4.3, we get:

$$\begin{aligned}
\chi(\{L_1, L_2, \ldots, \ell_n\}, E) &= \int_D q \cdot (\ell_2 \ldots \ell_n)_D - \int_S q \cdot (\ell_2 \ldots \ell_n)_S \\
&= \int_M q \cdot c_1(\mathcal{L}(D))\ell_2 \ldots \ell_n - \int_M q \cdot c_1(\mathcal{L}(S))\ell_2 \ldots \ell_n \\
&= \int_M q \cdot \ell_1 \ell_2 \ldots \ell_n
\end{aligned}$$

   as we sought to show.

When $n = 1$, this is in fact Riemann-Roch for curves! Let $M = C$ be a curve, L a line bundle with cohomology class $l \in H^2(C, \mathbb{Z})$. Then the χ-genus of L^{-1} is:

$$\chi(L^{-1}) = \int_C -l$$

Then: $\chi(L) = \chi(C) - \chi(C, L^{-1})$ and so we get:

$$\chi(C, L) = \chi(L) + \int_C l$$

Next by using theorem 3.2.3 we get:

$$\chi(C) = 1 - g$$

By classical algebraic topology, $\int_C l$ is called the *degree* of L, which (naturally) is equal to $\deg(L)$. Using Poincaré duality we get:

$$\chi(C, L) = \dim H^0(C, L) - \dim H^1(C, L) = \dim H^0(C, L) - \dim H^0(C, K \otimes L^{-1})$$

Combining it all together, we obtain:

$$\dim H^0(C, L) - \dim H^0(C, K \otimes L^{-1}) = 1 - g + \deg(L)$$

i.e., classical Riemann Roch! This result does not generalize as the correspondence between codimension 2 objects and elements of H^4 is not well-defined in the necessary way, which motivates inducting on line bundles.

Finally, an important result that will be used is that the virtual Euler characteristic characterizes these relations:

Theorem 3.4.9: Uniqueness Result

let $M, E, L_1, ..., L_r$ be a usual, $\mathcal{L} = \{L_1, ..., L_r\}$. Let G be a function associating each $(\mathcal{L}, E)_M$ to a $\mathbb{Q}$-power series in y and is independent of the order of L_i (i.e. these terms must all commute). Then if:

1. $G(M, E) = \chi_y(M, E)$
2. $$\begin{aligned} G(\{L_1, \cdots, L_r, A \otimes B\}, E)_M = \\ & G(\{L_1, \cdots, L_r, A\}, E)_M \\ & + G(\{L_1, \cdots, L_r, B\}, E)_M \\ & + (y-1)G(\{L_1, \cdots, L_r, A, B\}, E)_M \\ & - yG(\{L_1, \cdots, L_r, A, B, A \otimes B\}, E)_M \end{aligned}$$
3. If D is a (nonsingular) divisor of M and $L_1 = \mathcal{L}(D)$, then:

$$G(\mathcal{L}, E)_M = G(\{(L_1)_D, ..., (L_r)_D\}, E_D)_D$$

Then for all $(\mathcal{L}, E)_M$ with $r \geq 1$:

$$\chi_y(\mathcal{L}, E)_M = G(\mathcal{L}, E)_M$$

The same exact result holds when taking virtual characteristic.

***Proof*:**
We outline the result for the Euler-characteristic, leaving the virtual characteristic as an exercise. First, χ_M has properties (2) and (3) and therefore the function $\chi_M - G$ has properties (2) and (3). It is therefore sufficient to show that any function G' which satisfies (2), (3) and the additional property that:

$$G'(M, E) = 0$$

must be identically zero.

We proceed by induction. By theorem 3.2.11 there are divisors D and T of M such that $\mathcal{L}(D) = F_1 \otimes \mathcal{L}(T)$. Then (2) implies equation

$$\begin{aligned}
&\chi_M(\{\mathcal{L}(D), L_2, \cdots, L_r\}, E) \\
&\quad = \chi_M(\{L_1, \cdots, L_r\}, E) + \chi_M(\{\mathcal{L}(S), L_2, \cdots, L_r\}, E) \\
&\quad + (y-1)\chi_M(\{\mathcal{L}(S), L_1, \cdots, L_r\}, E) - y\chi_M(\{\mathcal{L}(D), \mathcal{L}(S), L_1, \cdots, L_r\}, E)
\end{aligned}$$

from theorem 3.4.8 with χ_M replaced by G'. This equation has five terms, and we have to prove that the second term is zero. The induction hypothesis and (3) imply that terms 1, 3, 4, 5 are zero. For $r = 1$, it is necessary to use $G' = 0$ to prove that terms 1, 3 are zero. Therefore term 2 is zero.

This shall be important when we show that the T-characteristic holding information about the Chern character and Todd class satisfies the same properties, and hence will have to be equal (comes back in theorem 6.2.4).

3.5 *Vanishing Theorems

In this section, we mention but not prove two important vanishing results that will let us sometimes reduce χ to H^0, therefore making it particularly simple to answer the Riemann Roch Problem:

Theorem 3.5.1: Kodaira Vanishing Theorem

Let L be a complex line bundle over a compact complex manifold M. If L^{-1} is positive, then the cohomology group $H^i(M, L)$ vanish for all $i \neq n$.

***Proof*:**
[26]

Using Serre duality, we get

Theorem 3.5.2: Kodaira Vanishing Theorem - Euler Characteristic

let L, M be as before and take $L \otimes K^{-1}$ for the canonical class K. If it is positive, then the groups $H^i(M, L)$ vanish for all $i > 0$, hence:

$$\dim h^0(M, L) = \chi(M, L)$$

***Proof*:**
[26]

4

Todd Class

In section 2.4, we defined the Chern character which gives:

$$\mathrm{ch}(E_1 \oplus E_2) = \mathrm{ch}(E_1) + \mathrm{ch}(E_2) \in H^*(M, \mathbb{Q})$$

In section 3.3, we saw that we can define the Euler characteristic for vector bundle and that:

$$\chi(M, E_1 \oplus E_2) = \chi(M, E_1) + \chi(M, E_2)$$

As the Euler class is given by the top Chern class (recall remark 1.2.1), this algebraic property gives the hope that we can directly generalize Chern's Theorem to:

$$\chi(M, E) = \int_M \mathrm{ch}(E)$$

However, this does not give the expected result:

Example 4.0.1: Incorrect Generalization

Let $E = \mathbb{1}$ be the trivial bundle. Then $\mathrm{ch}(\mathbb{1}) = 1$. Now consider $M = S^2$. Then:

$$2 = \chi(S^2) = \chi(S^2, \mathbb{1}) \neq \int_{S^2} \mathrm{ch}(S^2) = 0$$

showing we do not recover the Euler characteristic.

The problem is that we are not taking into considering the geometric information of M[1]. What would be desired is a "correction class" $C(TM)$ (C for correction) where TM is the tangent bundle where:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \smile C(TM)$$

[1] recall the Euler class is not defined directly as the Top Chern class on TM to take into account the geometric information of the given manifold, see [12]

This shall be the *Todd class* $\mathrm{td}(M)$ where $\mathrm{td}(M) = C(TM)$. What properties should $\mathrm{td}(-)$ require? For one, we must have:

$$\chi(M, \mathbb{1}) = \chi(M) = \int_M \mathrm{td}(M)$$

For another, as $\chi(M \oplus Y) = \chi(M) \cdot \chi(Y)$, $\mathrm{td}(M \oplus Y) = \mathrm{td}(M) \smile \mathrm{td}(Y)$, namely it is *multiplicative* on direct sums. Finally, we shall be able to reduce to proving the result on the complex projective spaces (see theorem 5.1.10), hence it is necessary to have $\mathrm{td}(\mathbb{C}\mathrm{P}^k)$ agree with $\chi(\mathbb{C}\mathrm{P}^k)$. In particular, as we know the line bundles over $\mathbb{C}\mathrm{P}^k$, we can deduce $\mathrm{td}(-)$ by making it agree with the values of $\chi(-)$.

Let us show the strategy in the one-dimensional case. First, we shall show that $\mathrm{td}(-)$ will correspond by a power series:

$$Q(x) = 1 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \cdots$$

which will return the appropriate polynomial of Chern classes given the dimension of $\mathbb{C}\mathrm{P}^n$ (exactly how is explain in section 4.1). The first term of this series gives the information for $\mathbb{C}\mathrm{P}^1$, hence let's find it by insuring $\mathrm{td}(\mathbb{C}\mathrm{P}^1) = \chi(\mathbb{C}\mathrm{P}^1)$. Recall that $T\mathbb{C}\mathrm{P}^1 = \mathcal{O}(2)$, the twisted line bundle, and $\chi(\mathbb{C}\mathrm{P}^1) = 1$. Thus we want: $\mathrm{td}(\mathbb{C}\mathrm{P}^1) = Q\left(c_1(\mathcal{O}(2))\right)$. It is easy to compute that

$$c_1(\mathcal{O}(2)) = h \in H^2(\mathbb{C}\mathrm{P}^1; \mathbb{Z}) \cong \mathbb{Z}[h]/(h^2)$$

as $h^2 = 0$, we get:

$$\begin{aligned}
\chi(\mathbb{C}\mathrm{P}^1) = 1 &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{td}(\mathbb{C}\mathrm{P}^1)) \\
&= \int_{\mathbb{C}\mathrm{P}^1} Q(c_1(T\mathbb{C}\mathrm{P}^1)) \\
&= \int_{\mathbb{C}\mathrm{P}^1} Q(2h) \\
&= \int_{\mathbb{C}\mathrm{P}^1} (1 + 2a_1 h) = 2a_1
\end{aligned}$$

Thus, it must be that $a_1 = \frac{1}{2}$, giving us:

$$Q(x) = 1 + \frac{1}{2}x + a_1 \frac{x^2}{2!} + \cdots$$

Notice that if we consider $\mathcal{O}(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d))$, then we get:

$$\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d)) = d + 1$$

and to compute

$$\mathrm{ch}(\mathcal{O}(d)) = e^{kh} = 1 + dh$$

as $h^2 = 0$. Thus:

$$\begin{aligned}\chi(\mathbb{C}\mathrm{P}^1, \mathcal{O}(d)) = d+1 &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile \mathrm{td}(\mathbb{C}\mathrm{P}^1)) \\ &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile Q(c_1(T\mathbb{C}\mathrm{P}^1)) \\ &= \int_{\mathbb{C}\mathrm{P}^1} \mathrm{ch}(\mathcal{O}(d)) \smile Q(2h) \\ &= \int_{\mathbb{C}\mathrm{P}^1} (1+2h) \smile (1+h) \\ &= \int_{\mathbb{C}\mathrm{P}^1} [1+(1+d)h] = d+1\end{aligned}$$

Giving equality in this special case. Repeating this process again for $\mathbb{C}\mathrm{P}^k$ for $k \in \mathbb{N}$, we shall get the next coefficients. In section 4.1, we go over the general algebraic method for finding such a power series, and in proposition 4.1.4 we shall show how we deduce the rest of this Todd power series. Overall, repeating this process will define $\mathrm{td}(-)$ and by construction it agrees on at least the Euler characteristic of $\mathbb{C}\mathrm{P}^n$ with the standard line bundles. This shall be enough in the proof of Hirzebruch Riemann Roch by taking advantage of the results on Cobordism and the splitting principle.

After defining the Todd class, it shall be combined with the Chern character to create the *T-characteristic* which shall be shown to behave well algebraically and has the desired properties for any vector bundle over a space $E \to M$.

4.1 Algebraic Build-up: Multiplicative sequences

The direct sum of bundles translates to a cup product of characteristic classes. This gives a natural algebraic relation that we know codify for future reference. Let B be a commutative ring, and define:

$$\mathcal{B} = B[z, p_1, p_2, ...]$$

Define a grading given by the following:

$$p_{j_1} p_{j_2} \cdots p_{j_r} \text{ has weight } j_1 + j_2 + \cdots + j_r$$

Then let $\mathcal{B}_k$ be the weight k-groups; note $\mathcal{B}_0 = B$ and $\mathcal{B}_r \mathcal{B}_s = \mathcal{B}_{r+s}$. Let $\pi(k)$ be all partitions of k. Then $\mathcal{B}_k$ is a B-module of rank $\pi(k)$.

Definition 4.1.1: Multiplicative Sequence

Let $\{K_j\}$ be a sequence of polynomials in indeterminates p_i where $K_0 = 1$ and $K_j \in \mathcal{B}_j$. Then the sequence $\{K_j\}$ is called a *multiplicative sequence* or *m-sequence* if whenever a power series is the product of two power series:

$$1 + p_1 z + p_2 z^2 + \cdots = (1 + p_1' z + p_2' z^2 + \cdots)(1 + p_1'' z + p_2'' z^2 + \cdots)$$

with indeterminates z, p_i', p_i'', then we get the identify:

$$\sum_{j=0}^{\infty} K_j(p_1, p_2, ...p_j) z^j = \left(\sum_{i=1}^{\infty} K_i(p_1', p_2', ...p_i') z^i \right) \left(\sum_{j=1}^{\infty} K_j(p_1'', p_2'', ...p_i'') z^j \right)$$

that is:

$$K\left(\sum_{i=0}^{\infty} q_i z^i \right) = \sum_{i=0}^{\infty} K_i(q_1, ..., q_i) z^i$$

where $q_i \in \mathcal{B}$ of weight i. In other words a sequence $\{K_j\}$ is called multiplicative if it induces an endomorphism on

$$B[p_1, p_2, ...]\,[[z]]$$

Given a multiplicative sequence $\{K_j\}$, we associate to it its *characteristic power series* as the image of $K(1+z)$:

$$K(1+z) = \sum_{j}^{\infty} K_j(1, 0, ..., 0) z^j = \sum_{j}^{\infty} b_j z^j$$

An important property of vector bundles is that they can be pulled back over a flag manifold so that they split into line bundles. This strategy shall be important to us, and hence we will be considering:

$$1 + p_1 z + \cdots + p_n z^n = \prod_{i}^{n} (1 + \beta_i z)$$

where β_i is a symmetric polynomial in the p_i's. By some elementary ring theory and theory of symmetric functions, we can interpret $\mathcal{B}$ as the ring of all symmetric functions in $\beta_1, ..., \beta_n$.

We can characterize m-sequences via 1-unit power series:

Proposition 4.1.2: Correspondence 1-unit Power Series and m-sequences

There is a bijective correspondence between m-sequences and formal power series with constant term 1.

***Proof*:**
First, let's show any m-sequence $\{K_j\}$ is completely determined by its characteristic power series

$$Q(z) = K(1+z)$$

Proof. By decomposing:

$$1 + p_1 z + \cdots + p_n z^n = \prod_i^n (1 + \beta_i z)$$

We get:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

Therefore any polynomial K_j with $j \leq m$ is determined as a symmetric polynomial in the β_i and hence as a polynomial in the p_i. This holds for arbitrary m, completing the proof. □

Next, let

$$Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i \qquad b_i \in B$$

be a formal power series. Then there is an associated m-sequence $\{K_j\}$ where $K(1+z) = Q(z)$.

Proof. Consider the usual decomposition:

$$1 + p_1 z + \cdots + p_n z^n = \prod_i^n (1 + \beta_i z)$$

and consider the product $\prod_{i=1}^{m} Q(\beta_i z)$ where β_i are symmetric functions. Then the coefficients z^j are homogeneous of weight j. Thus, it is in $K_j^{(m)}(p_1, \ldots, p_j)$. It follows that $K_j^{(m)}$ does not depend on m for $m \geq j$.

Now, define $K_j = K_j^{(m)}$ for $m \geq j$. I claim $\{K_j\}$ is the required m-sequence. Indeed, the earlier proof the multiplicative property is true if the indeterminates p_i', p_i'' are replaced by 0 for large values of i, and so $\{K_j\}$ is an m-sequence. Finally applying $m = 1$ in equation:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

shows that $K(1+z) = Q(z)$, completing the proof. □

Combining these, we get the correspondence

The simplest example would be $\{p_i\}$ which has the characteristic series $1 + z$. The notation p_i is due to the connection to Pontryagin classes. Recall Pontryagin classes are given in the $4i$th cohomology group. The HRR theorem shall work with Chern classes which are in the $2i$th cohomology gorups, and so we shall be working with the following will be a useful substitution. Let $c_0 = p_0 = 1$, $z = x^2$, and γ_i be such that $\beta_i = \gamma_i^2$. then:

$$\sum_{i=1}^{\infty} p_i(-z)^i \left(\sum_{j=0}^{\infty} c_j(-x)^j \right) \left(\sum_{i=0}^{\infty} c_i x^i \right)$$

Naturally, if $\{K_j(p_1, ..., p_j)\}$ is an m-sequence with $Q(z)$ as the characteristic power series, then let

$\{\widetilde{K}_j(c_1,...c_j)\}$ be an m-sequence where $\widetilde{Q}(x) = Q(x^2)$. Then:

$$K_j(p_1,...,p_j) = \widetilde{K}_{2j}(c_1,...,c_{2j})$$
and
$$0 = \widetilde{K}_{2j+1}(c_1,....,c_{2j+1})$$

Thus, the m-sequence in c_i with the characteristic power series $1+x^2$ is given by

$$1, 0, p_1, 0, p_2, ...$$

From this relations, we get:

$$\begin{aligned}
p_1 &= -2c_2 + c_1^2 \\
p_2 &= 2c_4 - 2c_3c_1 + c_2^2 \\
p_3 &= -2c_6 + 2c + 5c + 1 = 2c_4c_2 + c_3^2, ...
\end{aligned}$$

Let's now find a way to calculate the explicit calculation of the polynomials of m-sequences. Given $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$, consider the factorization:

$$1 + b_1 z + b_2 z^2 + \cdots + b_m z^m = (1+\beta_1' z)\cdots(1+\beta_m' z)$$

Then define the sum:

$$\sum (\beta_1')^{j_1} (\beta_2')^{j_2} \cdots (\beta_r')^{j_r} \qquad \text{where} \qquad j_1 \geq j_2 \geq \cdots \geq j_r \geq 1,\ \sum_{s=1}^{r} j_s = k \leq m$$

which denotes the symmetric function in the β_i' which is the sum of all pairwise distinct monomials obtained by permuting $(\beta_1',...,\beta_m')$ to the monomial $(\beta_1')^{j_1} (\beta_2')^{j_2} (\beta_r')^{j_r}$. The number of monomial in the sum is $m!/h$, h being the number of permutations living the aforementioned monomial fixed. Now:

Lemma 4.1.3: Computation of m-sequence

Let $\{K_j(p_1....,p_j)\}$ be the m-sequence corresponding to $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$. Then the coefficient of $p_{j_1} p_{j_2} \cdots p_{j_r}$ of K_s where $j_1 \geq j_2 \geq \cdots \geq j_r \geq 1$ and $\sum_{s=1}^{r} j_s = k$ is equal to:

$$\sum (j_1, j_2, ..., j_r)$$

***Proof*:**
it is computations that will be omitted.

Define:

$$Q(z) = \frac{\sqrt{z}}{\tanh\sqrt{z}} = 1 + \sum_{k=1}^{\infty} (-1)^{k-1} \frac{2^{2k}}{(2k)!} B_k z^k$$

where the B_k are the Bernoulli numbers. Note that in this case the coefficient ring is $B = \mathbb{Q}$. Furthermore, we shall have $B_k > 0$ skip the value of 1/2. Thus we get:

$$B_1 = \frac{1}{6},\ B_2 = \frac{1}{30},\ B_3 = \frac{1}{42},\ B-4 = \frac{1}{30},\ B_5 = \frac{5}{66},\ B_6 = \frac{691}{2730},\ B_7 = \frac{7}{6},\ B_8 = \frac{3617}{510}, ...$$

We shall denote the m-sequence of $Q(z)$ as $\{L_j(p_1, ..., p_j)\}$. Computing we get:

$$\begin{aligned}
L_1 &= \frac{1}{3}p_1 \\
L_2 &= \frac{1}{5\cdot 7}(7p_2 - p_1^2) \\
L_3 &= \frac{1}{3^3\cdot 5\cdot 7}(62p_3 - 13p_2p_1 + 2p_1^3) \\
&\vdots
\end{aligned}$$

We will want the following normalization: $L_k(p_1...., p_k) = 1$. To that end, if we let $p_i = \binom{2k+1}{i}$, then we get:

$$1 + p_1z + p_2z^2 + \cdots + p_kz^k = (1+z)^{2k+1} \bmod z^k + 1$$

From this we get:

Proposition 4.1.4: Coefficient of L-characteristic Series

Let $Q(z) = \frac{\sqrt{z}}{\tanh\sqrt{z}}$. Then for every k, the coefficient J_k of z^k in $(Q(z))^{2k+1}$ is equal to 1, and $Q(z)$ is the *only* power series with rational coefficient with this property.

***Proof*:**
Apply Cauchy's integral formula and compute (see [16, p.12] to see the exact computations)

The above is useful for Pontryagin classes. Let us find the equivalent for Chern classes. Let:

$$Q(x) = \frac{x}{1-e^{-x}} = 1 + \frac{1}{2}x + \sum_{k=1}^{\infty}(-1)^{k-1}\frac{B_k}{(2k)!}x^{2k}$$

be the characteristic polynomial. Then we get an m-sequence of polynomials $\{T_k\}$ called *Todd polynomials*. For future reference, we have the identity:

$$\frac{x}{1-\exp(-x)} = \exp\left(\frac{1}{2}x\right)\frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$$

The m-sequence for $Q(x) = \frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$ is given by:

$$A_1 = \frac{-2}{3},\ A_2 = \frac{2}{45}(-4p_2 + 7p_1^2),\ A_3 = \frac{-4}{3^3\cdot 5\cdot 7}(16p_3 - 44p_2p_1 + 31p - 1^3),\ \cdots$$

By using the translation from Pontryagin to Chern class lemma, we get:

$$T + k(c_1, ..., c_k) = \sum_{\substack{r,s\\ r+2s=k}} \frac{1}{2^{4s}r!}\left(\frac{1}{2}c_1\right)^r A_s(p_1, ..., p_s)$$

We can compute the first few values:

$$\begin{aligned}
T_1 &= \frac{1}{2}c_1 \\
T_2 &= \frac{1}{12}(c_2 + c_1^2) \\
T_3 &= \frac{1}{24}c_2c_1 \\
T_4 &= \frac{1}{720}(-c_4 + c_3c_1 + 3c_2^2 + 4c_2c_1^2 - c_1^4) \\
T_5 &= \frac{1}{1440}\left(-c_4c_1 + c_3c_1^2 + 3c_2^2c_1 - c_2c_1^3\right) \\
&\vdots
\end{aligned}$$

The reader should already notice how the Riemann Roch for curves and surfaces has part the geometric component captured within the 1st and 2nd Todd polynomial.

Again, by substituting $c_i = \binom{n+i}{i}$ defined by:

$$1 + c_1x + \cdots + c_nx^n = (1+x)^{n+1} \bmod x^{n+1}$$

gives

$$T_n(c_1, ..., c_n) = 1$$

Just like before, $Q(x)$ is the only power series with rational coefficients with the property that $Q(x))^{k+1} = 1$ (see [16, p.14], or just repeat the above results with the appropriate modifications).

Recall the Euler characteristic is called the *arithmetic genus* (following Hartshorne's vocabulary). This will motivate how evaluating characteristic series tangent bundles by selecting their Pontryagin/Chern classes shall be called the *genus* with respect to the characteristic series. Depending on the characteristic series, we shall give a name to the genus.

Definition 4.1.5: K-genus

Let M be an oriented n-dimensional manifold and $\{K_j(p_1, ..., p_j)\}$ be an m-sequence. Then:

$$\begin{cases} 0 & \dim_{\mathbb{R}} M \not\equiv 0 \bmod 4 \\ \int_M K_n(p_1, ..., p_n) & \dim_{\mathbb{R}} M = 4n \end{cases}$$

is called the *K-genus* of M, where $\int_M K_n(p_1, ..., p_n)$ represents taking the tangent bundle TM, finding the Pontryagin classes, and evaluating at the fundamental class M.

If we are working with Chern classes, then we would have $n = 2k$.

By what we've discussed earlier, the K-genus is multiplicative.

Proposition 4.1.6: K-genus is Multiplicative

$$K(M \times N) = K(M)K(N)$$

***Proof*:**
immediate form the properties of characteristic series.

4.1.1 Generating Function

As seen in chapter 3, we define a natural generating function χ_y combining the Euler characteristic χ^p. The idea is that we can figure out the coefficients of this generating function when plugin in different values of the dummy variable y. Let us define do the same for the T-characteristic. Let $B = \mathbb{Q}[y]$. Then consider the m-sequence $T_j(y; c_1, ..., c_j)$ with characteristic power series:

$$Q(y;x) = \frac{x(y+1)}{1-e^{-x(y+1)}} - yx = \frac{x(y+1)}{e^{x(y+1)}-1} + x$$

Lemma 4.1.7

For every n the coefficient of x^n in $(Q(y;x))^{n+1}$ is equal to $\sum_{i=0}^{n}(-1)^i y^i$, and $Q(y,x)$ is the only power series with coefficients in $\mathbb{Q}[y]$ which has this property.

***Proof*:**
Take substitution

$$c_i = \binom{n+1}{i}$$

then:

$$T_n(y; c_1, \dots, c_n) = 1 - y + y^2 - \dots + (-1)^n y^n.$$

Then as $Q(\frac{1}{y}; yx) = Q(y; -x)$:

$$y^n T_n(\frac{1}{y}; c_1, \dots, c_n) = (-1)^n T_n(y; c_1, \dots, c_n)$$

as we sought to show

The polynomial $T_n(y; c_1, \dots, c_n)$ can be written in a unique way in the form

$$T_n(y; c_1, \dots, c_n) = \sum_{p=0}^{n} T_n^p(c_1, \dots, c_n) y^p.$$

satisfying the same grading as χ_y, and the polynomials $T_n^p(c_1, \dots, c_n)$ satisfy

$$T_n^p(c_1, \dots, c_n) = (-1)^n T_n^{n-p}(c_1, \dots, c_n)$$

that is the “Poincaré duality”. Now, consider the factorization:

$$1 + c_1 x + \dots + c_n x^n = \prod_{i=1}^{n}(1 + \gamma_i x)$$

Then we have:

Proposition 4.1.8: Computing T_y-Polynomial

$$T_n^p(c_1, ..., c_n) = \kappa_n \left[\sum \exp(-\gamma_{j_1} - \cdots - \gamma_{j_r}) \prod_{i=1}^{n} \frac{\gamma_i}{1-\exp(-\gamma_i)} \right]$$

where the sum is over the $\binom{n}{p}$ pairwise combinations of distinct γ_i, and $\kappa_n[-]$ denotes the sum of all homogeneous polynomials of degree n (mirroring evaluating at the fundamental class in).

***Proof*:**
Indeed:

$$\begin{aligned}
\sum_{p=0}^{n} T_n^p y^p &= \kappa_n \left[\prod_{i=1}^{n} (1 + y\exp(-\gamma_i)) \frac{\gamma_i}{1-\exp(-\gamma_i)} \right] \\
&= \kappa_n \left[\prod_{i=1}^{n} \frac{(1 + y\exp(-(1+y)\gamma_i))}{1+y} \cdot \frac{(1+y)\gamma_i}{1-\exp(-(1+y)\gamma_i)} \right] \\
&= \kappa_n \left[\prod_{i=1}^{n} Q(y;\gamma_i) \right] = \sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) y^p
\end{aligned}$$

as we sought to show

Finally note that $Q(0;x) = \frac{x}{1-e^{-x}}$, $Q(-1;x) = 1 + x$ and $Q(1;x) = \frac{x}{\tanh x}$. Therefore:

$$T_n^0(c_1, \ldots, c_n) = T_n(c_1, \ldots, c_n) \text{ (Todd polynomial)},$$

Proposition 4.1.9: $y = -1$ and $y = 1$ Values

1. $\sum_{p=0}^{n} (-1)^p T_n^p(c_1, \ldots, c_n) = c_n$
2. $\sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) = \tilde{L}_n(c_1, \ldots, c_n)$, that is:

$$\sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) \begin{cases} 0 \text{ if } n \text{ is odd} \\ L_k(p_1, \ldots, p_k) \text{ if } n = 2k. \end{cases}$$

***Proof*:**
exercise

The first equation we see links to the Euler-characteristic as the Euler class is given by the top Chern class, while the second equation will be equal to the spin index as will be defined in chapter 5.

4.2 Todd Class

With the algebraic established, we can formally define the Todd class:

Definition 4.2.1: Todd Class

Let M be a complex smooth manifold, and E a continuous q-vector bundle on M. Let $c_i \in H^{2i}(M, \mathbb{Z})$ be its Chern classes. Then we can define the *total Todd class* of E by taking the Todd polynomials and taking:

$$\mathrm{td}(E) = \sum_{j=1}^{\infty} T_j(c_1, ..., c_j)$$

where $\{T_j(c_1, ..., c_j)\}$ is given by the characteristic series:

$$Q(x) = \frac{x}{1 - e^{-x}} = 1 = \frac{1}{2}x + \sum_{j=1}^{\infty}(-1)^{j-1}\frac{B_j}{(2j)!}x^{2j}$$

where B_j is the Bernoulli numbers.

The reader can verify that the direct product of manifolds $M \times Y$ is multiplicative on the Todd class. This means that if

$$0 \to E' \to E \to E'' \to 0$$

Then:

$$(1 + c_1't + \cdots + c_{q'}'t^{q'})(1 + c_1''t + \cdots + c_{q''}''t^{q''}) = 1 + c_1t + \cdots + c_qt^q$$

and so:

$$\mathrm{td}(E) = \mathrm{td}(E') \cdot \mathrm{td}(E'')$$

Mirroring the property of the Euler Characteristic.

To write down the m-sequence, recall that the m-sequence of the characteristic power series $Q(z) = \frac{2\sqrt{z}}{\sinh(2\sqrt{z})}$ has as the first few terms:

$$A_1 = \frac{-2}{3},\ A_2 = \frac{2}{45}(-4p_2 + 7p_1^2)\ A_3 = \frac{-4}{3^2 \cdot 5 \cdot 7}\left(16p_3 - 44p_2p_1 + 31p_1^3\right)$$

Then note that :

$$\frac{x}{1 - e^{-x}} = \exp\left(\frac{1}{2}x\right)\frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$$

By our earlier work:

$$\mathrm{td}(E \oplus E') = \mathrm{td}(E)\mathrm{td}(E')$$

In the special case where $q = 1$ and $c_1(E) = d \in H^2(M, \mathbb{Z})$, then:

$$\mathrm{td}(E) = \frac{d}{1 - e^{-d}}$$

Then for a general q-bundle we can apply the splitting principle. In terms of the Todd class, we get define:

$$\sum_{j=0}^{q} c_j x^j = \prod_{i=1}^{q}(1+\gamma_i x)$$

Thus, when dealing with a general complex vector bundle E and take its Chern roots γ_i:

$$\mathrm{td}(E) = \prod_{i=1}^{q} \frac{\gamma_i}{1-e^{-\gamma_i}}$$

Let us compare this to the usual results we get about the Chern class: it is the natural "dual" of the Todd class in the following way:

Theorem 4.2.2: Borel-Serre Identity

Let E be a continuous $\mathrm{GL}_q(\mathbb{C})$-bundle over M. Then:

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch}\left(\bigwedge^r E^*\right) = (\mathrm{td}(E))^{-1} c_q(E)$$

***Proof*:**
Proof: If $\sum_{j=0}^{q} c_j(E)x^j = \prod_{i=1}^{q}(1+\gamma_i x)$ by the properties of Chern classes

$$\mathrm{ch}\left(\bigwedge^r E^*\right) = \sum e^{-(\gamma_{i_1}+\cdots+\gamma_{i_r})}$$

where the sum is over all combinations $i_1, \ldots, i_r$ with $1 \le i_1 < \cdots < i_r \le q$. Therefore

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch} \bigwedge^r E^* = \prod_{i=1}^{q}(1-e^{-\gamma_i}) = (\gamma_1 \cdots \gamma_q) \prod_{i=1}^{q} \frac{1-e^{-\gamma_i}}{\gamma_i} = (\mathrm{td}(E))^{-1} c_q(E).$$

Recall that if we are talking about the Chern class of a complex manifold M, we are talking about the Chern class of its tangent bundle, and the K-genus is defined for such structures:

Definition 4.2.3: Todd Genus

Let M be an n-dimensional almost complex manifold. The *Todd genus* (or *T-genus*) is the number:

$$T(M_n) = T_n(c_1, ..., c_n)[M_n] = \kappa_n[\mathrm{td}(TM)] = \int_M \mathrm{td}(TM)$$

where $[M_n]$ represents the tangent bundle of M_n.

The *generalized Todd Genus* is given by $\{T_y(y; c_1, ..., c_n\}$ and:

$$T_y(M_n) = \sum_{p=0}^{n} T^p(M_n)y^p$$

is the *generalized Todd Genus.*

The generalized Todd genus is a natural generaliztion of the todd genus as:

$$T_0(M_n) = T^0(M_n) = T(M_n)$$

To important terms that we must evaluate is at the $y = -1$ and $y = 1$. For $y = -1$:

$$\begin{aligned} T_{-1}(M_n) &= \sum_{p=0}^{n} (-1)^p T^p(M_n) \\ &= \prod_i Q(-1; \gamma_i)[M_n] \\ &= \prod_i Q(1 + \gamma_i x)[M_n] \\ &\overset{!}{=} c_n[M_n] \\ &= \chi(M_n) \end{aligned}$$

where $\overset{!}{=}$ comes from ignoring the lower order terms because perfect pairing. Evaluating at 1 we by proposition 4.1.9 we get:

$$T_1(M_n) = \sum_{p=0}^{n} T^p(M_n) = \tau(M_n)$$

where τ is shorthand for the Pontryagin number as it will be equal to the index of M_n. By lemma 4.1.7, the T-genus is the only genus associated to an m-sequence with rational coefficients which takes the value 1 on every complex projective space $\mathbb{C}\mathrm{P}^n$, and the T_y-genus is the only genus associated to an m-sequence with coefficients in $\mathbb{Q}[y]$ which takes the value $1 - y + y^2 - \cdots + (-1)^n y^n$ on $\mathbb{C}\mathrm{P}^n$.

Proposition 4.2.4: Properties of Todd Genus

Let B be a commutative ring where $\mathbb{Q} \subseteq B$. Then

1. $\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = c_n$, for all n.
2. $T_n^{(0)}(c_1, \ldots, c_n) = \mathrm{td}(c_1, \ldots, c_n)(= T_n(c_1, \ldots, c_n))$.
3. $$T_n^l(c_1, ..., c_n) = (-1)^n T_n^{n-l}(c_1, ..., c_n)$$
4. $$\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ L_{\frac{n}{2}}(p_1, \ldots, p_{\frac{n}{2}}) & \text{if } n \text{ is even.} \end{cases}$$

The *total Todd class* of a vector bundle, E, is

$$\mathrm{td}(E) = \sum_{j=0}^{\infty} \mathrm{td}_j(c_1, \ldots, c_j)t^j = 1 + \frac{1}{2}c_1(E) + \frac{1}{12}(c_1^2(E) + c_2(E))t^2 + \frac{1}{24}(c_1(E)c_2(E))t^3 + \cdots$$

***Proof*:**
exercise

4.2.1 Virtual Todd Genus

Let us now translate the results over for splitting manifolds. Let $E \to M$ be complex vector bundle over a manifold, M^Δ the associate flag manifold such that the pullback along $f : M^\Delta \to M$ makes E split into linear bundles. The same applies even if $M = V$ is a variety with singularities. We can breakdown the pullback of E into a filtration:

$$f^*(E) = E_r \supseteq E_{r-1} \supseteq \cdots \supseteq E_1 \supseteq E_0 = 0$$

This shall afford us the ability to calculate the Todd class, and hence the Todd genus, using line bundles.

Let us start by relating the Chern class to a sub-manifold Let $V_{n-k} \subseteq M_n$ be a submanifold of codimension k and let $j : V_{n-k} \to M_n$ be an embedding. If M_n is almost complex, then we have the pullback of vector bundles;

$$j^*(TM_n) \cong TV_{n-k} \oplus N_{M/V} \tag{4.1}$$

where $N_{M/V}$ is the normal bundle of V with respect to M; let us label it ν. If $k = 1$, then we simply get:

$$c(\nu) = 1 + j^*(v)$$

where $v \in H^2(M_n; \mathbb{Z})$ is the cohomology class determined by the fundamental calls of V_{n-1}. By equation (4.1):

$$\begin{aligned} &1 + c_1(V_{n-1}) + c_2(V_{n-1}) + \cdots \\ &= j^*\left((1 + c_1(M_n) + c_2(M_n) + \cdots)(1 + v)^{-1}\right) \end{aligned}$$

Then we can define the T_y-genus of V_{n-1}. To that end, define the characteristic power series:

$$Q(y;x) = \frac{x}{R(y;x)}$$

where $R(y;x)$ is a shorthand for $\frac{e^{x(y+1)}-1}{e^{x(y+1)}+y}$. Then equation (4.1) implies:

$$\sum_{i=0}^{\infty} T_i(y; c-1(V_{n-1}), ..., c_i(V_{n-1})) = j^* \left(\frac{R(y;v)}{v} \sum_{i=0}^{\infty} T_i(y; c_1(M_n), ... \right)$$

and so the T_y-genus is:

$$T_y(V_{n-1}) = \kappa_n \left[R(y;v) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ..., c_j(M_n)) \right] = \kappa_n[(1-e^{-v})\mathrm{td}(TM)]$$

where the 2nd equality is an immediate application of equation (4.1). Now, by the splitting principal, we can filter TM, or any vector bundle E, where each intermediate step is linear. This motivates the following definition:

Definition 4.2.5: Virtual Todd Genus

Let $v_1, ..., v_r \in h^2(M_n; \mathbb{Z})$. Then:

$$T_y(\{v_1, ..., v_r\})_M = \kappa_n \left[R(y;v_1) \cdots R(y;v_r) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ...) \right]$$

Define the $T^p(\{v_1, ..., v_r\})$ to be the terms in the summand of T_y when expanded:

$$T_y(\{v_1, ..., v_r\})_M = \sum_{p=0}^{n-r} T^p(\{v_1, ..., v_r\})_M y^p$$

By proposition 4.1.8, the virtual Todd genus is a polynomial of degree $n-r$ in y with rational coefficients. if $r > n$, it is 0. If $r = n$:

$$T_y(\{v_1, ..., v_n\})_M = v_1 v_2 \cdots v_n[M_n]$$

The virtual Todd genus respects the same properties as the Todd genus, namely:

$$T(\{v_1, ..., v_r\})_M = T_0(\{v_1, ..., v_r\})_M = T^0(\{v_1, ..., v_r\})_M$$

and

$$T^p(\{v_1, ..., v_r\})_M = (-1)^{n-r} T^{n-r-p}(\{v_1, ..., v_r\})_M$$

And by definition:

$$T_y(V_{n-1}) = T_y(\{v\})_M$$

for appropriate v and were $V_{n-1} \hookrightarrow M_n$ is a codimension 1 manifold. Let us now work towards showing that the virtual Todd genus satisfies some of the algebraic propertites of theorem theorem 3.4.9. Let

$$R(x) = \frac{e^{ax-1}}{e^{ax}+y}$$

for some intermediate a, y. Then:

$$R(u+v) = R(u) + R(v) + (y-1)R(u)R(v) - yR(u)R(v)R(u+v)$$

Letting $a = 1+y$, we get an equation for $R(y;x)$. Then as T_y decomposes into products of such equations we get:

Theorem 4.2.6: Functional Equation of T_y

Let $v_1, ..., v_r$, u, v be defined as before. Then:

$T_y(\{v_1, ..., v_r, u+v\}) =$
$T_y(\{v_1, ..., v_r, u\}) + T_y(\{v_1, ..., v-r, v\}) + (y-1)T_y(\{v_1, ...v_r\})(u,v)_y T_y(\{v_1, ..._r, u, v, u+v\})$

If $r = 0$:

$$T_y(\{u+v\}) = T_y(\{u\}) + T_y(\{v\}) - (y-1)T_y(\{u,v\}) - yT_y(\{u,v,u+v\})$$

As $y = -1$ gives the Euler characteristic, we see that it satisfies:

$$T_{-1}(\{u+v\}) = T_{-1}(\{u\}) + T_{-1}(\{v\}) - 2T_{-1}(\{u,v\}) + T_{-1}(\{u,v,u+v\})$$

When $y = 0$, we get the Todd genus satisfies:

$$T(\{u+v\}) = T(\{u\}) + T(\{v\}) - T(\{u,v\})$$

and when $y = 1$, we shall see that the spin index will satisfy the equation:

$$T_1(\{u+v\}) = T_1(\{u\}) + T_1(\{v\}) - T_1(\{u,v,u+v\})$$

4.3 T-characteristic – combining Algebraic and Geometric information

So far, we saw that the Todd class captures some information about the underlying space M. We shall want to capture some information about the bundle that is given by the Chern character. Combining these two together, we get:

Definition 4.3.1: T-characteristic

Let M be a compact complex manifold, E a complex vector bundle. Then

$$T(M,E) = \kappa[\mathrm{ch}(E)\mathrm{td}(TM)] = \int_M \mathrm{ch}(E) \cdot \mathrm{td}(M)$$

Is called the *T-characteristic* of E over M

Note that if $E = \mathbb{1}$, then $T(M, \mathbb{1}) = \kappa[\mathrm{ch}(\mathbb{1})\mathrm{td}(TM)] = \kappa[\mathrm{td}(TM)]$ showing that we reduce to only holding information about M. As $\mathbb{C}^*$-bundles over M_n are classified by elements $d \in H^2(M_n, \mathbb{Z})$, we shall sometimes write $T(M_n, d)$ for $T(M_n, E)$ where E is associated to d. By the results in section 4.2.1:

Proposition 4.3.2: Separating Manifold and Bundle

$$T(M,d) = T(M) - T(\{-d\})_M$$

***Proof*:**
this can also be proven directly:

$$\begin{aligned}
&= T(M_n) - T(-d)_M \\
&= \kappa_n[\mathrm{td}(TM)] - \kappa_n[(1-e^d)\cdot \mathrm{td}(TM)] \\
&= \kappa_n[\mathrm{td}(TM) - (1-e^d)\cdot \mathrm{td}(TM)] \\
&= \kappa_n[\big(1-(1-e^d)\big)\cdot \mathrm{td}(TM)] \\
&= \kappa_n[\big(1-1+e^d\big)\cdot \mathrm{td}(TM)] \\
&= \kappa_n[e^d\cdot \mathrm{td}(TM)] \\
&= T(M,d)
\end{aligned}$$

Notice how this mirrors the same property as the Euler characteristic (proposition 3.4.2). If E is a complex vector bundle of dimension q, E' of dimension q', both over M_n, then using the above equation we get:

$$T(M_n, E\oplus E') = T(M_n,E) + T(M_n,E')$$

Then by the properties of the characters we get:

$$T(M\times N, \pi_M^*(E)\otimes \pi_N^*(\eta)) = T(M,E)T(N,\eta)$$

Let us extend this to T_y-characteristics and show that it is additive and multiplicative in the same way. First, by proposition 4.1.8:

$$T\left(M_n, \bigwedge^p (T^*M)\right) = T^p(M_n)$$

Then $\wedge^p(T^*M)\otimes E$, so $T(M_n, \lambda^p(T^*M)) = T^p(M_n)$. By taking:

$$T_y(M_n,E) = \sum_{p=0}^{n} T^p(M_n,E)y^p$$

Thus we can re-write the T-characteristic as:

$$T(M_n,E) = \kappa_n\left[\mathrm{ch}(E)\mathrm{ch}\left(\bigwedge^p(T^*M)\right)\mathrm{td}(TM)\right]$$

Then by applying the same proof techniques for proposition proposition 4.1.8 we get:

$$T_y(M_n,E) = \kappa_n\left[\left(\sum_{i=1}^{q} e^{(1+y)d_i}\right)\left(\sum_{j=0}^{n} T_j(y;c_1,\ldots,c_j)\right)\right]$$

Notice that:

$$T_{-1}(M_n, E) = q \cdot \chi(M_n)$$

recovering the Euler-Poincaré characteristic of M_n, and multiplying by q as it is the dimension of E. Similarly, generalizing proposition 4.2.4 we get:

$$T^p(M_n, E) = (-1)^n T^{n-p}(M_n, E^*)$$

and if $p = 0$:

$$T(M_n, E) = (-1)^n T\left(M_n, \bigwedge^n (T^*M) \otimes E^*\right)$$

Recovering the duality of χ^p. Next, by extending $\mathrm{ch}(-)$ so that it works on generating functions:

$$\mathrm{ch}_{(y)}(E) = e^{(1+y)\gamma_1} + \cdots + e^{(1+y)\gamma_q}$$

where γ_i are the Chern roots of E. Then:

$$\mathrm{ch}_{(y)}(E \oplus E') = \mathrm{ch}_{(y)}(E) + \mathrm{ch}_{(y)}(E') \qquad \text{and} \qquad \mathrm{ch}_{(y)}(E \otimes E') = \mathrm{ch}_{(y)}(E) \cdot \mathrm{ch}_{(y)}(E')$$

and so combining with the results previous results we get:

$$T_y(M_n, E \oplus E') = T_y(M_n, E) + T_y(M_n, E')$$

and

$$T_y(M \times V, \pi_M^*(E) \otimes \pi_V^*(\eta)) = T_y(M, E) T_y(V, \eta)$$

Let us now develop this to the define the virtual T-characteristic. Motivated by the case for the Euler characteristic (definition 3.4.1), we define the following:

Definition 4.3.3: Virtual T-characteristic

$$T_y(\{v_1, ..., v_r\}, E)_M = \kappa_n \left[\mathrm{ch}_{(y)}(E) \prod_{i=0}^{r} R(y; v_i) \sum_{j=0}^{\infty} T_j(y; c_1(M_n), ...) \right]$$

Proposition 4.3.4: Properties of Virtual T-characteristic and Submanifolds

1. Let V_{n-1} be an almost complex submanifold of M_n ,$v \in H^2(M_n, \mathbb{Z})$ the cohomology class determined by V_{n-1} and $j : V_{n-1} \hookrightarrow M_n$ be its embedding into M_n. Let $v_2, ..., v_r \in H^2(M_n, \mathbb{Z})$ and E a q-dimensional complex bundle over M_n. Then:
$$T_y(\{j^*v_2, ..., j^*v_r\}, j^*(E))_V = T_y(\{v, v_2, ..., v_r\}, E)_M$$
2. the functional equation in theorem 4.2.6 extends to the virtual T-characteristic.
3. $T_y(\{v_1, ..., v_r\}, E)$ is a $n-r$ degree polynomial in y with rational coefficients, and if $r = n$:
$$T_y(\{v_1, ..., v_n\}, E) = q \cdot \int_M v_1 \cdots v_n$$
4. $y^{n-r}T_{\frac{1}{y}}(\{v_1, ..., v_r\}, E)_M = (-1)^{n-r}T_y(\{v_1, ..., v_r\}, E)_M$
5. If M is an n-dimensional smooth manifold, L is a line bundle over M with chern class $1+v$, $v \in H^2(M, \mathbb{Z})$, E is a q-dimensional vector bundle over M, and $v_1, ..., v_r \in H^2(M, \mathbb{Z})$, then:
$$\begin{aligned}&T_y(\{v_1, ..., v_r\}, E)_M = \\ &\quad T_y(\{v_1, ..., v_r\}, E)_M + T_y(\{v_1, ..., v_r\}, E \otimes L^{-1})_M + yT_y(\{v_1, ..., v_r, v\}, E \otimes L^{-1})_M\end{aligned}$$
6. following the above convention and letting $V \subseteq M$ be a codimension 1 submanifold:
$$T^p(M_n, E) = T^p(V_{n-1}, j^*(E)) + T^p(M_n, E \otimes L^{-1}) + T^{p-1}(V, j^*(E \otimes L^{-1}))$$

***Proof*:**
definition unpacking.

Let us apply this to a split manifold. Let M be a compact almost complex split manifold of complex dimension n so that $c(M) = \prod_{i=1}^{n}(1 + a_i)$ for appropriate a_i. Then the Todd genus $T(M)$ can be expressed in terms of virtual indices:

Proposition 4.3.5: Todd Genus in Terms of Virtual T-characteristic

Let M be a complex split manifold, TM split into $A_1, ..., A_n$. Then:
$$(1+y)^n T(M) = \sum_{l=0}^{n} y^l \sum_{1 \le i_1 < \cdots < i_l \le n} T_y(\{a_{i_1}, \ldots, a_{i_l}\})_M$$

***Proof*:**

$$\begin{aligned}
\kappa_n\left[\prod_{i=1}^{n}(1+yR(y;a_i))\prod_{i=1}^{n}Q(y;a_i)\right] &= \kappa_n\left[\prod_{i=1}^{n}(Q(y;a_i)+a_iy)\right] \\
&= \kappa_n\left[\prod_{i=1}^{n}\frac{(1+y)a_i}{1-\exp(-(1+y)a_i)}\right] \\
&= (1+y)^n\kappa_n\left[\prod_{i=1}^{n}\frac{a_i}{1-\exp(-a_i)}\right] \\
&= (1+y)^nT(M)
\end{aligned}$$

4.4 Todd Genus Invariant Under Splitting

Let $j : M^\Delta \to M$ be the pullback on which TM splits. Let us now show that $T(M) = T(M^\Delta)$, that is the Todd genus is invariant under splitting. Essentially, if $\mathbf{F}(n)$ is the appropriate flag manifold, then $T(\mathbf{F}(n)) = 1$ we shall show

$$T(M^\Delta) = T(M)T(\mathbf{F}(n)) = T(M)$$

Algebraic Preliminaries for Flag Manifold having Trivial Todd Genus

Let $\mathbb{F}$ be a field of characteristic 0, and let $c_1, \ldots, c_n$ be indeterminates over $\mathbb{F}$. Take the field extension $\mathbb{F}(c_1, \ldots, c_n)(\gamma_1, \ldots, \gamma_n)$ where the c_i are the elementary symmetric functions of the γ_j. The $n!$ elements $\gamma_1^{a_1}\gamma_2^{a_2}\cdots\gamma_{n-1}^{a_{n-1}}$ with $0 \leq a_i \leq n-i$ form a basis for this extension.

Any formal power series P in $\gamma_1, \ldots, \gamma_n$ with coefficients in $\mathbb{F}$ can be uniquely expressed as:

$$P = \sum_{a_1,\ldots,a_{n-1}} e_{a_1,\ldots,a_{n-1}}\gamma_1^{a_1}\cdots\gamma_{n-1}^{a_{n-1}}$$

where the coefficients $e_{a_1,\ldots,a_{n-1}}$ are power series in $c_1, \ldots, c_n$.

Definition 4.4.1: Indicator

The *indicator* of a formal power series P in the variables $\gamma_1, \ldots, \gamma_n$ is defined as:

$$e(P) = (-1)^{n(n-1)/2}e_{n-1,n-2,\ldots,1}$$

where $e_{n-1,n-2,\ldots,1}$ is the coefficient of the basis element $\gamma_1^{n-1}\gamma_2^{n-2}\cdots\gamma_{n-1}^{1}$.

The indicator can be calculated with the formula:

$$e(P) = \frac{\sum_{s\in S_n}\operatorname{sign}(s)\cdot s(P)}{\prod_{i>j}(\gamma_i-\gamma_j)}$$

where the denominator is the Vandermonde determinant.

Lemma 4.4.2: Invariance of Indicator

If a polynomial P is invariant under the interchange of γ_i and γ_j for some $i \neq j$, then its indicator $e(P)$ is zero.

Lemma 4.4.3: Indicator of a Product

Let $P = \prod_{i>j} \frac{\gamma_i - \gamma_j}{1-e^{-(\gamma_i - \gamma_j)}}$. Then the indicator is $e(P) = 1$.

The Invariance

The flag manifold $\mathbf{F}(n) = GL(n, \mathbb{C})/\mathcal{T}(n, \mathbb{C})$, where $\mathcal{T}(n, \mathbb{C})$ is the subgroup of upper triangular matrices, is a complex analytic split manifold. Its cohomology ring is given by [12]:

$$H^*(\mathbf{F}(n), \mathbb{Z}) \cong \mathbb{Z}[\gamma_1, \dots, \gamma_n]/I^+(c_1, \dots, c_n)$$

where I^+ is the ideal generated by the elementary symmetric functions of the γ_i of positive degree. The total Chern class of its tangent bundle is:

$$c(T\mathbf{F}(n)) = \prod_{i>j}(1 + \gamma_i - \gamma_j)$$

Theorem 4.4.4: Todd Genus of the Flag Manifold

The Todd genus of the flag manifold $\mathbf{F}(n)$ is 1.

***Proof*:**
This result is derived from lemma 4.4.3 and the definition of the Todd genus:

$$T(\mathbf{F}(n)) = \kappa_n \left[\prod_{i>j} \frac{\gamma_i - \gamma_j}{1 - e^{-(\gamma_i - \gamma_j)}} \right] = 1$$

where κ_n denotes evaluation on the fundamental cycle.

Thus:

Theorem 4.4.5: T-characteristic Pullback Invariant

Let E be a q-dimensional complex vector bundle over a compact n-dimensional complex manifold M. Let M^Δ be the associated fiber bundle with the flag manifold $\mathbf{F}(q)$ as fiber, and let $\varphi : M^\Delta \to M$ be the projection. For any complex line bundle C over M, the T-characteristic is multiplicative:

$$T(M^\Delta, \varphi^* C) = T(M, C) \cdot T(\mathbf{F}(q)) = T(M, C)$$

More generally given a collection $\{\ell_1, ..., \ell_r\} \subseteq H^2(M, \mathbb{Z})$:

$$T(\{\varphi^*\ell_1, \dots, \varphi^*\ell_r\}, \varphi^* C)_{M^\Delta} = T(\{\ell_1, \dots, \ell_r\}, C)_M$$

***Proof*:**
The proof relies on expressing the T-characteristic as an integral over the total space and using the indicator $e(P)$ from section 4.4 to evaluate the contribution from the fiber, which is shown to be 1 by lemma 4.4.3 (see [16] for the details)

theorem 4.4.5 is a crucial tool for relating the invariants of a manifold to those of its associated split manifold. By applying this result to the split manifold M^Δ associated with M, we arrive at the main integrality theorem.

Theorem 4.4.6: Integrality Theorem

Let M be a compact almost complex manifold of complex dimension n.

1. The Todd genus $T(M)$ multiplied by 2^n is an integer.
2. More generally, for any classes $b_1, \dots, b_r \in H^2(M, \mathbb{Z})$, the virtual Todd genus $T(b_1, \dots, b_r)_M$ multiplied by 2^{n-r} is an integer.

***Proof*:**
Theorem 4.4.5 implies that $T(M) = T(M^\Delta)$. For a split manifold, $2^n T(M^\Delta)$ can be expressed as a sum of virtual indices by applying proposition 4.3.5 with $y = 1$, completing the proof.

Theorem 4.4.7: Multiplicative Property of the Generalized Todd Genus

The multiplicative property extends to the generalized T_y-genus. For a fiber bundle M^Δ over M with fiber $\mathbf{F}(q)$, we have:

$$T_y(M^\Delta) = T_y(M) \cdot T_y(\mathbf{F}(q))$$

This property also holds for fiber bundles where the fiber is a complex projective space $\mathbb{C}\mathrm{P}^{q-1}$. Specifically, if M^Δ is a fiber bundle over M with fiber $\mathbb{C}\mathrm{P}^{q-1}$ associated with a (q)-dimensional complex vector bundle, then:

$$T_y(M^\Delta) = T_y(M) \cdot T_y(\mathbb{C}\mathrm{P}^{q-1})$$

***Proof*:**
Combine the above results with section 4.3.

5

Reduction via Cobordism

In this section, the necessary cobordism concepts are covered to show the spin index M of a complex manifold is a cobordant invariant and is defined using Pontryagin classes. Due to the length of this paper, the key proof showing the isomorphism between the geometric and algebraic cobordism ring will be omitted; the proof can be found in [16, section 7].

Note that the Hirzebruch Riemann Roch was generalized by Atiyah and Singer to the *Atiyah-Singer Index Theorem*, see [11]. In [11], they replaced the use of the 4-periodicity of Pontryagin classes with the 4-periodicity of *Bott periodicity* (defined in my note in [12]) allowing for a broader application to topological spaces. This paper sticks to the cobordic method for its connections to recent development in mathematics (ex. derived algebraic cobordism [14], connection to generalized Poincaré Conjecture [10], etc.)

5.1 (Oriented) Cobordism Ring

By Stoke's Theorem, if ω is a form on an orientable manifold M with boundary ∂M, then:

$$\int_{\partial M} \omega = \int_M d\omega$$

Thus, figuring out the possible manifolds N such that $N = \partial M$ is a worth-while endeavor. With a bit of elementary effort, it can be shown that $N = \mathbb{R}\mathrm{P}^2$ is not the boundary of any 3-manifold M, and so the answer is false in general. However, $\mathbb{R}\mathrm{P}^2$ is not orientable, so does the restriction to orientable manifolds guarantee the existence? It is easy to see it is not the case by just complexifying and considering $\mathbb{C}\mathrm{P}^2$. Thus this spurs the question of trying to classify manifolds "up to boundary". This leads to the following notion:

Definition 5.1.1: Cobordant

let M, N be two topological n manifolds. Then M, N are *cobordant* if there exists a manifold C such that

$$\partial C = M \sqcup N$$

If $M = \partial C$, then M is said to be null-cobordant (cobordant to the empty set).

The cobordism operation defines a natural ring structure (addition by disjoint union, multiplication under direct product), inspiring to find which ring it is isomorphic to as well as find generators of the ring. This culminates in two important results:

1. The cobordism ring Ω is isomorphic to $\mathbb{Q}[x_1, x_2, ..., x_n, ...]$ where $[\mathbb{C}\mathrm{P}^i] \mapsto x_i$

2. The *spin index* of Riemann manifold (defined in section 5.2) is a homomorphism from the cobordism ring to $\mathbb{Q}$

The key result this gives us is that it represents all cobordism equivalence classes using Pontryagin classes. Recall that the T_1-characteristic is equal to a combination of Pontryagin classes (proposition 4.1.9). Theorem 6.2.1 will show that χ_1 is equal to the spin index which will be shown to be a cobordism invariance, thus the spin index is a key part in connection the two characteristics.

In this section, the manifolds shall be real manifolds. Let V^n be an oriented compact differentiable manifold where n is the dimension. As V is compact and orientable, it is a cycle, and by cohomology theory $H^n(V^n, \mathbb{Z})$ is generated by one element. The value of the n-dimensional cohomology class x on the fundamental cycle V^n is denoted $x[V^n]$. If the coefficients of the cohomology group are in A, then:

$$x \in H^n(V^n, A) \qquad x[V^n] \in A$$

We can extend this by tensoring by a group B so that :

$$x[V^n] \in A \otimes B \qquad x \in H^n(V^n, A) \otimes B \cong H^n(V^n, B)$$

For our purposes, let $n = 4k$ and let $p_i \in H^{4i}(V^{4k}, \mathbb{Z})$ be the Pontryagin classes of V^{4k} (i.e. of its tangent bundle). We get a natural grading by the indices i so that given $k = j_1 + j_2 + \cdots + j_r$ we get the product $p_{j_1} \cdots p_{j_r}$ and an integer (as $A = \mathbb{Z}$):

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

Let $\pi(k)$ be the number of distinct partitions of k.

Definition 5.1.2: Pontryagin Numbers and Chern Number

The *Pontryagin numbers* are the $\pi(k)$ numbers :

$$p_{j_1}\cdots p_{j_r}[V^{4k}]$$

where $n = 4k$. If n is not divisible by 4, then the Pontryagin number is 0.

Similarly the *Chern numbers* of a complex manifold M (with an almost complex structure so that we are given a particular orientation) are all:

$$c_{j_1}\cdots c_{j_r}[V^{2k}]$$

where $n = j_1 + j_2 + \cdots + j_r$ and the j_i run through all partitions.

By definition of the Euler class, if M_n is an oriented almost complex manifold, $c_n[M_n] = \chi(M_n, TM_n)$, showing that the Pontryagin/Chern numbers are "generalizations" of the Euler characteristic. We shall focus on Pontryagin numbers as we shall need them for the classification result.

We can define an equivalence relation where two manifolds $M \sim_p N$ if they have the same dimension and are the same Pontryagin number. Note that if the dimension is not divisible by 4, all manifolds are in the same equivalence relation given by $\sim_p$ as they are all 0.

If M, N are disjoint, the Pontryagin numbers of $M \sqcup N$ is the sum of the Pontryagin numbers of M, N; this comes straight from the properties of characteristic classes. Let us show the same works for the product of manifolds. Now let M, N be smooth $n = 4k$ dimensional manifolds. Then by classical differential geometry we have $T(M \times N) = \pi_M^*(TM) \times \pi_N^*(TM)$. Then:

$$(\pi_M^*(x)\pi_N^*(y))[M \times N] = x[M] \cdot y[N] \qquad \forall x \in H^n(M, \mathbb{Z}) \otimes B,\ y \in H^m(N, \mathbb{Z}) \otimes B$$

Thus, we can extend this results to characteristic sequences of Pontryagin classes. Considering the Pontryagin classes of M, N and $M \times N$:

$$\begin{gathered} 1, p_1, p_2, \ldots \\ 1, p_1', p_2', \ldots \\ 1, p_1'', p_2'', \ldots \end{gathered}$$

By the properties of characteristic classes we get:

$$1 + p_1'' + p_2'' + \cdots = (1 + p_1 + p_2 + \cdots)(1 + p_1' + p_2' + \cdots)$$

Or, as a generating function:

$$\sum_{k=0}^{\infty} p_k'' z^k = \sum_{i=0}^{\infty} \pi_M^*(p_i) z^i \sum_{j=0}^{\infty} \pi_N^*(p_j') z^j \text{ mod torsion.}$$

Thus, define the following:

Definition 5.1.3: Algebraic K-genus

Let $\{K_j\}$ be an m-sequence and let M be a 4ℓ-manifold. Then the *algebraic K-genus* of M is evaluated is:

$$K(M) = K_\ell(p_1, ..., p_\ell)[M] = \int_M K_\ell(p_1, ..., p_\ell)$$

where $p_1, ..., p_\ell$ are the Pontryagin classes of TM.

Proposition 5.1.4: Properties of Algebraic Genus

Let $\{K_\ell\}$ be an m-sequence, M, N be oriented 4ℓ-manifolds. Then:

1. $K(M \sqcup N) = K(M) + K(N)$
2. $K(M \times N) = K(M) \cdot K(N)$
3. $K(-M) = -K(M)$ where $-M$ is the reverse orientation put on M.

Thus, $K(-)$ respects the equivalence relation $\sim_p$.

***Proof*:**
(1) and (2) follow from the above discussion, (3) follows elementary cohomology.

Definition 5.1.5: Algebraic Oriented Cobordism Ring

Let:

$$\widetilde{\Omega}^n = \left\{[M]_p \;\middle|\; M \text{ an oriented } n\text{-manifold}\right\}$$

where the subscript p is dropped when it is clear. This ring is called the *algebraic cobordism ring*. By proposition 5.1.4, we can define the graded ring:

$$\widetilde{\Omega}^* = \bigoplus_{n=0}^{\infty} \widetilde{\Omega}^n$$

where:

$$\widetilde{\Omega}^n \widetilde{\Omega}^m \subseteq \widetilde{\Omega}^{nm}$$

Note that $\widetilde{\Omega}^n = 0$ if $4 \nmid n$ and $\widetilde{\Omega}^*$ is a commutative graded (torsion-free) ring.

The ring is called “algebraic” as it was defined using the algebraic properties of Pontryagin classes, we shall soon define the “geometric” cobordant ring that is the equivalence classes of cobordant manifolds and show they are isomorphic. For now, let us characterize the algebraic structure of the algebraic cobordism ring: the goal is to show that

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, ...]$$

To that end, we must find which elements of $\widetilde{\Omega}$ can represent each x_i. There is no natural choice, hence we define candidate sequences of elements given by manifolds $V^4, V^8, V^{12}, ...$ that can be chosen to form this basis. We shall that the complex projective spaces of even dimension shall satisfy these required conditions. Let us start by recalling that the Pontryagin class of V^{4k} can be written in indeterminate z and decomposed into:

$$1 + p_1 z + \cdots + p_k z^k = \prod_{i=1}^{k}(1 + \beta_i z)$$

Define:

$$s(V^k) = (\beta_1^k + \cdots + \beta_k^k)[V^{4k}]$$

By some symmetry function theory, we can think of this as needing the appropriate homogeneity and dimension properties, and hence if they are nonzero for each V^{4k} they are a desired list of representatives:

Definition 5.1.6: Basis Sequence

A sequence of manifold $V^4, V^8, V^{12}, ...$ is called a *basis sequence* if:

$$s(V^{4k}) \neq 0$$

Theorem 5.1.7: Basis Sequence Correspondence

Let $\{V^{4k}\}$ be a basis sequence and B a [commutative] ring containing $\mathbb{Q}$. Then the set of sequences $\{a_k\}_k$ where $a_k \in B$ are in bijective correspondence to the set of m-sequences $\{K_j(p_1, ..., p_j)\}$ with coefficients in B where $K(V^{4k}) = a_k$.

***Proof*:**
By proposition 4.1.2, it suffices to show that there is exactly one power series $Q(z)$ such that, for each V^{4k} in the sequence with Pontryagin where:

$$a_k = K_k[V^{4k}],$$

where K_k is the coefficient of z^k in $\prod_{i=1}^{k} Q(\beta_i z)$. This equation can be written

$$a_k = s(V^{4k})b_k + \text{polynomial in } b_1, b_2, \ldots, b_{k-1} \text{ of weight } k$$

The above polynomials depends only on V^{4k} and has integer coefficients. Then the coefficients b_k can be deduced via induction.

Theorem 5.1.8: Complex Projective Forms Basis Sequence

Let $\mathbb{C}\mathrm{P}^{2k}$ be the $2k$-dimensional complex projective space. Then they form a basis sequence for which

$$s(\mathbb{C}\mathrm{P}^{2k}) = 2k + 1$$

***Proof*:**
Let $h \in H^2(\mathbb{C}\mathrm{P}^{2k}, \mathbb{Z})$ be a generator. Then the Pontryagin class of $\mathbb{C}\mathrm{P}^{2k}$ is $(1+h^2)^{2k+1}$. The m-sequence of the power series $1+z^k$ defines a genus which for V^{4k} has the value $s(V^{4k})$ and which takes the value $2k+1$ on $\mathbb{C}\mathrm{P}^{2k}$, completing the proof.

Let us now consider $\widetilde{\Omega} \otimes \mathbb{Q}$, so we have rational coefficients: $\frac{a}{b}[V^{4k}]$. Certainly the Pontryagin number, K-genus, and $s(V^{4k})$ are all still well-defined (where the K-genus is well-defined as we required B to contain $\mathbb{Q}$). Now, the Pontryagin number of an element of $\widetilde{\Omega} \otimes \mathbb{Q}$ can be a non-integer. This new ring can still be represented using a basis sequence:

Theorem 5.1.9: Basis Sequence With Rational Numbers Correspondence

Let $\{V^{4k}\}$ be a basis sequence of oriented manifolds, and let $(j) = (j_1, ..., j_r)$ be a partition of k. Let $V_{(j)} = V^{4j_1} \times \cdots \times V^{4j_r}$. Then every $\alpha \in \widetilde{\Omega} \otimes \mathbb{Q}$ can be uniquely represented by the sum:

$$\alpha = \sum_{(j) \in \pi(j)} r_{(j)}[V_{(j)}] \qquad r_{(j)} \in \mathbb{Q}$$

where the sum goes over all partitions of k. Furthermore, given a system $\{a_{(j)}\}$ of rational numbers, there exists an element $\alpha \in \widetilde{\Omega} \otimes \mathbb{Q}$ where

$$p_{(j)}[\alpha] = \alpha_{(j)}$$

***Proof*:**
This is a linear algebra result. It suffices to prove that a sum $\sum r_{(j)}(V_{(j)})$ over all partitions (j) of k is zero if and only if $r_{(j)} = 0$ for each (j). Suppose that $\sum r_{(j)}(V_{(j)}) = 0$. Let $q_1, q_2, q_3, \ldots$ be a sequence of indeterminates. Then we can find for each integer $t \geq 0$ an m-sequence which takes the value q_k^t on V^{4k}. This implies that

$$\sum_{(j)} r_{(j)} q_{(j)} = 0,$$

where $q_{(j)}$ denotes the product $q_{j_1} q_{j_2} \ldots q_{j_r}$ for $(j) = (j_1, j_2, \ldots, j_r)$. Since the $q_{(j)}$ are pairwise distinct. But then using the Vandermonde determinant we can see that each $r_{(j)}$ vanishes, completing the proof.

Theorem 5.1.10: Isomorphism of Algebraic Cobordism Ring

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, ...]$$

***Proof*:**
Let $\{V^{4k}\}$ be an arbitrary sequence of oriented manifolds. What's left to show is that:

1. Given $\alpha = \sum_{(j)} r_{(j)}[V_{(j)}]$ then $s(\alpha) = r_k s[V^{4k}]$. Let $\{K_j\}$ be the m-sequence of the power series $1+z^k$. This m-sequence takes the value $s(\alpha)$ on elements $\alpha \in \Omega^{4k} \otimes Q$ and the value 0 on elements of $\Omega^{4j} \otimes Q$ with $1 \leq j < k$.

2. If for all k, every element of $\alpha \in \widetilde{\Omega}^{4k} \otimes \mathbb{Q}$ can be represented as a sum $\alpha = \sum_{(j)}[V_{(j)}]$, then $\{V^{4k}\}$ is a basis sequence. For the sake of contradiction, suppose that for some k, $s(V^{4k}) = 0$. Then, by (1), $s(\alpha) = 0$ for all $\alpha \in \Omega^{4k} \otimes Q$. But by theorem 5.1.8, $s(\mathbb{C}\mathrm{P}^{2k}) = 2k + 1$ – a contradiction.

Thus, we have a injective/surjective ring homomorphism, giving an isomorphism.

Lastly, the K-genus can be characterized as as representing homomorphism from the algebraic cobordism ring to $\mathbb{Q}$:

Theorem 5.1.11: Characterizing Rational Homomorphisms

Let $\varphi : \widetilde{\Omega} \otimes \mathbb{Q} \to \mathbb{Q}$ be a homomorphism. Then there exists an m-sequence K such that $\varphi(-) = K(-)$.

***Proof*:**
The proof in theorem 5.1.10 applies to any basis sequence with the appropriate properties; using this one can create an m-sequence such that $K[V^{4k}] = \varphi(x_i)$.

This fully encodes the algebraic information. Let us now define the geometric counter-part

Definition 5.1.12: Geometric Oriented Cobordism Ring

Two oriented smooth n-dimensional manifolds M^n, N^n are *cobordant* if there exist a manifold C such that

$$\partial C \cong (M^n \sqcup -N^n)$$

where $-N^n$ is the reverse orientation; if such a C exists it defines an equivalence relation $M \sim_c N$ which respects disjoint union, reverse orientation, and direct product; $\sqcup, -, \times$. Then Ω^n is the collection of such equivalence classes and:

$$\Omega = \bigoplus_n \Omega^n$$

is the *geometric cobordism ring.*

Definition 5.1.13: Geometric Genus

Let Ω be the oriented geometric cobordism ring. Then a *Geometric genus* (with respect to B) is a B-homomorphism:

$$\varphi : \Omega \to B$$

After showing Ω and $\widetilde{\Omega}$ are isomorphic, we shall see that all geometric genii are representable by an m-sequence .

Certainly this ring is graded, and it is *anti-commutative*, namely:

$$\alpha \cdot \beta = (-1)\beta \cdot \alpha \qquad \alpha \in \Omega^n,\ \beta \in \Omega^m$$

Theorem 5.1.14: Algebraic - Geometric Cobordism Ring Isomorphism

$$\Omega \cong \widetilde{\Omega}$$

***Proof*:**
see [16, section 7]

5.2 Intersection Form: Spin Index and Hirzebruch Signature Theorem

On any $4k$-dimensional manifold, there is a natural generalization of the symplectic form known as the *intersection form*: given a choice of $x, y \in H^{2k}(M, \mathbb{R})$, Define:

$$Q(x, y) = x \smile y[M^n]$$

As $x \smile y \in H^4k$:

$$(x \smile y) = (-1)^{k^2}(y \smile x) = (y \smile x)$$

showing that the form is *symmetric* (rather than anti-symmetric), which is why we are working with 4-dimensional manifolds. Thus, the eigenvalues of the corresponding matrix representations are always *real* and there always exists an orthogonal basis. If e_p and e_n represents the number of positive and negative eigenvalues, then:

Definition 5.2.1: Spin Index

$$\tau(Q) = e_p - e_n$$

The most important result about the index is that it is a *genus*:

Theorem 5.2.2: Spin Index is Genus

Let τ be the index function. Then:

1. $\tau(M \sqcup N) = \tau(M) + \tau(N)$
2. $\tau(M) \times N) = \tau(M) \cdot \tau(N)$
3. $\tau(-M) = -\tau(M)$
4. If M is nullcobordant, $\tau(M) = 0$

***Proof*:**
The proof for (2), (3), (4) are long; a summary is given (see [15] for the details)

1. follows by definition.

2. Let $M^{4k} = V^n \times W^m$. Then

$$H^{2k}(M^{4k},\mathbb{R}) \cong \sum_{s=0}^{2k} H^s(V^n,\mathbb{R}) \otimes H^{2k-s}(W^m,\mathbb{R}). \tag{5.1}$$

Let $x, y \in H^{2k}(M^{4k},\mathbb{R})$ be *orthogonal* if $xy[M^{4k}] = 0$. Introduce bases $\{v_i\}$ for $H^s(V^n,\mathbb{R})$ and $\{w_j\}$ for $H^t(W^m,\mathbb{R})$ such that $v_i^s v_j^{n-s}[V^n] = \delta_{ij}$ for $s \neq \frac{n}{2}$ and $w_i^t w_j^{m-t}[W^m] = \delta_{ij}$ for $t \neq \frac{m}{2}$.

Now, consider the group $A = H^{\frac{n}{2}}(V^n,\mathbb{R}) \otimes H^{\frac{m}{2}}(W^m,\mathbb{R})$, taking $A = 0$ if n and m are odd. Then A is orthogonal to the subgroup B of $H^{2k}(M^{4k},\mathbb{R})$, which consists of all elements of the summation in equation (5.1) in which no elements of A occur. As a basis for the group B, take:

$$\{v_i^s \otimes w_j^{2k-s}\}, (0 \leq s \leq n, s \neq \frac{n}{2})$$

then:

$$\begin{aligned}(v_i^s \otimes w_j^{2k-s})(v_{i'}^{s'} \otimes w_{j'}^{2k-s'})[M^{4k}] &= \pm 1 \text{ if } s + s' = n, i = i', j = j' \\ &= 0 \text{ otherwise.}\end{aligned}$$

It follows that, with respect to this basis, the restriction of the bilinear form $xy[M^{4k}]$ to B is represented by a matrix with blocks

$$\pm\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

down the diagonal and zero elsewhere. Therefore the index of the restriction of $xy[M^{4k}]$ to B is 0. Since A and B are orthogonal, $\tau(M^{4k})$ is equal to the index $\tau(A)$ of the restriction of the bilinear form $xy[M^{4k}]$ to A. There are now two cases to consider. If n and m are not divisible by 4 then $\tau(A) = 0$. If n and m are divisible by 4 then certainly: $\tau(A) = \tau(V^n) \cdot \tau(W^m)$.

3. Suppose that V^{4k} is the oriented boundary of M^{4k+1}, and that $j : V^{4k} \to M^{4k+1}$ is the embedding. Consider the following commutative diagram (exercise to show it's commutative, use Poincré duality):

$$\begin{tikzcd} H^{2k}(M^{4k+1},\mathbb{R}) \arrow[r, "j^*"] \arrow[d, "\cong"] & H^{2k}(V^{4k},\mathbb{R}) \arrow[r] \arrow[d, "i"] & H^{2k+1}(M^{4k+1} \bmod V^{4k},\mathbb{R}) \arrow[d, "\cong"] \\ H_{2k+1}(M^{4k+1} \bmod V^{4k},\mathbb{R}) \arrow[r] & H_{2k}(V^{4k},\mathbb{R}) \arrow[r, "j_*"] & H_{2k}(M^{4k+1},\mathbb{R}) \end{tikzcd}$$

Let A^{2k} be the image of j^* in $H^{2k}(V^{4k},\mathbb{R})$ and let K_{2k} be the kernel of j_* in $H_{2k}(V^{4k},\mathbb{R})$. Then A^{2k} is dual to $H_{2k}(V^{4k},\mathbb{R})/K_{2k}$ under the duality between $H^{2k}(V^{4k},\mathbb{R})$ and $H_{2k}(V^{4k},\mathbb{R})$. On the other hand, the diagram implies that, for $x \in H^{2k}(V^{4k},\mathbb{R})$,

$$x \in A^{2k} \Leftrightarrow i(x) \in K_{2k}.$$

Therefore, if $b_{2k} = \dim H_{2k}(V^{4k},\mathbb{R})$ is the $2k$-th Betti number of V,

$$\dim A^{2k} = \dim K_{2k} = b_{2k} - \dim K_{2k}$$

and

$$\dim A^{2k} = \frac{1}{2} b_{2k}.$$

If $x = j^* y \in A^{2k}$, and v is the fundamental cycle of V^{4k} then $x^2[V^{4k}] = (j^* y^2)[v] = (y^2)[j_* v] = 0$. Therefore, the cone

$$\{x \in H^{2k}(V^{4k}, \mathbb{R}); x^2[V^{4k}] = 0\}$$

contains the linear subspace A^{2k} of dimension $\frac{1}{2}b_{2k}$. It follows that the bilinear form $xy[V^{4k}]$ has $p^+ = p^-$ and hence that $\tau(V^{4k}) = 0$, completing the proof.

Theorem 5.2.3: Hirzebruch Index Theorem

Let M be a real differentiable oriented manifold, and let $\tau(M)$ be its index. Then:

1. if $\dim_{\mathbb{R}} M \not\equiv 0 \bmod 4$, then
$$\tau(M) = 0$$
2. if $\dim_{\mathbb{R}} M \equiv 0 \bmod 4$, then:
$$\tau(M) = L_k(p_1, ..., p_k)[M]$$

***Proof*:**
We must show they match on the generators $\mathbb{C}\mathrm{P}^n$. By theorem 5.1.11, it suffices to show that $\tau(M_{4k}) = 1$ since by proposition 4.1.4:

$$L_k(p_1, ..., p_k)[\mathbb{C}\mathrm{P}^{2k}] = 1$$

Take as basis sequence $M_{4k} = \mathbb{C}\mathrm{P}^{2k}$. Recall that index is defined by the intersection form $Q(x, y) = \int_{\mathbb{C}\mathrm{P}^{2k}} x \smile y$ on the middle cohomology group $H^{2k}(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z})$. The cohomology ring of complex projective space is:

$$H^*(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z}) \cong \mathbb{Z}[h]/\langle h^{2k+1}\rangle$$

where $h \in H^2(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z})$ is the Poncaré dual to a hyperplane. The middle cohomology group $H^{2k}(\mathbb{C}\mathrm{P}^{2k}; \mathbb{Z})$ one-dimensional real vector space with the single basis element $\{h^k\}$.

The matrix representing the intersection form Q in this basis is a 1×1 matrix whose only entry is $Q(h^k, h^k)$. Computing its value:

$$Q(h^k, h^k) = \int_{\mathbb{C}\mathrm{P}^{2k}} h^k \smile h^k = \int_{\mathbb{C}\mathrm{P}^{2k}} h^{2k}$$

By the definition of the generator h, the integral of the top-degree class h^{2k} over the fundamental class of the manifold is 1. The matrix representation is there fore 1 by 1 and is simply (1). Thus:

$$\tau(\mathbb{C}\mathrm{P}^{2k}) = e^+ - e^- = 1 - 0 = 1$$

as we sought to show.

Thus, by theorem 5.1.11, there is an m-sequence that represents this map:

Theorem 5.2.4: Index Correspondence M-sequence

The intext τ is represented by the m-sequence:

$$\frac{\sqrt{z}}{\tanh \sqrt{z}}$$

***Proof*:**
Follows immediately from theorem 5.2.3 and proposition 4.1.4.

5.3 Index and Splitting Principle

Just like the Euler Characteristic and T-characteristic, the index $\tau(M)$ can be written in terms of all 1-dimensional submanifolds by considering a filtration $V_r \subseteq V_{r-1} \subseteq \cdots \subseteq V_1 \subseteq M$ each submanifold being codimension 1 with respect to the next one. For the following results, we require a bit more theory of Chern classes:

Theorem 5.3.1: 1st Chern Class of Codimension 1 Complex Manifold

Let $j : X \to Y$ be the embedding of an oriented compact real $(m-2)$-dimensional differentiable manifold X in an oriented compact m-dimensional differentiable manifold Y. Let $h_X \in H^2(Y, \mathbb{Z})$ be the cohomology class defined, with respect to the given orientations, by the $(m-2)$-dimensional homology class represented by X. Let $N_{X/Y}$ be the normal bundle of X in Y. Then

$$c_1(N_{X/Y}) = j^*(h_X)$$

***Proof*:**
[16, theorem 4.8.1]

Theorem 5.3.2: Relating Chern and Pontryagin Classes

Let ξ be a $\mathbf{U}(q)$-bundle over M. Then

$$\begin{aligned}\tilde{p}(\varrho(\xi)) &= 1 - p_1(\varrho(\xi)) + p_2(\varrho(\xi)) - p_3(\varrho(\xi)) + \cdots \\ &= (1 + c_1(\xi) + c_2(\xi) + \cdots)(1 - c_1(\xi) + c_2(\xi) - \cdots).\end{aligned}$$

***Proof*:**
[16, theorem 4.5.1]

Theorem 5.3.3: Relation For Almost Complex Manifolds

The Chern classes c_i of the almost complex manifold M are related to the Pontryagin classes p_i of M (regarded as a differentiable manifold) by the equation

$$\tilde{p} = \sum_{i=0}^{\infty}(-1)^i p_i = \sum_{i=0}^{\infty} c_i \sum_{j=0}^{\infty}(-1)^j c_j.$$

***Proof*:**
[16, theorem 4.6.1]

5.3.1 Virtual index To Intersection Theory Objects

Let M^n be a compact oriented smooth manifold, $j : V^{n-k} \to M^n$ an embedding of the oriented submanifold V^{n-k}. Let $N_{M/V}$ be the normal bundle of V with respect to M. if j^* is the cohomology pullback

Then by some basic Riemannian geometry we have that we can embed TV into TM and we have:

$$TM = TV \oplus NV$$

where $NV := N_{M/V}$ is the normal bundle. Thus, of $\mathfrak{p}$ is the total Pontryagin class:

$$j^*\mathfrak{p}(TM) = \mathfrak{p}(TV) \cdot \mathfrak{p}(NV) \text{ mod torsion}$$

By theorem 5.3.3:

$$\mathfrak{p}(NV) = j^*(1 + v^2)$$

Thus:

$$\mathfrak{p}(V) = j^*\left((1 + v^2)^{-1}\mathfrak{p}(M)\right)$$

Now, taking the L-genus and taking advantage of the multiplicative and naturality property we get:

$$L(j^*(TM) = j^*(L(TM)) = L(TV \oplus NV) = L(TV) \cdot T(NV)$$

Thus we get:

$$j^*(L(TM)) = L(TV)) \cdot L(NV)$$

Moving things around:

$$L(TV) = j^*(L(TM)) \cdot L(NV)^*$$

Since we know the normal bundle has a single Pontryagin root $\gamma = v^2$, we get:

$$L(NV) = \frac{v}{\tanh(v)}$$

Finally, as $L(TM) = \sum_i^{\infty} L_i(TM) = \sum_i^{\infty} L_i(\mathfrak{p}_1(TM), ..., \mathfrak{p}_i(TM))$, we get:

$$L(TV) = \sum_{i=0}^{\infty} L_i(\mathfrak{p}_1(V), ..., \mathfrak{p}_i(V)) = j^*\left[\frac{\tanh(v)}{v}\sum_{i=0}^{\infty} L_i(\mathfrak{p}_1(M), ...\mathfrak{p}_i(M)\right]$$

Using this, we take advantage of Hirzebruch's Signature Theorem to get $\tau(V)$. First, by Poincaré duality, if $x \in H^{n-2}(M, \mathbb{Z}) \otimes \mathbb{Q}$, then:

$$j^*(x)[V] = vx[M]$$

Now, by Hirzebruch's signature theorem:

$$\tau(V) = \int_M \tanh \sum_{i=0}^{\infty} L_i(p_1(M), ..., p_i(M))$$

Using the computational methods available to us we get:

$$\begin{aligned}
n &= 2 \qquad \tau(V^0) = \int_{M^2} v \\
n &= 6 \qquad \tau(V^4) \int_{M^6} \frac{1}{3}(-v^3 + p_1 v) \\
n &= 10 \qquad \tau(V^8) = \int_{M^{10}} \frac{1}{45}(6v^5 - 5p_1 v^3 + (7p_2 - p_1^2)v)
\end{aligned}$$

Now here is where interpreting Chern classes naturally in intersection theorem comes in. Say we have a collection $v_1, ..., v_r \in H^2(M, \mathbb{Z})$. v_1 is associated to V^2. Now say v_2 is restricted to V^{n-2} and is associated to $V^{n-4} \subseteq V^{n-2}$. Finally, v_r is restricted to $V^{n-2(r-1)}$ and is associated to V^{n-2r}. Then if $j : V^{n-2r} \to M^n$, then:

$$j^*(x)[V^{n-2r}] = v_1 v_2 \cdots v_r x[M^n]$$

Then applying the L-power series successfully, we get the index formula:

$$\tau(V^{n_2 r}) = \int_{M^n} \tanh v_1 \tanh v_2 \cdots \tanh v_r \sum_{i=0}^{\infty} L_i(p_1(M^n), ..., p_i(M^n))$$

Then for sure by [12], it was proven that every 2-dimensional integral cohomology class of a compact oriented differentiable manifold M^n can be represented by a submanifold $V^{n-2} \subseteq M^n$. By applying this result iteratively, we can justify the above equation. Thus, $\tau(V^{n-2r})$ depends only on the (unordered) choice of cohomology classes $v_1, v_2, ..., v_r$. Thus, we can write:

Definition 5.3.4: Virtual Index

$$\tau(v_1, ..., v_r) = \int_{M^n} \tanh v_1 \tanh v_2 \cdots \tanh v_r \sum_{i=0}^{\infty} L_i(p_1(M^n), ..., p_i(M^n))$$

Every virtual index occurs as the index of a submanifold of M^n and is thus an integer.

Recall tanh satisfies:

$$\tanh(u + v) = \frac{\tanh(u) + \tanh(v)}{1 + \tanh(u)\tanh(v)}$$

or as Hirzebruch writes it:

$$\tanh(u + v) = \tanh(u) + \tanh(v) - \tanh(u)\tanh(v)\tanh(u + v)$$

This immediately gives:

Theorem 5.3.5: Functional Equation of Virtual Index

The virtual index is a function which associates an integer $\tau(v_1, v_2, \ldots, v_r)$ to each (unordered) r-ple $(v_1, v_2, \ldots, v_r)$ of 2-dimensional integral cohomology classes of a compact oriented differentiable manifold M^n. The function τ is zero if $n - 2r \not\equiv 0$ modulo 4, if $2r > n$, or if one of the classes v_i is zero. It satisfies the functional equation

$$\tau(v_1, \ldots, v_r, u + v) = \tau(v_1, \ldots, v_r, u) + \tau(v_1, \ldots, v_r, v) - \tau(v_1, \ldots, v_r, u, v, u + v)$$

If $n = 4k + 2$:

$$\tau(u + v) = \tau(u) + \tau(v) - \tau(u, v, u + v)$$

6

Riemann Roch Theorems

In this final chapter, we shall prove the Hirzebruch Riemann Roch. We start off with giving a proof of the classical Riemann-Roch theorem for curves and also a simple extension to the Riemann-Roch for vector bundles over curves as it provides insight on the algebra of increasing the dimension of the bundle beyond the dimension of the base space. We shall then start the proof of Hirzebruch Riemann Roch for line bundles by connection the χ_y-characteristic to the T_y-characteristic by showing they are both equal to the spin index τ when $y = 1$, and shall force their coefficients to be equal and hence also agree at $y = -1$. It is then extended to Hirzebruch Riemann Roch for vector bundles by one again taking advantage of the splitting principle, giving the final complete proof.

The next section will be dedicated to very rapidly going over the Grothendieck Riemann Roch. As a detailed proof would require build-up in intersection theory, the proof shall assume all the necessary background knowledge that can be found in [20].

The final section covers references to papers that further generalized the Riemann-Roch theorem in various different analytic, geometric, and algebraic directions.

6.1 Classical Riemann Roch Proofs

Let us start with the most classical version of the Riemann-Roch Theorem from an algebraic perspective, namely it will presented similarly to the proof in Hartshorne (see [34]) with some extra details added as compared to Hartshorne so that we don't reference some of the heavy machinery he built up (though Serre Duality is essentially unavoidable in this approach). By the correspondence between Riemann surfaces and algebraic curves, there is a purely geometric-analytic proof of this (this can be found in my notes in [32] or in A. Griffiths, Phillip excellent book *Introduction to Algebraic Curves* [28])

Theorem 6.1.1: RR For Riemann Surface with Line Bundles

Let M be a compact Riemann surface and $\mathcal{L}$ a complex analytic line bundle on M. Then

1. There exists a Cartier divisor D such that $\mathcal{L} \cong \mathcal{O}_M(D)$ so that $\deg(\mathcal{L}) := \deg(D)$
2. The Riemann-Roch equation holds:

$$\dim_{\mathbb{C}} H^0(M, \mathcal{L}) - \dim_{\mathbb{C}} H^0(M, \omega_M \otimes \mathcal{L}^\vee) = \deg(\mathcal{L}) + 1 - g$$

where $g = \dim H^0(M, \omega_M) = \dim H^1(M, \mathcal{O}_M)$ is the genus of M.

***Proof*:**
Any compact Riemann surface M is an algebraic variety, in particular a curve, and by [3, section 6.1] there is an embedding $M \hookrightarrow \mathbb{P}^N_{\mathbb{C}}$ for some N. By theorem 3.2.10 we have that $\mathcal{L}$ is isomorphically pushed forward through these maps and is an algebraic line bundle (potentially tensored with another algebraic bundle). As M is a variety, by [3, section 4.3] there exists a (Weil) divisor D such that $\mathcal{L} = \mathcal{O}_M(D)$. Let us show $\deg(\mathcal{L}) = \deg(D)$ is well-defined, i.e. independent of linear equivalence. Let $f \in \mathcal{M}(M)$. Then as shown in [32, section 6.3] there exists a branch covering $f : M \to \mathbb{P}^1_{\mathbb{C}}$ where:

$$\#(f^{-1}(0) = \#(f^{-1}(\infty)) = \text{deg. of map}$$

Hence:

$$\deg(f)\#(f^{-1}(0) - \#(f^{-1}(\infty)) = 0$$

Thus, $D \sim E$ implies $\deg(D) = \deg(E)$, hence $\deg(\mathcal{L})$ is well-defined

For the 2nd point, if $D = 0$, then $\mathcal{O}_M(0) = \mathcal{O}_M$, and $H^0(M, \mathcal{O}_M) = \mathbb{C}$ so $l(0) = 1$ and $i(0) = g$ and

$$l(0) - i(0) = 1 - g = \deg(0) + 1 - g$$

by [32, section 6.3], hence assume $D \neq 0$ (though the beginning of this discussion doesn't need this assumption). First use Serre Duality to get:

$$H^0(M, \omega_M \otimes \mathcal{L}^\vee) \cong H^1(M, \mathcal{L})^\vee$$

Reducing the right hand side to the Euler characteristic $\chi(M, \mathcal{O}_M(D))$. To get the right hand side, we take advantage of additivity of χ. For any point $P \in M$, we get an exact sequence:

$$0 \to \mathcal{O}_M(-P) \to \mathcal{O}_M \to \kappa_P \to 0$$

when κ_P is the skyscraper sheaf:

$$(\kappa_P)_x = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

Note that as this sheaf is flasque (flabby), H^i disappears for $i > 0$ so $\chi(M, \kappa_P) = 1$. Furthermore, as each point of D is disjoint, $\chi(M, \kappa_D) = n$.

Tensoring with $\mathcal{O}_M(D)$, and applying our theory from [3, section 4.3] we get:

$$0 \to \mathcal{O}_M(D - P) \to \mathcal{O}_M(D) \to \kappa_P \otimes \mathcal{O}_M(D) \to 0$$

Thus, by additivity of χ:

$$\chi(M, \mathcal{O}_M(D) = \chi(M, \kappa_P \otimes \mathcal{O}_M(D)) + \chi(M, \mathcal{O}_M(D-P))$$

Now, if $D > 0$, we can pick a point P appearing in D. Then $\deg(D-P) = \deg(D) - 1$. We can thus do induction on $\deg(D)$. If $\deg(D) = 0$, then $\mathcal{O}_M(D)$ has holomorphic functions, which as M is compact is only constant functions, and by the Brill-Noether Reciprocity ([32]) we have $i(D) = g$, giving the base case. Now suppose the result holds for $\deg(D) = n-1$ and consider $\deg(D) = n$. Then $\deg(D-P) = n-1$ from which we get by the inductive hypothesis

$$\deg(D-P) + 1 - g = \chi(M, \mathcal{O}_M(D-P))$$

adding back in P, we get:

$$\deg(D) + (-1+1) + 1 - g = \chi(M, \mathcal{O}_M(D))$$

giving the desired result.

For arbitrary D, we can split $D = D^+ - D^-$ where $D^+, D^- \geq 0$. Then we get a short exact sequence:

$$0 \to \mathcal{O}_M \to \mathcal{O}_M(D^-) \to \kappa_{D^-} \to 0$$

Then by additivity;

$$\chi(M, \mathcal{O}_M(D^-)) = \chi(M, \mathcal{O}_M) + \chi(M, \kappa_{D^-}) = 1 - g + \deg(D^-)$$

Now, tensor with $\mathcal{O}_M(D)$ to get the short exact sequence:

$$0 \to \mathcal{O}_M(D) \to \mathcal{O}_M(D + D^-) \to \kappa_{D^-} \to 0$$

where the last term is the same (as can be verified point-wise). Applying additivity of χ again we get:

$$\chi(M, \mathcal{O}_M(D + D^-)) = \chi(M, \mathcal{O}_M(D)) + \chi(M, \kappa_{D^-}) = \chi(M, \mathcal{O}_M(D)) + \deg(D^-)$$

As $D + D^- = D^+$, re-arranging we get:

$$\chi(M, \mathcal{O}_M(D)) = -\deg(D^-) + \chi(M, \mathcal{O}_M(D^+))$$

Applying our earlier result we get:

$$\chi(M, \mathcal{O}_M(D)) = -\deg(D^-) + \deg(D^+) + 1 - g \deg(D) + 1 - g$$

as we sought to show.

As $\mathcal{L}$ is a line bundle, we can take its value of the first Chern class $c_1(\mathcal{L}) \in H^2(M; \mathbb{Z})$. By letting $[M] \in H - 2(M; \mathbb{Z}) \cong \mathbb{Z}$ be the fundamental class, then:

$$\deg(\mathcal{L}) = c(\mathcal{L})[M] \in \mathbb{Z}$$

Then theorem 6.1.1 becomes:

$$\begin{aligned}\chi(M,\mathcal{L}) &= c_1(\mathcal{L})[M] + 1 - g \\ &= \left[c_1(\mathcal{L}) + \frac{1}{2}(2-2g)\right][M] \\ &= \left[c_1(\mathcal{L}) + \frac{1}{2}c_1(TM)\right][M]\end{aligned}$$

By Bass-Serre Cancellation (see [12, section 5.1]) or by some result from the theory of characteristic (see [12, section 3.2]), higher-dimensional vector bundles are trivial; in particular as $\dim_{\mathbb{C}} M = 1$, any vector bundle of dimension $n > 1$ will have trivial information cohomologically. Let us show this explicitly to further motivate HRR.

Theorem 6.1.2: Atiyah-Serre On Vector Bundles

Let M be either a compact complex manifold or an algebraic variety; in either case of dimension $d = \dim_{\mathbb{C}} M$. Let E be a rank $r > d$ vector bundle on M that is smooth or algebraic whether M is a manifold of variety respectively [a]. Then there is a trivial bundle $\mathbb{I}^{r-d}$ of rank $r - d$ that fits into the short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

and $\mathrm{rk}(E'') = d$.

[a] If algebraic, assume $\Gamma_{\mathrm{alg}}(M, \mathcal{O}_M(E)) \to E_x$ is surjective for all x;i.e. it is generated by its global sections

***Proof*:**
If E is smooth, then by partitions of unity it is always generated by its global sections. As algebraic vector bundles need not be fine, we must assume they are (for example $\mathcal{O}_{\mathbb{P}^1}(-1)$ does not satisfy this condition). Without loss of generality, we shall work over $\overline{k} = \mathbb{C}$.

Then, for each $x \in E$, $\Gamma(M, \mathcal{O}_M(E)) \twoheadrightarrow E_x$ surjective onto the fiber E_x. Then we can choose a finite dimensional subspace $W(x) \subseteq \Gamma(M, \mathcal{O}_M(E))$ as $W(x) \to E_x$ surjective. Then either $\mathbb{C}$- or Z-near x, $\Gamma(M, \mathcal{O}_M(E))$ surjects onto the fibers in the neighbourhood (namely, the determinant is smooth/regular, and hence we can follow a similar argument in both cases). As M is [quasi]-compact, there exists a finite open cover with this property.

Label these choices $W(i)$. Taking all their generators, we have a finite dimensional subspace $W \subseteq \Gamma(M, \mathcal{O}_M(E))$ such that $W \twoheadrightarrow E_x$ surjects for all $x \in M$ (namely $\sigma \mapsto \sigma(x)$ surjects). Let:

$$\ker(x) = \ker(W \to E_x)$$

Then as $\dim E_x = r$,

$$\dim \ker(x) = \dim W - r$$

As the dimension is constant, $\dim \ker(x)$ is independent of M, thus we can take their union over M. To be precise, let $\mathbb{P}(\ker(x)) \subseteq \mathbb{P}(W)$ be the projectivizations of $\ker(x) \subseteq W$ (i.e. collection of all lines). Then consider

$$Z = \cup_{x \in M} \mathbb{P}(\ker(x))$$

and takes its Z-closure if necessary. Then:

$$\dim Z = \dim M + \dim W - r - 1 = \dim W + d - r - 1$$

where the -1 comes from the fact that $\dim(\mathbb{P}(V)) = \dim V - 1$. Hence, we have:

$$\operatorname{codim}(Z \hookrightarrow \mathbb{P}(W)) = r - d$$

Then it is simple algebraic/affine geometry to see there exists a projective (linear) subspace $T \subseteq \mathbb{P}$ with $\dim T = r - d - 1$ such that

$$T \cap Z = \emptyset$$

Let $T = \mathbb{P}(S)$ for some subspace $S \subseteq W$ (so $\dim S - r - d$). Then by construction:

$$M \times S = M \times \mathbb{C}^{r-d} = \mathbb{I}^{r-d}$$

From this, we can define a map $\mathbb{I}^{r-d} \hookrightarrow E$ sending $(x, s) \in s(x) \in E$. As $T \cap Z = \emptyset$, $s(x)$ is never zero, and hence $\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$ has full rank at each point. Letting $E'' = E/\operatorname{im}(\mathbb{I}^{r-d} \hookrightarrow E)$, we get a vector bundle of dimension d and a short exact sequence:

$$0 \to \mathbb{I}^{r-d} \to E \to E'' \to 0$$

as we sought to show.

Now, as shown in chapter 2 we have that $c_i(\mathbb{I}^r) = 0$ for all i (it has trivial cohomology), so by additivity of characteristic classes we have that $c_1(E) = c_1(E'')$. As $c_1(E) = c_1\left(\bigwedge^r E\right)$, we get:

$$c_1(E) = c_1\left(\bigwedge^{\operatorname{rk}(E'')} E''\right)$$

Using this, we can prove the following generalization:

Theorem 6.1.3: RR For Riemann Surface With Vector Bundles

Let M be a compact Riemann surface and E a complex vector bundle over M of rank r. Then:

$$\chi(M, E) = c_1(E) + r(1 - g)$$

where $c_1(E)$ is the first Chern class of E.

***Proof*:**

Start with embedding M in $\mathbb{P}^N$ via an algebraic equation and pushforward E so that it is an algebraic vector bundle. Then E need not be generated by global holomorphic functions, however by twisting it enough

$$E(n) := E \otimes \mathcal{O}_M(n) \cong E \otimes \mathcal{O}_M^{\otimes n}$$

we shall have a very ample sheaf, and hence $E(n)$ shall have enough global section (see [3, section 4.4] for the details). From this, we can apply theorem 6.1.2:

$$0 \to \mathbb{I}^{r-1} \to E(n) \to E'' \to 0$$

where $\mathrm{rk}(E'') = 1$. Then twisting it by $\mathcal{O}_M(-n)$ we get:

$$0 \to \coprod_{r-1} \mathcal{O}_M(-n) \to E \to E''(-n) \to 0 \tag{6.1}$$

Then:

$$\chi(M, \mathcal{O}_M(E)) = \chi(M, E''(-n)) + \chi\left(\coprod_{r-1} \mathcal{O}_M(-n)\right)$$

As $\mathrm{rk}(E''(-n)) = 1$, by ordinary Riemann-Roch we have:

$$\chi(M, E''(-n)) = c_1(E''(-n)) + 1 - g$$

giving us:

$$\chi(M, \mathcal{O}_M(E)) = c_1(E''(-n)) + 1 - g + \chi\left(\coprod_{r-1} \mathcal{O}_M(-n)\right)$$

From this, we shall proceed with inducting on the rank r. The base case $r = 1$ was theorem 6.1.1. Assume Riemann Roch holds for $r - 1$, so that by the inductive hypothesis:

$$\chi\left(\coprod_{r-1} \mathcal{O}_M(-n)\right) = c_1\left(\coprod_{r-1} \mathcal{O}_M(-n)\right) + (r-1)(1-g)$$

Then:

$$\chi(M, \mathcal{O}_M(E)) = c_1(E''(-n)) + 1 - g + c_1\left(\coprod_{r-1} \mathcal{O}_M(-n)\right) + (r-1)(1-g)$$

Then we can combine the Chern classes by additivity on equation (6.1), and so;

$$\chi(M, \mathcal{O}_M(E)) = c_1(E) + r(1-g)$$

as we sought to show.

We shall show that:

$$\chi(M, \mathcal{O}_M(E)) = c_1(E) + \frac{\mathrm{rk}(E)}{2} c_1(TM)$$

6.2 Hirzebruch-Riemann-Roch Theorem

Let us now proceed to prove the main result. The first section builds up the important connections between the three main functions: the χ_y-characteristic, the T_y-characteristic and the spin index. These two shall equal to the Euler characteristic when $y = -1$ and the spin index when $y = 1$. We have showed that the T_y-characteristic is invariant under pullback, and so we shall have to do the same for the χ_y-characteristic. In the next section, we combine these results to show the result holds over split manifolds, then over general projective varieties with the trivial bundle, then over line bundle, and finally over vector bundles.

6.2.1 Index, T_y-characteristic, χ_y-characteristic

If with $y = 1$, then we get the following key result:

Theorem 6.2.1: Euler Characteristic and Index

$$\chi_1(M) = \sum_{p=0}^{n} \chi^p(M) = \sum_{p,q} (-1)^q h^{p,q}(M)$$

is equal to the *spin index* $\tau(M)$

Throughout this proof, let M_n represent an n-dimensional manifold, and M represent the same manifold when it is not necessary to remember the dimension.

***Proof*:**
This proof will be very rapid, a more detailed exposition can be found in [16, theorem 15.8.2].

If n is odd then by proposition 3.3.2:

$$\chi^p(M_n) = (-1)^n \chi^{n-p}(M_n) = -\chi^{n-p}(M_n)$$

Thus $\sum_{p=0}^{n} \chi^p(M_n) = 0$. On the other hand, $\tau(M_n) = 0$ for odd n. Thus for n odd the theorem is true for arbitrary compact complex manifolds.

Now suppose n is even and $z_j = x_{2j-1} + ix_{2j}$ are local complex coordinates. In this proof, we shall use the orientation given by the natural order $dx_1 \wedge dx_2 \wedge \cdots \wedge dx_{2n}$. Often in the literature on Kähler manifolds, the orientation for M_n is given by $dx_1 \wedge dx_3 \wedge \cdots \wedge dx_{2n-1} \wedge dx_2 \wedge dx_4 \wedge \cdots \wedge dx_{2n}$ (ex. see [17]). The two orientations differ by a sign $(-1)^{\frac{n(n-1)}{2}}$. Without loss of generality, assume $n = 2m$ to avoid sign issues.

Let us now decompose $H^n(M, \mathbb{R})$ into the orthogonal subspaces on which it will be easy to compute τ. Let $B^{p,q}$ be the complex vector space of harmonic forms of type (p, q). By theorem 3.3.5, the fundamental form ω is a harmonic form of type $(1, 1)$ whose product with any other harmonic form is again harmonic. Define a homomorphism

$$L : B^{p,q} \to B^{p+1,q+1}$$

by associating to each form $\alpha \in B^{p,q}$ the form $L(\alpha) = \omega\alpha \in B^{p+1,q+1}$. Then, since ω is real, $\overline{L\alpha} = L\bar{\alpha}$. Recall there is an anti-isomorphism:

$$\star : B^{p,q} \to B^{n-p,n-q}$$

where $\star(\alpha) = \overline{\star(\bar{\alpha})} = \star(\bar{\alpha})$. From it, define the homomorphism:

$$\Lambda : B^{p,q} \to B^{p-1,q-1}$$

defined by $\Lambda = (-1)^{p+q} \star L \star$. Then $\Lambda = (-1)^{p+q} \star L \star$ and $\overline{\Lambda\alpha} = \Lambda\bar{\alpha}$.

The kernel of Λ is denoted by $B_0^{p,q}$ and called the subspace of effective harmonic forms of type (p, q). Then:

$$\Lambda L^k : B_0^{p-k,q-k} \to B^{p-1,q-1}$$

$(p + q \leq n, k \geq 1)$ is (up to a non-zero scalar factor) equal to L^{k-1} and

$$L^k : B_0^{p-k,q-k} \to B^{p,q}$$

$(p+q \leq n)$ is injective.

For $p+q \leq n$, by taking advantage of the Hodge decomposition and the commutativity of $\star$, we get:

$$B^{p,q} = B_0^{p,q} \oplus LB_0^{p-1,q-1} \oplus \cdots \oplus L^r B_0^{p-r,q-r}$$

where $r = \min(p,q)$. Define $B_k^{p,q} = L^k B_0^{p-k,q-k}$. The elements of $B_k^{p,q}$ are called harmonic forms of type (p,q) and class k. Next, for $\varphi \in B_k^{(p,q}$ where $p+q = n$:

$$\star\varphi = (-1)^{q+k}\bar{\varphi}$$

so $\bar{\varphi} \in B_k^{q,p}$. Now, by using theorem 3.1.7 with the above results, the cohomology group $H^n(M_n, \mathbb{C})$ can be decomposed into:

$$H^n(M_n, \mathbb{C}) = \sum_{\substack{p+q=n \\ k \leq \min(p,q)}} B_k^{p,q}.$$

The summands in the direct sum decomposition are mutually orthogonal with respect to the scalar product:

$$(\alpha, \beta) = \int_{M_n} \alpha \wedge \star\beta$$

Indeed, the scalar product can be non-zero only if $\alpha \wedge \star\beta$ is of type (n,n). Therefore $B_k^{p,q}$, $B_{k'}^{p',q'}$ are orthogonal for $(p,q) \neq (p',q')$. If $\alpha \in B_k^{p,q}$ and $\beta \in B_{k'}^{p,q}$ for $k > k'$ and $p+q = n$ then

$$(\alpha, \beta) = (L^k\alpha_0, L^{k'}\beta_0) \text{ with } \alpha_0, \beta_0 \text{ effective } (\Lambda\alpha_0 = \Lambda\beta_0 = 0).$$

Since L and Λ are adjoint operators, $(L\alpha, \varphi) = (\alpha, \Lambda\varphi)$, and therefore, by

$$(\alpha, \beta) = (\alpha_0, \Lambda^k L^{k'}\beta_0) = 0$$

Now, the real cohomology groups $H^n(M_n, \mathbb{R})$ can be identified with the real vector space of real harmonic forms. In particular, there is the direct sum decomposition:

$$H^n(M_n, \mathbb{R}) = \sum_{\substack{p+q=n \\ k \leq p \leq q}} E_k^{p,q}$$

where $E_k^{p,q}$ is the real vector space of real harmonic forms α which can be written in the form $\alpha = \varphi + \bar{\varphi}$ with $\varphi \in B_k^{p,q}$ (and hence $\bar{\varphi} \in B_k^{q,p}$). By definition of $\tau(M_n)$, we have:

$$Q(\alpha, \beta) = \int_{M_n} \alpha \wedge \beta \quad (\alpha, \beta \in H^n(M_n, \mathbb{R})).$$

By taking the results of the complex, the real vector space summands in the real decomposition are mutually orthogonal with respect to this quadratic form. Now $\star\varphi = (-1)^{q+k}\bar{\varphi}$ implies that the quadratic form $(-1)^{q+k}Q(\alpha, \beta)$ is positive definite when restricted to $E_k^{p,q}$. Therefore

$$\tau(M_n) = \sum_{\substack{p+q=n \\ k \leq p \leq q}} (-1)^{q+k} \dim_{\mathbb{R}} E_k^{p,q}$$

Clearly $\dim_{\mathbb{R}} E_k^{p,q} = 2\dim_{\mathbb{C}} B_k^{p,q}$ for $p < q$. If $n = 2m$ then $\dim_{\mathbb{R}} E_k^{m,m} = \dim_{\mathbb{C}} B_k^{m,m}$.
Therefore

$$\tau(M_n) = \sum_{\substack{p+q=n \\ k\leq \min(p,q)}} (-1)^{q+k} \dim_{\mathbb{C}} B_k^{p,q}$$

Now, let $h^{p,q} = \dim_{\mathbb{C}} B^{p,q}$. Then:

$$h^{p-k,q-k} - h^{p-k-1,q-k-1} = \dim_{\mathbb{C}} B_k^{p,q} \text{ for } p+q \leq n.$$

Since $h^{r,s} = h^{s,r} = h^{n-r,n-s}$:

$$h^{p-k-1,q-k-1} = h^{p+k+1,q+k+1} \text{ for } p+q = n.$$

Combining this all, we get;

$$\begin{aligned}
\tau(M_n) &= \sum_{\substack{k\geq 0 \\ p+q=n}} (-1)^{q-k} h^{p-k,q-k} + \sum_{\substack{k\geq 0 \\ p+q=n}} (-1)^{q+k+1} h^{p+k+1,q+k+1} \\
&= \sum_{p+q\leq n} (-1)^q h^{p,q} + \sum_{p+q>n} (-1)^q h^{p,q} \\
&= \sum_{p,q} (-1)^q h^{p,q}
\end{aligned}$$

as we sought to show.

Theorem 6.2.2: T-Characteristic and Index

$$T_1(M_n) = \sum_{p=0}^{n} T^p(M_n) = \tau(M_n)$$

is equal to the *spin index* $\tau(M)$

***Proof*:**
Let $Q(y;x)$ be the associated power series for T_y. Then:

$$Q(1;x) = \frac{x(2e^{2x} - (e^{2x}-1))}{e^{2x}-1} = \frac{x(e^{2x}+1)}{e^{2x}-1} = x\coth(x) = \frac{x}{\tanh(x)}$$

which as we saw in section 5.3.1 is the power series associated to the spin index.

By combining theorem 6.2.2 with proposition 4.3.5 and apply theorem 5.3.5 we get:

$$2^n T(M) = \sum_{l=0}^{n} \sum_{1\leq i_1<\cdots<i_l\leq n} \tau(a_{i_1},\ldots,a_{i_l})_M$$

From this we get:

Theorem 6.2.3: Splitting and Virtual Todd Genus

Let M be a compact almost complex split manifold, and let $\ell_1, \ldots, \ell_r$ be elements of $H^2(M, \mathbb{Z})$. The virtual Todd genus $T(\ell_1, \ldots, \ell_r)_M$ multiplied by 2^{n-r} is an integer.

Next, recall theorem 3.4.9 which shows that χ_y is the unique function with the desired properties. In section 4.3 and section 4.2.1, every condition of the virtual T_y-characteristic was shown to satisfy the conditions for theorem 3.4.9 except $T_y(\{L_1, ..., L_r\}, E) = \chi_y(\{L_1, ..., L_r\}, E)$. With the results on the spin index, we have:

Theorem 6.2.4: Virtual Characteristics Equivalence

Let M be a projective variety and $L_1, ..., L_r$ complex line bundles over M with cohomology classes $\ell_1, ..., \ell_r \in H^2(M, \mathbb{Z})$. Then:

$$\chi_1(\{L_1, ..., L_r\})_M = T_1(\{L_1, ..., L_r\})_M = \tau(\{\ell_1, ..., \ell_r\})_M$$

***Proof*:**
By the above theorems we have $\chi_1(M) = T_1(M) = \tau(M)$. By taking $V \hookrightarrow M$ and considering the characteristic and spin index of V, it is easy to show the equality holds. Thus, the virtual indexes are equal.

Next, we present a theorem by Borel that shows that the Euler characteristic is invariant under pullback to a split manifold:

Theorem 6.2.5: Euler-characteristic Invariant Under Splitting Pullback

Let M be a Kähler manifold, E a complex vector bundle over M, M^Δ the split manifold with projection $\varphi : M^\Delta \to M$. Let F be any vector bundle over M^Δ. Then:

$$\chi_y(M^\Delta, \varphi^*(E)) = \chi_y(M, E)\chi_y(F)$$

If $F = \mathbf{F}(q)$ then

$$\chi_y(M^\Delta, \varphi^*(E)) = \chi_y(M, E)$$

***Proof*:**
The second quality comes from $\chi(\mathbf{F}(q)) = 1$. For the first equation, the proof is long and would require knowledge of spectral sequence. For an introduction to spectral sequences, I recommend [8], and for the proof see [16, Appendix 2].

6.2.2 Hirzebruch Riemann Roch

Let M be a projective variety, $T(M)$ be the Todd genus, and $\chi(M)$ be the arithmetic genus. We shall show that $T(M) = \chi(M)$.

Theorem 6.2.6: Result over split manifold

Let M be an projective variety which is also a complex split manifold. Then:

$$\chi(M) = T(M)$$

***Proof*:**
Let $m = \dim M$, and $A_1, ..., A_m$ the complex diagonal line bundles defined over M. Then by proposition 4.3.5:

$$(1+y)^m T(M) = \sum_{l=0}^{m} y^l \sum_{1 \leq i_1 < \cdots < i_l \leq n} T_y(a_{i_1}, ... a_{i_l})_M$$

where the T_y are polynomials, and by theorem 3.4.6

$$(1+y)^m \chi(M) = \sum_{l=0}^{m} y^l \sum_{i_1 < \cdots < i_l} \chi(A_{i_1}, \ldots, A_{i_l})_M$$

As M is algebraic, by theorem 3.4.8 χ_y is a polynomial. Then if the two agree on some $y \neq -1$, we are done. By choosing $y = 1$, theorem 6.2.1 and theorem 6.2.2 show they are equate, completing the proof.

Now, as we know we can pullback any vector bundle E over M to via the map $M^\Delta \to M$ where M^Δ is the flag manifold associated to $\mathbf{F}(q)$, by theorem 4.4.7 we have $T(M) = T(M^\Delta)$. We showed the same for the topological Euler characteristic, we now show the same for the vector bundle:

Theorem 6.2.7: Euler-characteristic Invariant Under Splitting

Let E be a complex $\mathrm{GL}_q(\mathbb{C})$-bundle over a projective variety M, M^Δ the splitting pullback with flag bundle $\mathbf{F}(q)$. Let φ be the map and $\varphi^*(M)$ be the pullback. Then

$$\chi(M, E) = \chi(M^\Delta, E)$$

***Proof*:**
First, theorem 3.2.10 implies that M^Δ is algebraic. Let φ be the projection from M^Δ on to M. The bundle $\varphi^* E$ over M^Δ splits: let $E_1, \ldots, E_q$ be the corresponding $\mathbb{C}^*$-bundles and let γ_i be the image of $c_1(E_i)$ under the natural homomorphism $H^2(M^\Delta, \mathbb{Z}) \to H^2(M^\Delta, \mathbb{C})$. By theorem 3.3.5 the cohomology classes γ_i are of type $(1, 1)$. The cohomology homomorphism φ^* maps $H^*(M, \mathbb{C})$ injectively into $H^*(M^\Delta, \mathbb{C})$, and since φ is a complex analytic map, φ^* maps cohomology classes of type (p, q) to cohomology classes of type (p, q). As $H^*(M^\Delta, \mathbb{C})$ is generated by $\varphi^* H^*(M, \mathbb{C})$ and the γ_i (see [12]), since each γ_i is of type $(1, 1)$, all cohomology classes of type $(0, p)$ in $H^*(M^\Delta, \mathbb{C})$ must lie in $\varphi^* H^*(M, \mathbb{C})$. Therefore $h^{0,p}(M^\Delta) = h^{0,p}(M)$ and hence $\chi(M) = \chi(M^\Delta)$, as we sought to show.

Theorem 6.2.8: Hirzebruch Riemann Roch for Line Bundles

Let M be a projective variety. Then:

$$\chi(M) = T(M)$$

If L is a line bundle, then:

$$\chi(M, L) = \int_M \mathrm{ch}(L) \cdot \mathrm{td}(M)$$

***Proof*:**
Let M^Δ be the split manifold. Then:

$$\chi(M) \overset{\text{th } 6.2.7}{=} \chi(M^\Delta) \overset{\text{th } 6.2.6}{=} T(M^\Delta) \overset{\text{th } 4.4.5}{=} T(M)$$

For adding the line bundle, consider $L_1, ..., L_r$ complex line bundles over M. Then by theorem 6.2.4

$$\chi(\{L_1, ..., L_r\})_M = T(\{L_1, ..., L_r\})_M$$

For $r = 1$, $L_1 = L$, $\chi(L)_M = T(L)_V$. By proposition 3.4.2:

$$\chi(L) = \chi(M) - \chi(M, L^{-1})$$

and by proposition 4.3.2

$$T(L) = T(M) - T(M, L^{-1})$$

replacing L by L^{-1} As $T(M) = \chi(M)$ and $T(L) = \chi(L)$, we manipulate and get:

$$\chi(M, L) = T(M, L)$$

as we sought to show.

Finally:

Theorem 6.2.9: Hirzebruch Riemann Roch

Let M be a projective variety and E a complex vector bundle (equiv. a coherent sheaf). Then:

$$\chi(M, E) = \int_M \mathrm{ch}(E) \cdot \mathrm{td}(M)$$

***Proof*:**
let E be a complex vector bundle over M, and take the usual $\varphi : M^\Delta \to M$ such that $\varphi^*(E)$ splits. Then by theorem 4.4.5

$$T(M, E) = T(M^\Delta, \varphi^*(E))$$

By theorem 6.2.5:

$$\chi(M, E) = \chi(M^\Delta, \varphi^*(E))\chi(\mathbf{F}(q)) = \chi(M^\Delta, \varphi^*(E))$$

Thus, we can reduce line bundles, in particular given $\varphi^*(E)$ splits into $A_1, ..., A_q$, by use of the properties of T shown in section 4.3:

$$T(M^\Delta, \varphi^*(E)) = \sum_{i=1}^{q} T(M^\Delta, A_i)$$

and by theorem 3.3.7:

$$\chi(M^\Delta, \varphi^*(E)) = \sum_{i=1}^{q} \chi(M^\Delta, A_i)$$

As M^Δ is projective by theorem 2.2.3, by theorem 6.2.8 for all $1 \leq i \leq q$:

$$\chi(M^\Delta, A_i) = T(M^\Delta, A_i)$$

But then combining everything we get:

$$\chi(M, E) = T(M, E)$$

as we sought to show.

6.3 *Grothendieck Riemann Roch

This section is simply for exposure and does not go into the necessary background, but it does provide an outline of the proof of Grothendieck Riemann Roch.

The Grothendieck-Riemann-Roch (GRR) theorem is a generalization of Hirzebruch Riemann Roch as a property of proper morphisms. Recall the commutative diagram presented at the end of section 2.4:

$$\begin{tikzcd} K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(Y) \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TY)"] \\ A_*(X;\mathbb{Q}) \arrow[r, "f_*"] & A_*(Y;\mathbb{Q}) \end{tikzcd}$$

The Hirzebruch Riemann Roch can be thought of as:

$$\begin{tikzcd} K(X) \arrow[r, "f_!"] \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TX)"'] & K(\mathrm{pt}) \arrow[d, "\mathrm{ch}(-)\mathrm{td}(TY)"] \\ A_*(X;\mathbb{Q}) \arrow[r, "f_*"] & \mathbb{Q} \end{tikzcd}$$

as we will demonstrate in this section. This section provides a brief overview of the theorem and its proof, following the exposition in Ravi Vakil's notes on intersection theory [19]. The main idea in Hirzebruch's proof of using the equivalence between the Euler characteristic and T-characteristic going through the spin index τ is replaced by showing any proper morphism f can be decomposed into a closed immersion and a projection. The closed immersion will in fact reduce further to only needing to prove the result on $\mathbb{P}^n \to \mathrm{pt}$ which exactly mirrors the Hirzebruch Riemann Roch proof, mimicking the strategy done in the introduction of chapter chapter 4. The proof using closed immersion is where the virtual results are replaced. The codimension information is still present but generalized into intersection with *normal cones* (think of the construction of a vector bundle from a locally free sheaf, $\mathrm{Spec}_X(\mathrm{Sym}(\mathcal{F}))$, and replace $\mathrm{Sym}(\mathcal{F})$ with R which can hold "singular" information).

Build-up

The key objects in the Grothendieck-Riemann-Roch theorem are defined in a purely algebraic context. The *Grothendieck group of vector bundles*, denoted $K^0(X)$, is the abelian group generated by the isomorphism classes of vector bundles (locally free sheaves of finite rank) on a scheme X where the group operation $[E] = [E'] + [E'']$ is given for every $0 \to E' \to E \to E'' \to 0$ of vector bundles. The tensor product of vector bundles gives $K^0(X)$ a commutative ring structure. Similarly, the *Grothendieck group of coherent sheaves*, denoted $K_0(X)$, is the abelian group generated by isomorphism classes of coherent sheaves on X, with the same relation for short exact sequences. If X is a smooth variety, the natural map $K^0(X) \to K_0(X)$ is an isomorphism, and we denote the group by $K(X)$.

For a proper morphism $f : X \to Y$, take the pushforward map $f_* : K_0(X) \to K_0(Y)$. Pushing forward a coherent sheaf $\mathcal{F}$ via f does not generally preserve exactness, namely the functor f_* is only left-exact (see [34], [3]). The failure of exactness is measured by the higher direct image sheaves, $R^i f_* \mathcal{F}$. For a short exact sequence of coherent sheaves $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ on X, applying the pushforward functor yields a long exact sequence of coherent sheaves on Y:

$$0 \to f_*\mathcal{F}' \to f_*\mathcal{F} \to f_*\mathcal{F}'' \to R^1 f_*\mathcal{F}' \to R^1 f_*\mathcal{F} \to \cdots$$

To ensure that the pushforward is a homomorphism on the Grothendieck group, it must preserve the defining additive relation. This is achieved by defining the pushforward via an alternating sum (see [34]):

$$f_*([\mathcal{F}]) = \sum_{i \geq 0} (-1)^i [R^i f_* \mathcal{F}]$$

With this definition, the long exact sequence ensures that $f_*([\mathcal{F}]) = f_*([\mathcal{F}']) + f_*([\mathcal{F}''])$ in $K_0(Y)$.

This pushforward generalizes the Euler-Poincaré characteristic: by taking $Y = \mathrm{Spec} k$ to be a point, then $K_0(Y) \cong \mathbb{Z}$ (a coherent sheaf on a point is a finite-dimensional vector space, and its class is its dimension). For a proper scheme X over k, the higher direct image sheaves $R^i f_* \mathcal{F}$ correspond to the cohomology groups $H^i(X, \mathcal{F})$. Specifically, $R^i f_* \mathcal{F}$ is a sheaf on a point, so its class in $K_0(\mathrm{Spec} k)$ is simply $\dim_k H^i(X, \mathcal{F})$. Therefore, the pushforward to a point is:

$$f_*([\mathcal{F}]) = \sum_{i \geq 0} (-1)^i \dim_k H^i(X, \mathcal{F}) = \chi(X, \mathcal{F})$$

Next, the intersection-theoretic Chern classes are defined as operators on the Chow groups. For a vector bundle E on a scheme X, the *i-th operational Chern class*, $c_i(E)$, is a homomorphism

$$c_i(E) \cap - : A_k(X) \to A_{k-i}(X)$$

for all k. These classes satisfy properties analogous to those of the topological Chern classes used in the Hirzebruch-Riemann-Roch theorem, such as the Whitney sum formula. For smooth varieties, these operational classes correspond to the classes defined via topology (see [20]).

The *Chern character* is a ring homomorphism $\mathrm{ch} : K(X) \to A_*(X) \otimes \mathbb{Q}$. It is defined to be additive over direct sums, which makes it compatible with the Grothendieck group structure. For a vector bundle E with Chern roots $\gamma_1, \cdots, \gamma_r$ (formal symbols such that $c_k(E)$ is the k-th elementary symmetric polynomial in the γ_i), the Chern character is

$$\mathrm{ch}(E) = \sum_{i=1}^{r} e^{\gamma_i} = \mathrm{rank}(E) + c_1(E) + \frac{1}{2}(c_1(E)^2 - 2c_2(E)) + \cdots$$

This definition extends to all elements of $K(X)$. Its key properties are:

- $\mathrm{ch}([E] + [F]) = \mathrm{ch}(E) + \mathrm{ch}(F)$
- $\mathrm{ch}([E] \otimes [F]) = \mathrm{ch}(E) \cdot \mathrm{ch}(F)$

The *Todd class* of a vector bundle E with Chern roots $\gamma_1, \cdots, \gamma_r$ is

$$\mathrm{td}(E) = \prod_{i=1}^{r} \frac{\gamma_i}{1 - e^{-\gamma_i}} \in A_*(X) \otimes \mathbb{Q}.$$

The Todd class is multiplicative on short exact sequences: if $0 \to E' \to E \to E'' \to 0$ is exact, then $\mathrm{td}(E) = \mathrm{td}(E') \cdot \mathrm{td}(E'')$. Thus, the Chern character and Todd class are defined verbatim to how they were defined in this paper up to change of mathematical domain.

Grothendieck Riemann Roch

The commutative diagram in the beginning of this section can be translated into the following:

Theorem 6.3.1: Grothendieck-Riemann-Roch

Let $f : X \to Y$ be a proper morphism of smooth quasi-projective varieties. Then for any $\alpha \in K(X)$, the following equality holds in $A_*(Y) \otimes \mathbb{Q}$:

$$\mathrm{ch}(f_![\alpha]) \cdot \mathrm{td}(TY) = f_*(\mathrm{ch}([\alpha]) \cdot \mathrm{td}(TX)).$$

Let us reduce this proof in the way stated at the beginning:

Proposition 6.3.2: Reduction to Projections and Immersions

To prove the Grothendieck-Riemann-Roch theorem for any proper morphism $f : X \to Y$ between smooth varieties, it is suffices to prove GRR for the cases where f is a closed immersion and where f is a projection.

***Proof*:**
The GRR theorem can be stated as the commutativity of the diagram involving the transformation $T_Z(\alpha) = \mathrm{ch}(\alpha) \cdot \mathrm{td}(TZ)$. If we have two morphisms $f : X \to Z$ and $g : Z \to Y$, the pushforward is functorial, so $(g \circ f)_* = g_* \circ f_*$ for both K-theory and Chow groups. If the diagram commutes for f and g, then

$$T_Y \circ (g \circ f)_* = T_Y \circ g_* \circ f_* = g_* \circ T_Z \circ f_* = g_* \circ f_* \circ T_X = (g \circ f)_* \circ T_X.$$

So the diagram also commutes for the composition $g \circ f$. Any proper morphism $f : X \to Y$

between smooth varieties can be factored as a closed immersion $i : X \hookrightarrow \mathbb{P}^n \times Y$ followed by a projection $p : \mathbb{P}^n \times Y \to Y$. Thus, proving the theorem for closed immersions and projections suffices.

Proposition 6.3.3: GRR for Projections

The Grothendieck-Riemann-Roch theorem holds for the projection $\pi : \mathbb{P}^n \times Y \to Y$.

***Proof*:**
We need to show that for any $\alpha \in K(\mathbb{P}^n \times Y)$, $\pi_*(T_{\mathbb{P}^n \times Y}(\alpha)) = T_Y(\pi_* \alpha)$. By the Künneth formula, it suffices to check for elements of the form $\alpha = \beta \otimes \gamma$, where $\beta \in K(\mathbb{P}^n)$ and $\gamma \in K(Y)$. The tangent bundle splits as $T\mathbb{P}^n \times Y \cong \pi_1^* T\mathbb{P}^n \oplus \pi_2^* TY$, so the Todd class is multiplicative: $\mathrm{td}(T\mathbb{P}^n \times Y) = \pi_1^* \mathrm{td}(T\mathbb{P}^n) \cdot \pi_2^* \mathrm{td}(TY)$. The Chern character is also multiplicative for exterior products. The right side of the GRR formula becomes:

$$\begin{aligned} \pi_*(T_{\mathbb{P}^n \times Y}(\beta \otimes \gamma)) &= \pi_*(\mathrm{ch}(\beta \otimes \gamma) \cdot \mathrm{td}(T_{\mathbb{P}^n \times Y})) \\ &= \pi_*(\pi_1^*(T_{\mathbb{P}^n}(\beta)) \cdot \pi_2^*(T_Y(\gamma))) \\ &\overset{!}{=} \pi_*(\pi_1^*(T_{\mathbb{P}^n}(\beta))) \cdot T_Y(\gamma) \end{aligned}$$

where $\overset{!}{=}$ comes from intersection theory (see ex. [19]). The term $\pi_*(\pi_1^*(T_{\mathbb{P}^n}(\beta)))$ is the pushforward of a class from $\mathbb{P}^n$ to a point, which is the usual integration by De Rham Theory. The left side of the GRR formula is:

$$\begin{aligned} T_Y(\pi_*(\beta \otimes \gamma)) &= T_Y(\pi_*(\beta) \cdot \gamma) \\ &= \mathrm{ch}(\pi_*(\beta)) \cdot T_Y(\gamma) \end{aligned}$$

Comparing the two sides, the GRR formula for the projection π reduces to showing that $\mathrm{ch}(\pi_*(\beta)) = \int_{\mathbb{P}^n} T_{\mathbb{P}^n}(\beta)$ for any $\beta \in K(\mathbb{P}^n)$. This is precisely the Hirzebruch-Riemann-Roch theorem for $\mathbb{P}^n$ (pushing forward to a point), which can be verified directly (recall the computations done in the beginning of chapter 4; they are exactly this!). To recall the result of that section and generalizing a bit, for a line bundle $\mathcal{O}(k)$, $\pi_*[\mathcal{O}(k)] = \chi(\mathbb{P}^n, \mathcal{O}(k)) = \binom{n+k}{n}$. Then:

$$\binom{n+k}{n} = \int_{\mathbb{P}^n} \mathrm{ch}(\mathcal{O}(k)) \cdot \mathrm{td}(T\mathbb{P}^n),$$

Since the classes of line bundles generate $K(\mathbb{P}^n)$, the theorem holds for projections.

Proposition 6.3.4: GRR for Closed Immersions

The Grothendieck-Riemann-Roch theorem holds for any closed immersion $i : X \hookrightarrow Y$ of smooth varieties.

This is where the virtual machinery developed in this paper is replaced with the machinery of intersection theory.

***Proof*:**
This is an overview of the proof in [19] so that the reader gets a sense of the development of the machinery of intersection theory.

First, Vakil proves a special case where X is embedded as the zero section in a projective bundle over itself. Then, the general case is reduced to this special case by using the deformation to the normal cone (one of the key idea of intersection theory).

Step 1 – The special case: $i : X \hookrightarrow Y = \mathbb{P}(N \oplus \mathcal{O}_X)$
Let N be a vector bundle on X, and consider the immersion i of X as the zero section into the projective bundle $Y = \mathbb{P}(N \oplus \mathcal{O}_X)$. The normal bundle of this immersion is $N_{X/Y} \cong N$. For any vector bundle $\mathcal{E}$ on X, one can construct a Koszul resolution of $i_*\mathcal{E}$ on Y. This resolution allows for an explicit computation of $\mathrm{ch}(i_*\mathcal{E})$ in terms of the Chern classes of $\mathcal{E}$ and the bundles involved in the geometry of the projective bundle Y. An algebraic manipulation, relying on the properties of Chern characters and Todd classes under short exact sequences, shows that the GRR formula holds in this specific situation. The formula essentially reduces to an identity involving the Todd class of the normal bundle N.

Step 2 – Deformation to the normal cone
For a general closed immersion $i : X \to Y$, the deformation to the normal cone is a flat family $M \to \mathbb{P}^1$ whose fiber over $0 \in \mathbb{P}^1$ is Y and whose fiber over $\infty \in \mathbb{P}^1$ is the special fiber $M_\infty = \mathrm{Bl}_X(Y) \cup_E \mathbb{P}(C_{X/Y} \oplus \mathcal{O}_X)$, where E is the exceptional divisor of the blow-up of Y along X. The key insight is that because the fibers are rationally equivalent, any intersection-theoretic formula that holds on one fiber must hold on the other (which matches with the intuition for flat families as presented in [34], along with the added rational-equivalent requirement).

Now, if the formula holds for the immersions into the special fiber M_∞, it must hold for the original immersion $i : X \to Y$. The special fiber consists of two pieces glued along the exceptional divisor E. The GRR formula is additive over unions, and the contribution from the blow-up part will cancels, leaving only the contribution from the piece $\mathbb{P}(C_{X/Y} \oplus \mathcal{O}_X)$.

This reduces the problem to proving GRR for the immersion of X into $\mathbb{P}(C_{X/Y} \oplus \mathcal{O}_X)$, in particular this reduces the general problem to the special case proved in Step 1, completing the proof of the Grothendieck-Riemann-Roch theorem.

6.4 ** Current Generalization Results

Since the publication of the Hirzebruch-Riemann-Roch in 1954, many generalizations have taken place. In the purely algebraic perspective, Grothendeick proved the Grothendieck-Riemann-Roch in 1957. In 1963, Atiyah and Singer proved the *Atiyah-Singer Index Theorem* (ASIT), a generalization in a more analytic context by taking advantage of the fact that the Hodge operator $\overline{\partial}$ is an *elliptic operator* and uses *Bott periodicity* to replace the more constricted use-case of cobordism (notice they are both 4-periodic theories).

These two results have since been pushed further. ASIT has been pushed as far being well-posed on quasi-conformal manifolds (a generalization of a conformal structure to a more topological setting) by Connes, Sullivan, and Teleman in 1994 ([22]). Some variations also work for other types of operators, for example heat equation (which is a parabolic operator) [2].

For the generalizations of GRR, the result has been pushed as far as Noetherian schemes by Navarro,

A. and Navarro, J. in 2017 ([5]). The result in intersection theory strongly suggest that these results can be generalized to derived schemes, and indeed in 2012 a series of author have proved exactly this ([27]). From a geometric point of view, the results were extended as far as non-compact orbifolds in 2016 ([1])

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Bibliography

[1] Gerard Freixas i Montplet and Anna von Pippich. "Riemann-Roch isometries in the non-compact orbifold setting". In: (Sept. 22, 2016). DOI: 10.48550/arXiv.1604.00284. URL: http://arxiv.org/abs/1604.00284.

[2] Peter B. Gilkey and Steven George Krantz. *Invariance Theory: The Heat Equation and the Atiyah-Singer Index Theorem*. eng. 2nd ed. Studies in Advanced Mathematics Ser. Chapman and Hall/CRC, 2018. ISBN: 978-0-8493-7874-4 978-1-351-43643-4.

[3] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.

[4] *Hodge theory*. eng. Mathematical notes. Princeton university press, 2014. ISBN: 978-0-691-16134-1.

[5] A. Navarro and J. Navarro. "On the Riemann-Roch formula without projective hypothesis". May 30, 2017. DOI: 10.48550/arXiv.1705.10769. URL: http://arxiv.org/abs/1705.10769.

[6] C.-Y. Jean Chan. "A Corresondence between Hilbert Polynomial and Chern Polynomials over Projective Spaces". 2004. DOI: 10.48550/arXiv.math/0406588. URL: http://arxiv.org/abs/math/0406588.

[7] John M. Lee. *Introduction to Smooth Manifolds*. Vol. 218. Graduate Texts in Mathematics. Springer, 2003. ISBN: 978-0-387-95448-6 978-0-387-21752-9. URL: http://link.springer.com/10.1007/978-0-387-21752-9.

[8] Allen Hatcher. *Spectral Sequences*. Cornell, NA. URL: https://pi.math.cornell.edu/~hatcher/AT/ATch5.pdf.

[9] Allen Hatcher. *Algebraic Topology*. URL: https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[10] Ming Yang. "Cobordism Theory and Poincare Conjecture". In: (Apr. 27, 2010). DOI: 10.48550/arXiv.0901.2375. URL: http://arxiv.org/abs/0901.2375.

[11] M. F. Atiyah and I. M. Singer. "The index of elliptic operators on compact manifolds". en. In: 69 (3 1963), pp. 422–433. ISSN: 0273-0979, 1088-9485. DOI: 10.1090/S0002-9904-1963-10957-X. URL: https://www.ams.org/bull/1963-69-03/S0002-9904-1963-10957-X/.

[12] Nathanael Chwojko-Srawley. *Everything You Need to know about K-theory.* Vol. 1. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_K-Theory.pdf.

[13] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology.* 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.

[14] Toni Annala. "Derived Algebraic Cobordism". In: (Mar. 22, 2022). DOI: 10.48550/arXiv.2203.12096. URL: http://arxiv.org/abs/2203.12096.

[15] Réné Thom. "Quelques propriétés globales des variétés différentiables". In: 28 (1964), pp. 17–86.

[16] Friedrich Hirzebruch. *Topological Methods in Algebraic Geometry.* 1st ed. Springer, 1995. URL: https://link-springer-com.myaccess.library.utoronto.ca/book/10.1007/978-3-642-62018-8.

[17] W. V. D. Hodge. *A Special Type of K"ahler Manifold.* Vol. 1. 1951, pp. 104–117.

[18] John M. Lee. *Introduction to Riemannian Manifolds.* Vol. 176. Graduate Texts in Mathematics. Springer International Publishing, 2018. ISBN: 978-3-319-91754-2 978-3-319-91755-9. URL: http://link.springer.com/10.1007/978-3-319-91755-9.

[19] Ravi Vakil. "Intersection Theory". en. In: (NA). URL: https://math.stanford.edu/~vakil/245/.

[20] William Fulton. *Intersection Theory.* en. Springer-Verlag, 1984. ISBN: 978-0-387-12176-5.

[21] Jean Gallier and Stephen Shatz. *Complex Algebraic Geometry.* Vol. 1. Feb. 25, 2011.

[22] Alain Connes, Dennis Sullivan, and Nicolas Teleman. "Quasiconformal mappings, operators on hilbert space, and local formulae for characteristic classes". en. In: 33 (4 Oct. 1, 1994), pp. 663–681. ISSN: 00409383. DOI: 10.1016/0040-9383(94)90003-5. URL: https://linkinghub.elsevier.com/retrieve/pii/0040938394900035.

[23] Wei-Liang Chow. "On Compact Complex Analytic Varieties". In: 71 (4 1949), pp. 893–914. ISSN: 0002-9327. DOI: 10.2307/2372375. URL: https://www.jstor.org/stable/2372375.

[24] Lars Valerian Ahlfors. *Complex analysis: an introduction to the theory on analytic functions of one complex variable.* eng. 3. ed. International series in pure and applied mathematics. McGraw-Hill, 2007. ISBN: 978-0-07-000657-7.

[25] Claire Voisin and Leila Schneps. *Hodge theory and complex algebraic geometry I.* eng. Cambridge studies in advanced mathematics. Cambridge University Press, 2002. ISBN: 978-0-511-61534-4.

[26] Pierre Deligne and Luc Illusie. "Relèvements modulop2et décomposition du complexe de de Rham". In: 89 (2 June 1, 1987), pp. 247–270. ISSN: 1432-1297. DOI: 10.1007/BF01389078. URL: https://doi.org/10.1007/BF01389078.

[27] Parker Lowrey and Timo Schürg. "Grothendieck-Riemann-Roch for derived schemes". In: (Aug. 30, 2012). DOI: 10.48550/arXiv.1208.6325. URL: http://arxiv.org/abs/1208.6325.

[28] Phillip A. Griffiths. *Introduction to Algebraic Curves.* Vol. 78. Mathematical Monographs. American Mathematical Society.

[29] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra.* 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.

[30] Nathanael Chwojko-Srawley. *Everything You Need To Know About Differential Geometry.* 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_differential_geometry.pdf.

[31] Allen Hatcher. *Vector Bundles & K-Theory Book*. 2.0. 2017. URL: https://pi.math.cornell.edu/~hatcher/VBKT/VBpage.html.

[32] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.

[33] Jack Hall. "GAGA theorems". May 17, 2022. DOI: 10.48550/arXiv.1804.01976. URL: http://arxiv.org/abs/1804.01976.

[34] Robin Hartshorne. *Algebraic geometry*. Vol. 52. Springer Science & Business Media, 2013. URL: https://books.google.com/books?hl=en&lr=&id=7z4mBQAAQBAJ&oi=fnd&pg=PR13&dq=info:XErvue4ovgwJ:scholar.google.com&ots=OR51pvoipT&sig=NcjWQsiJwE9XL79fBWQTMMub5WM.